

Intermediate Microeconomic Theory Lecture Notes

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Ch0. Math Review

Exogenous vs. Endogenous Variable

- Exogenous variable
 - A variable whose value is taken as given in the analysis of an economic system.
 - Because it's given in the model, a decision maker cannot choose or decide the value.
- Endogenous variable
 - A variable whose value is determined within the economic system being studied
 - The control variables of the decision maker
- Why the exogenous vs. endogenous variable is a big deal?
 - This is because economists focused on the causal relationship

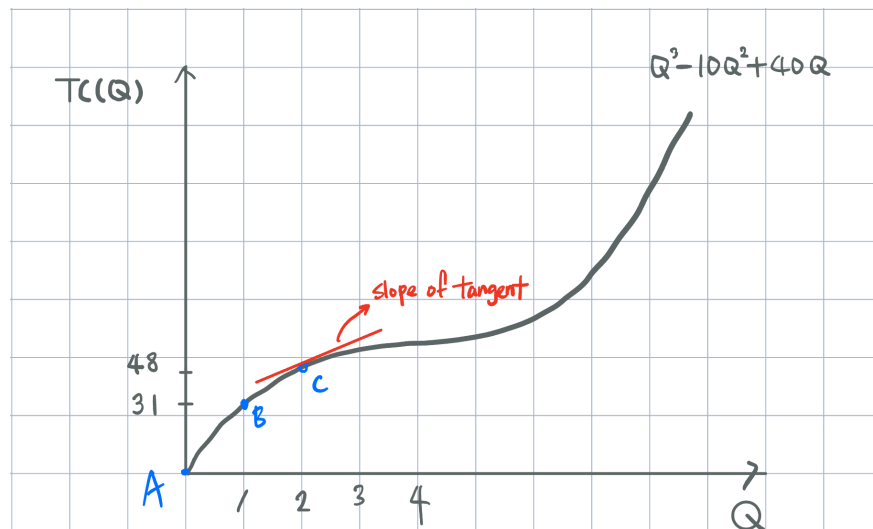
Constrained Optimization

- Constrained optimization
 - An analytical tool for making the best(optimal) choice, taking into account any possible limitations or restrictions on the choice
 - (1) **Objective function:** The relationship that a decision maker seeks to maximize or minimize.
 - (2) **Constraints:** The restrictions or limits imposed on a decision maker in a constrained optimization problem.
 - e.g) Consumer's level of satisfaction when one purchases F units of food and C units of clothing is measured by the product FC . The consumer has a fixed monthly income I . Each unit of food costs P_F and each unit of clothing costs P_C

$$\begin{aligned} & \max FC \\ & \text{subject to } P_F F + P_C C \leq I \\ & \text{exogenous variables: } I, P_F, P_C \\ & \text{endogenous variables: } F, C \end{aligned}$$

Marginal Reasoning

- What is a margin?
 - How a dependent variable changes as a result of adding one unit of an independent variable
- $Q^{Cake} = f(P^{cake})$
 - Q^{Cake} : Quantity of cake (Dependent/endogenous variable)
 - $f(P^{cake})$: Price of cake (Independent/exogenous variable)
- e.g. $C(Q) = Q^3 - 10Q^2 + 40Q$: Total cost function
 - Arc marginal cost ($B \rightarrow C$): $\frac{\Delta C}{\Delta Q}$
 - * If we measure the quantity produced increased by 1 unit from 1 to 2, the total cost increased by $\frac{48}{31} = 1.548$
 - Marginal Cost: $\frac{dC}{dQ}$ (First Derivative/Slope of tangent)
 - * Marginal cost at $Q = 2$ is $\frac{dC}{dQ}|_{Q=2} = 3Q^2 - 20Q + 40 = 12 - 40 + 40 = 12$
 - Average Cost: $\frac{TC}{Q}$
 - * Average cost at $Q = 2$ is $\frac{48}{2} = 24$



Calculus Review

Power Rule If $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

Example: If $f(x) = x^3$, then $f'(x) = 3x^{3-1} = 3x^2$.

Sum Rule If $f(x) = g(x) + h(x)$, then $f'(x) = g'(x) + h'(x)$.

Example: If $f(x) = 3x^2 + 2x$, then $f'(x) = (3 \cdot 2x^{2-1}) + (2 \cdot 1) = 6x + 2$.

Product Rule If $f(x) = g(x)h(x)$, then $f'(x) = g'(x)h(x) + g(x)h'(x)$.

Example: If $f(x) = x^2 \cdot \sin(x)$, then $f'(x) = (2x \cdot \sin(x)) + (x^2 \cdot \cos(x))$.

Quotient Rule If $f(x) = \frac{g(x)}{h(x)}$, then $f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2}$.

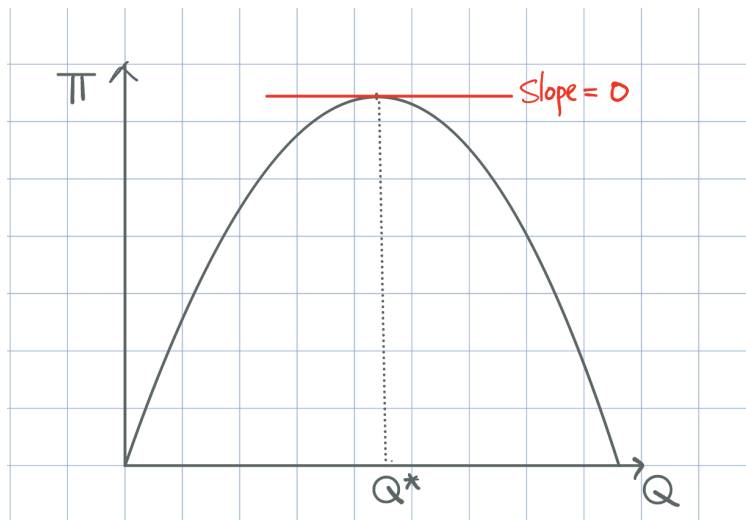
Example: If $f(x) = \frac{x^2}{x+1}$, then $f'(x) = \frac{(2x \cdot (x+1) - x^2 \cdot 1)}{(x+1)^2}$.

Chain Rule If $f(x) = g(h(x))$, then $f'(x) = g'(h(x))h'(x)$.

Example: If $f(x) = (x^2 + 1)^3$, then $f'(x) = 3(x^2 + 1)^2 \cdot (2x)$.

Finding Maximums and Minimums

- e.g. $\Pi(\text{profits}) = f(Q)$
 - FOC(First order condition): $\frac{d\Pi}{dQ} = 0 \rightarrow Q^*$
 - * By setting the FOC, we can find the Q^* which makes the slope of the tangent is equal to zero, and at this Q^* , the function can be maximized or minimized



- e.g. $\Pi = Q^2 - 6Q + 1$
 - FOC: $\frac{d\Pi}{dQ} = 2Q - 6 = 0 \rightarrow Q^* = 3$
- e.g. $13Q_1 - 2(Q_1)^2 + Q_1Q_2 + 8Q_2 - 2(Q_2)^2$
 - FOC: $\frac{d\Pi}{dQ_1} = 0 \rightarrow 13 - 4Q_1 + Q_2 = 0$
 - FOC: $\frac{d\Pi}{dQ_2} = 0 \rightarrow Q_1 + 8 - 4Q_2 = 0$
 - $Q_1^* = 4, Q_2^* = 3$

Ch2. Demand and Supply Analysis

Perfectly Competitive Market

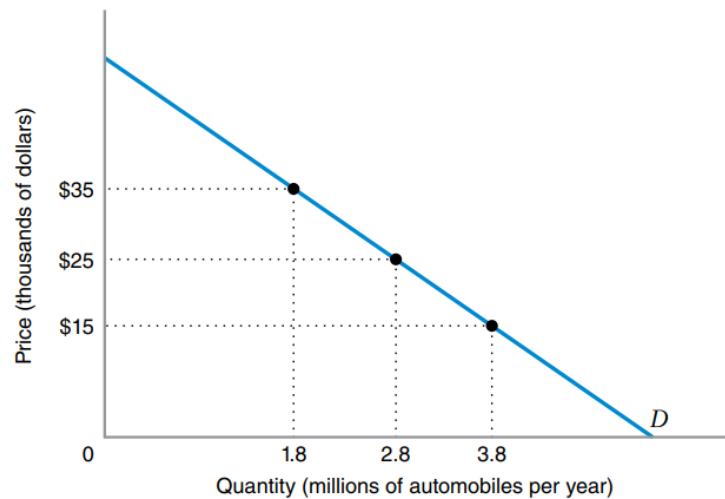
- Perfectly competitive Market assumptions
 - (1) Fragmented market: Large numbers of buyers and sellers are all **price-takers**.
 - (2) Undifferentiated/Homogeneous Products: consumers perceive the product(goods and services) to be identical so they don't care who they buy it from.
 - (3) Perfect information about price: everyone knows the price and compares the prices instantly.
 - (4) Equal access to resource: everyone has access to the same technology and inputs, free entry into the market so if it is profitable for new firms to enter into the market, they will.
- Market demand function
 - It indicates how the quantity of goods demanded by the sum of all consumers in the market depends on various factors

$$Q^d = Q(P, P_o, I, \dots)$$

$$Q^d = Q(P)$$

$$Q^d = a - bP + cM + dA$$

- Market demand curve
 - A curve that shows us the quantity of goods that consumers are willing to buy at different prices
 - The demand curve focuses only on the relationship between the price of the good and the quantity of the good demanded. When we draw the demand curve, we imagine that **all other factors that affect the quantity demanded are fixed**.
 - With market demand, we take the price of a good or service as an exogenous and quantity as an endogenous variable

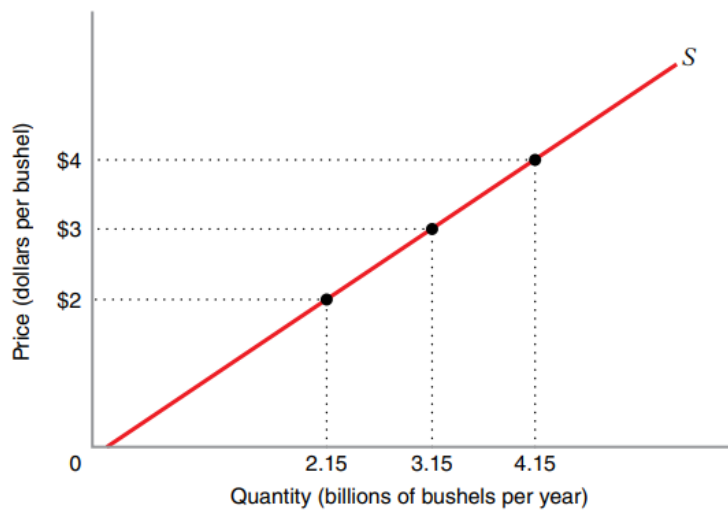


- Law of demand
 - The inverse relationship between the price of a good and the quantity demanded when all other factors that influence demand are held fixed.
- Demand shifters
 - Price of related goods(substitutes, complements), income, number of buyers, tastes, expectations
- Market supply function
 - It indicates how the quantity of goods supplied by the sum of all producers in the market depends on various factors

$$Q^s = Q(P, P_o, w, r, \dots)$$

$$Q^s = Q(P)$$

- Law of supply
 - The positive relationship between price and quantity supplied, when all other factors that influence supply are held fixed.
- Supply shifters
 - Price of related products, input prices, number of sellers, technology, expectations



Price Elasticity of Demand

- Elasticity
 - How much one variable changes(in percentage term) with a 1 percent change in a different variable, either increase or decrease
 - $\epsilon_{x,y} = \frac{\%changeX}{\%changeY} = \frac{endogenous}{exogenous}$
- Price Elasticity of Demand
 - A measure of the rate of percentage change of quantity demanded with respect to price, holding all other determinants of demand constant.

$$\epsilon_{Q^d,P} = \frac{\text{percentage change in quantity}}{\text{percentage change in price}}$$

$$\epsilon_{Q^d,P} = \frac{\% \Delta Q^d}{\% \Delta P} = \frac{\Delta Q^d / Q^d}{\Delta P / P} = \frac{\Delta Q^d}{\Delta P} \cdot \frac{P}{Q^d} = \frac{\partial Q^d}{\partial P} \cdot \frac{P}{Q^d}$$

- e.g. $\epsilon_{Q^d,P} = \frac{-3}{1}$: with 1% increase in price, there is a 3% decrease in quantity demanded(**Don't forget to mention the direction!**)

Value of ϵ	Classification	Meaning
$\epsilon = 0$	Perfectly inelastic	Q_d is completely insensitive to price
$\epsilon < 1 $	inelastic	Q_d is relatively insensitive to price
$\epsilon = 1 $	unitary elastic	% change in Q_d is equal to % change in price
$\epsilon > 1 $	elastic	Q_d is relatively sensitive to price
$\epsilon > \infty $	Perfectly elastic	Q_d become zero or infinity when any change in price

Table 1: Price Elasticity of Demand

- Why Elasticity instead of slope?
 - $\epsilon_{Q^d,P} = \frac{\partial Q^d}{\partial P} \cdot \frac{P}{Q^d} = \text{slope} \cdot \frac{P}{Q^d}$
 - The slope measures the absolute change in quantity demanded brought about by a one-unit change in price.
 - By contrast, the price elasticity of demand measures the percentage change in quantity demanded brought about by a 1 percent change in price.
 - Thus, the price elasticity of demand expresses changes in prices and quantities in common terms. This allows us to compare the sensitivity of quantity demanded to price across different goods or different countries.

- Constant elasticity demand curve
 - A demand curve in the form $Q = aP^\epsilon$. The term ϵ is the price elasticity of demand along this curve
 - e.g. $QP^2 = 200$ then $Q = 200P^{-2}$. So the elasticity $\epsilon = -2$ and is constant at any given Q and P
- Determinants of the price elasticity of demand
 - (1) Demand tends to be more price elastic when there are good **substitutes** for a product.
 - (2) Demand tends to be more price elastic **when a consumer's expenditure on the product is large**(either in absolute terms or as a fraction of total expenditures).
 - (3) Demand tends to be less price elastic when the product is seen by consumers as being a **necessity**.
 - (4) Demand tends to be more elastic for a particular brand of a good, than for the good in general(scope)

Exercise. If $Q^d = 400 - 10p$ and $p = 30$, what is the elasticity of demand with respect to the $p=30$?

Other Elasticities

- Income elasticity of demand
 - The ratio of the percentage change of quantity demanded to the percentage change of income, holding price, and all other determinants of demand constant.

$$\epsilon_{Q^d, I} = \frac{\% \Delta Q^d}{\% \Delta I} = \frac{\Delta Q/Q}{\Delta I/I} = \frac{\Delta Q}{\Delta I} * \frac{I}{Q} = \frac{\partial Q}{\partial I} * \frac{I}{Q}$$

- Cross-price elasticity of demand
 - The ratio of the percentage change of the quantity of one good demanded with respect to the percentage change in the price of another good.

$$\epsilon_{Q_i, P_j} = \frac{\% \Delta Q_i}{\% \Delta P_j} = \frac{\Delta Q_i/Q_i}{\Delta P_j/P_j} = \frac{\Delta Q_i}{\Delta P_j} * \frac{P_j}{Q_i} = \frac{\partial Q_i}{\partial P_j} * \frac{P_j}{Q_i}$$

$$\epsilon_{Q_i, P_j} = \begin{cases} > 0 & \text{demand substitutes} \\ < 0 & \text{demand complements} \end{cases}$$

- Price Elasticity of Supply
 - The percentage change in quantity supplied for each percent change in price, holding all other determinants of supply constant.

$$\epsilon_{Q^s, P} = \frac{\text{percentage change in quantity}}{\text{percentage change in price}}$$

$$\epsilon_{Q^s, P} = \frac{\% \Delta Q^s}{\% \Delta P} = \frac{\Delta Q^s/Q^s}{\Delta P/P} = \frac{\Delta Q^s}{\Delta P} * \frac{P}{Q^s} = \frac{\partial Q^s}{\partial P} * \frac{P}{Q^s}$$

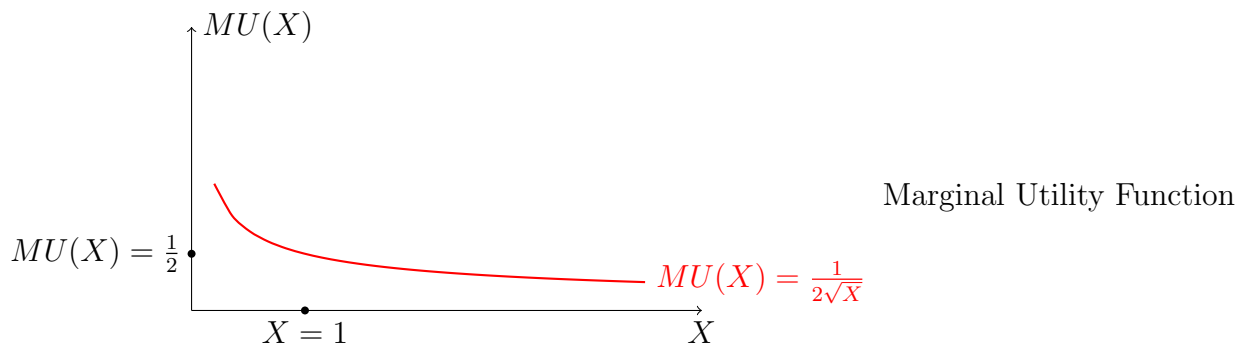
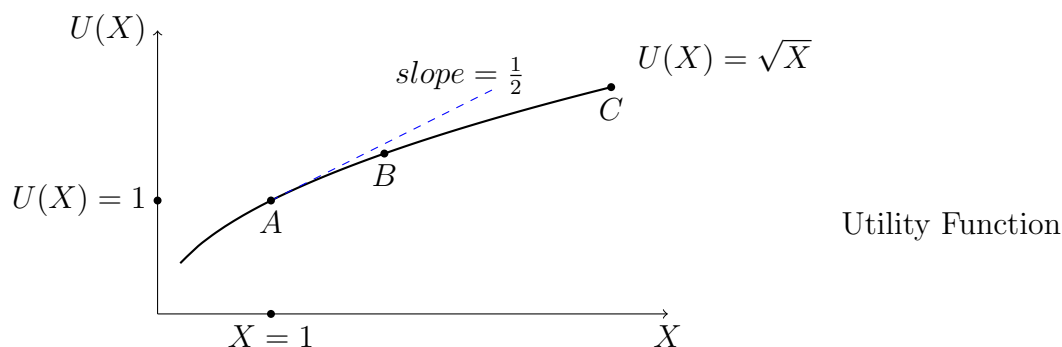
Ch3. Consumer Preferences and the Concept of Utility

Representations of Preferences

- Consumer preferences
 - Indications of how a consumer would **rank** any two possible baskets, assuming the baskets were available to the consumer at no cost.
 - basket(or bundle): A combination of goods and services that an individual might consume
 - * e.g. {coffee, beef, shoes, electricity, water, ...}
- Assumptions about preferences
 - (1) **Complete**: That is, the consumer is able to rank any two baskets. For baskets A and B , for example, the consumer can state her preferences according to one of the following possibilities:
 - she prefers basket A to basket B
 - she prefers basket B to basket A
 - she is **indifferent** between, or equally happy with, baskets A and B
 - (2) **Transitive**: The consumer makes choices that are consistent with each other. If she prefers basket A to B , and basket B to C . We can expect her to prefer basket A to C .
 - (3) **Monotonic**: The more is better. If a basket with more of at least one good and no less of any good is preferred to the original basket, property of non-satiation.
- Ordinal & Cardinal Ranking
 - **Ordinal Ranking**: Ranking that indicates whether a consumer prefers one basket to another. But does not contain quantitative information about the intensity of that preference $\Rightarrow$ Would not tell us how much more she/he likes A than B
 - **Cardinal Ranking**: A quantitative measure of the intensity of a preference for one basket over another.

- Utility Function

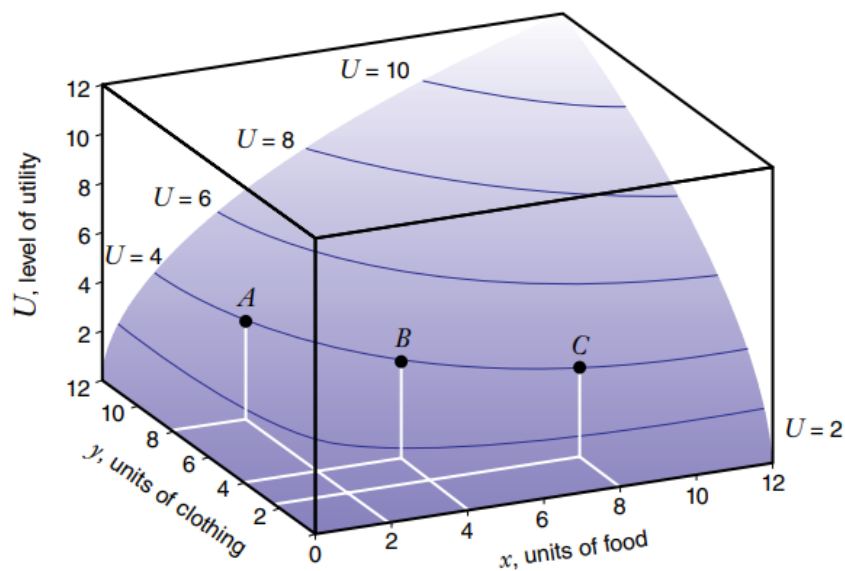
- A function that measures the level of satisfaction a consumer receives from any basket of goods and services (Ordinal concept.)
- e.g. $U = F(x_1, x_2, x_3, \dots, x_n)$: x are quantities of n goods that might be consumed in a period
- e.g. $U(y) = y^5$: total utility of consuming muffins. y = weekly consumption of muffins.
- e.g. $U(X) = \sqrt{X}$



- Marginal Utility (MU)

- The rate at which total utility changes as the level of consumption rises
- $MU_x = \frac{\Delta U}{\Delta x}$: Slope at points give marginal utility (MU), each additional unit makes a person happy but by less than the previous unit
- Graphically, the marginal utility at a particular point is represented by the slope of a line that is tangent to the utility function at that point.

- Principle of Diminishing Marginal Utility
 - The principle that after some point, as consumption of a good increases, the marginal utility of that good will begin to fall.
 - The more of something we consume, the less additional satisfaction we get from additional consumption.
 - (1) Total utility and marginal utility cannot be plotted on the same graph
 - (2) The marginal utility is the slope of the (total) utility function.
 - (3) The relationship between total and marginal functions holds for other measures in economics.
- Marginal Utility of multiple goods
 - The marginal utility of any one good is the rate at which total utility changes as the level of consumption of that good rises, holding constant the levels of consumption of all other goods.
 - e.g. $U = \sqrt{XY}$
 - $\Rightarrow MU_x = \frac{\Delta U}{\Delta x} \Big|_{y \text{ is held constant}} = \frac{\sqrt{y}}{2\sqrt{x}}$
 - $\Rightarrow MU_y = \frac{\Delta U}{\Delta y} \Big|_{x \text{ is held constant}} = \frac{\sqrt{x}}{2\sqrt{y}}$



- Indifference curve
 - A curve connecting a set of consumption baskets that yield the same level of satisfaction to the consumer.
 - Each curve represents baskets yielding the same level of utility to him/her

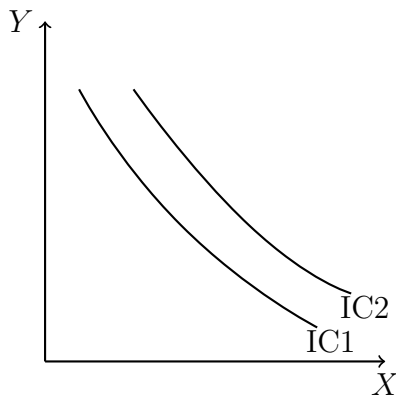


Figure 1: Indifference Curves

- Properties of indifference curves
 - (1) When the consumer likes both goods, all the indifference curves have a negative slope $\Rightarrow$ normal goods
 - (2) Indifference curves cannot intersect $\Rightarrow$ transitivity
 - (3) Every consumption basket lies on one and only one indifference curve
 - (4) Indifference curves are not "thick"

- The Marginal Rate of Substitution(MRS_{xy})
 - The rate at which the consumer will give up one good to get more of another, **holding the level of utility constant**
 - When two goods(x, y) have positive marginal utilities, the trade-off that the consumer is willing to make between the two goods is illustrated by the **slope** of the indifference curve.
 - e.g. if $MRS_{xy} = \frac{2}{1} = 2$, then the slope of the indifference curve is -2
 $\Rightarrow$ It means, to get one more unit of x, he/she should give up two units of y, holding utility constant

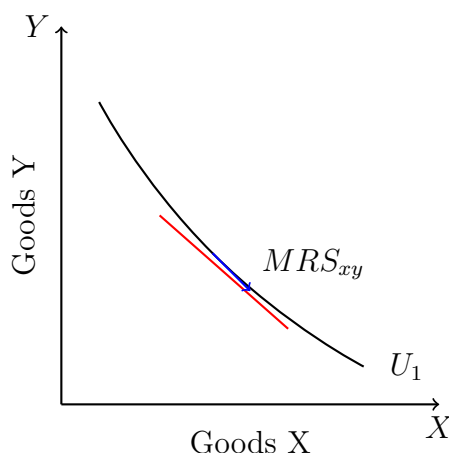


Figure 2: Marginal Rate of Substitution (MRS)

- MRS_{xy} mathematical understanding
 - MRS_{xy} at any point is the negative of the slope of the indifference curve at that point
 - $\Delta U = MU_x(\Delta x) + MU_y(\Delta y) = 0$
 - $MRS_{xy} = \frac{MU_x}{MU_y} = -\frac{\Delta y}{\Delta x} |_{\text{holding utility constant}}$

Exercise. If $U(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$, what is the $MRS_{x,y}$?

Exercise. If $U(x, y) = 2x + 4y$, what is the $MRS_{x,y}$?. And please interpret the result.

- Diminishing Marginal Rate of Substitution

- A feature of consumer preferences for which the marginal rate of substitution of one good for another good diminishes as the consumption of the first good increases along an indifference curve
- For many goods, MRS_{xy} diminishes as the amount of x increases along an indifference curve
- If MRS_{xy} diminishes as the consumer increases x along an indifference curve, then the slope of the indifference curve must be getting flatter (less negative) as x increases.

- Special Preference 1: **Cobb-Douglas**

- Cobb-Douglas Utility function: A function of the form $U = Ax^\alpha y^\beta$, where U measures the consumer's utility from x units of one good and y units of another good and where A , α , and β are positive constants.
 - * The marginal utilities (MU_x, MU_y) are positive for both goods $\Rightarrow$ the more is better assumption is satisfied.
 - * Since the marginal utilities are both positive, the indifference curves will be downward sloping.
 - * The Cobb-Douglas utility function also exhibits a diminishing marginal rate of substitution $\Rightarrow$ an indifference curve being convex towards the origin.
- e.g. $U(x, y) = x^{\frac{1}{2}} \cdot y^{\frac{1}{2}}$

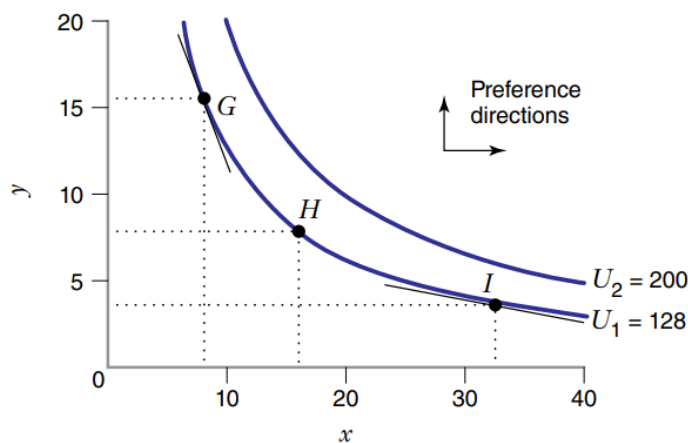


Figure 3: $U(x, y) = xy$

- Special Preference 2: **Perfect substitutes**

- Two goods such that the marginal rate of substitution of one good for the other is constant; therefore, the indifference curves are straight lines.
- $U = aX + bY$
 - * $MU_X = a$, $MU_Y = b$ (a, b are positive constant)
 - * $MRS_{x,y} = \frac{a}{b}$, the slope of the indifference curve is $-\frac{a}{b}$
 - * The $MRS_{x,y}$ of perfect substitutes is constant, not diminishing, and it can touch the axis but still have positive utility
- e.g. $U(P, W) = P + 2W$. If a consumer likes both pancakes and waffles and is always willing to substitute two pancakes for one waffle.

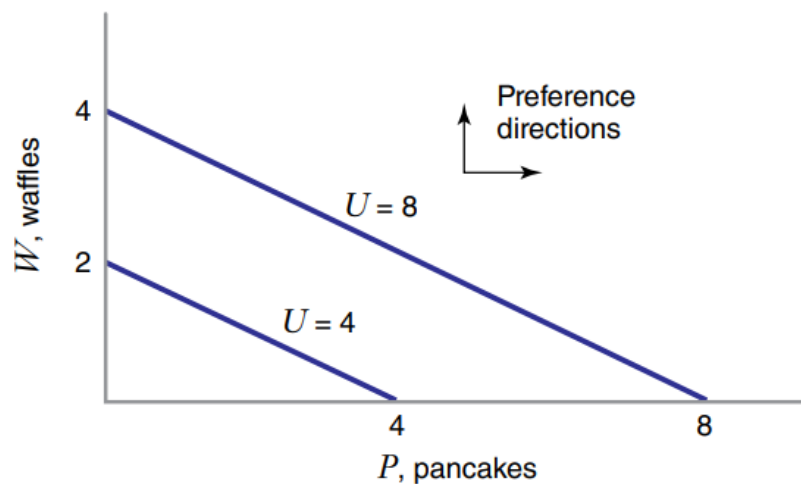
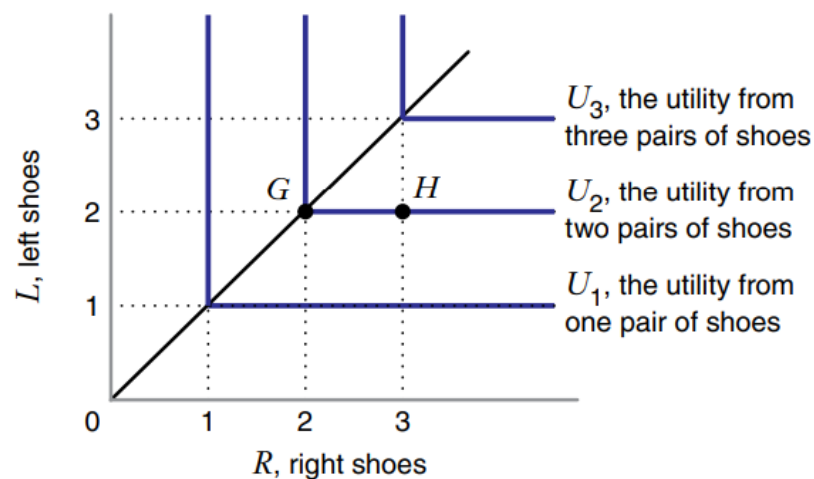


Figure 4: $U(P, W) = P + 2W$

- Special Preference 3: **Perfect Complements**

- Two goods that the consumer always wants to consume in fixed **proportion(ratio)** to each other
- $U(x,y)=\min(Ax,By)$
 - * $A, B > 0$
 - * At corner points $Ax = By$.
- e.g. A pair of shoes

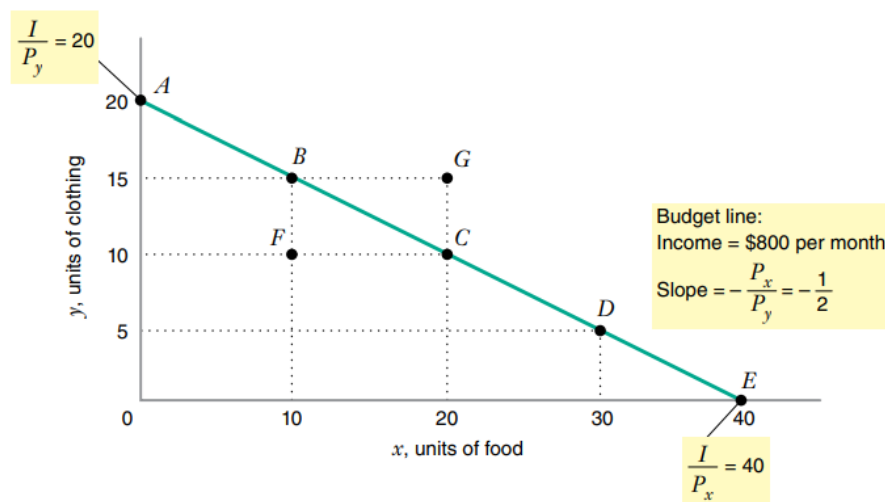


Exercise. If there is a consumer always consume 8oz of coffee(x) and 1 oz of sugar(y). Write the utility function of this case and draw the graph of the indifference curve.

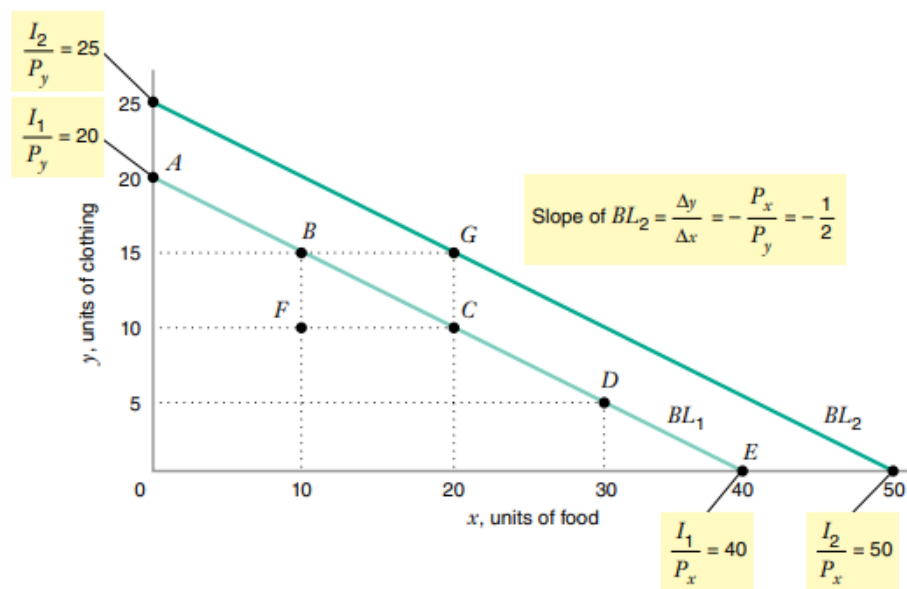
Ch4. Consumer Choice

The Budget Constraint

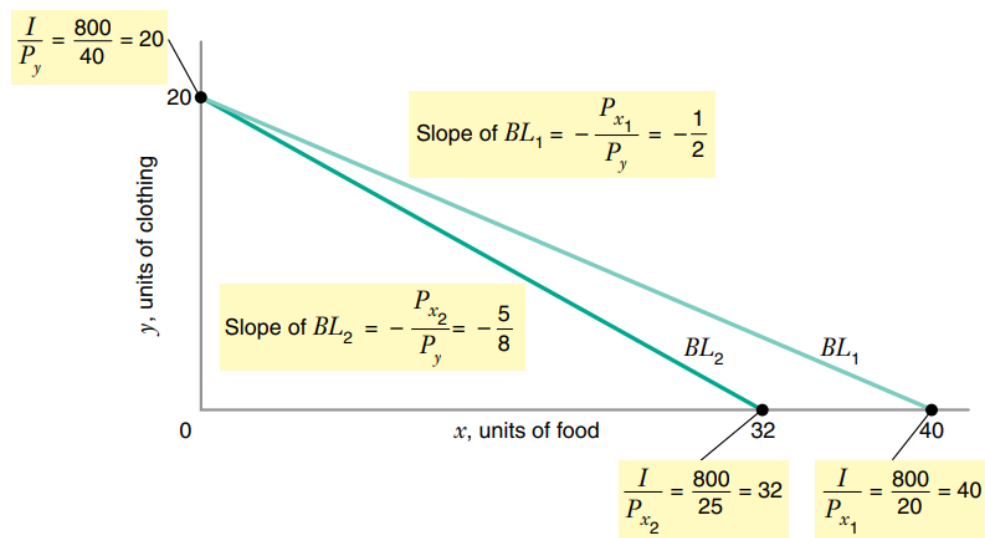
- Budget Constraint
 - The set of baskets that a consumer can purchase with a limited amount of income
- Budget Line
 - The set of baskets that a consumer can purchase when spending all of his or her available income
 - $P_x x + P_y y = I$
 - * Horizontal intercept: $\frac{I}{P_x}$
 - * Vertical intercept: $\frac{I}{P_y}$
 - * Slope of budget line: $-\frac{\frac{I}{P_y}}{\frac{I}{P_x}} = -\frac{P_x}{P_y}$
 - e.g. $I = 800, P_x = 20, P_y = 40$
 - * budget line: $20x + 40y = 800$
 - * intercepts: $\frac{800}{40} = 20, \frac{800}{20} = 40$
 - * slope: $-\frac{20}{40} = -\frac{1}{2}$



- How does a **change in income** affect the budget line?
 - When income rises, the set of choices available to the consumer will increase.
 - An increase in income shifts the budget line outward in a parallel fashion.
 - A decrease in income shifts the budget line inward
 - e.g. $I = 1000, P_x = 20, P_y = 40$
 - * budget line: $20x + 40y = 1000$
 - * intercepts: $\frac{1000}{40} = 25, \frac{1000}{20} = 50$
 - * slope: $-\frac{20}{40} = -\frac{1}{2}$



- How does a **price change** affect the budget line?
 - An increase in the price of one good moves the intercept on that good's axis toward the origin (rotate)
 - A decrease in the price of one good would move the intercept on that good's axis away from the origin (rotate)
 - e.g. $I = 800, P_x = 25, P_y = 40$
 - * budget line: $25x + 40y = 800$
 - * intercepts: $\frac{800}{40} = 20, \frac{800}{25} = 32$
 - * slope: $-\frac{25}{40} = -\frac{5}{8}$

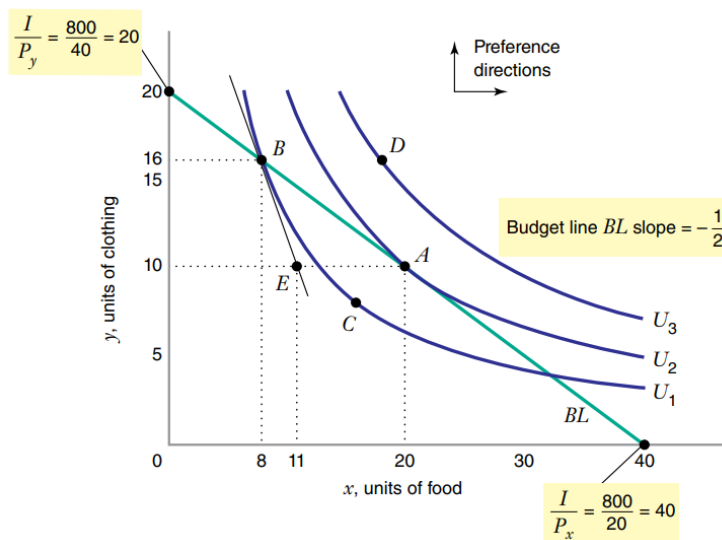


Optimal Choice

- Optimal Choice
 - If we assume that a consumer makes purchasing decisions rationally and we know the consumer's preferences and budget constraints, we can determine the consumer's optimal choice.
 - Consumer choice of a basket of goods under two conditions
 - (1) Maximizes satisfaction(utility) while
 - (2) Allowing him to live within his budget constraint.
 - An optimal consumption basket(choice) must be located on the budget line.
 - * The unspent income could be used to increase satisfaction(utility) with the purchase of additional goods.
- Optimal Choice problem for a consumer

$$\begin{aligned} &\max U(x, y) \\ &\text{subject to } P_x x + P_y y \leq I \end{aligned}$$

- $\max U(x, y)$: "choose x and y to maximize utility"
- $P_x x + P_y y \leq I$: "the expenditures on x and y must not exceed the consumer's income I "
- e.g. $I = 800, P_x = 20, P_y = 40, U(x, y) = xy$
 - * Optimal basket: A
 - * Unaffordable basket: D
 - * Non-optimal baskets: E, C



- Case1: Solving **Cobb-Douglas** Function

- Step 0: Set up the consumer's problem

$$\begin{aligned} \text{Max} U(X, Y) &= AX^\alpha Y^\beta : \text{Objective Function} \\ \text{s.t. } P_X X + P_Y Y &= I : \text{Budget Constraint} \end{aligned}$$

- Step 1: Rewrite the constraint in terms of one of the variables

$$Y = \frac{I}{P_Y} - \frac{P_X}{P_Y} X$$

- Step 2: Substitute the Y into the objective function

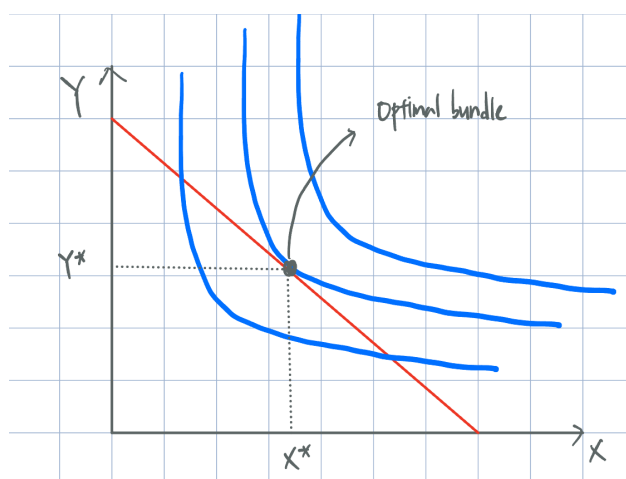
$$\text{Max} U(X) = AX^\alpha \left[\frac{I}{P_Y} - \frac{P_X}{P_Y} X \right]^\beta$$

- Step 3: F.O.C

$$\frac{\partial U}{\partial X} = 0 \Rightarrow \text{find } X^*$$

- Step 4: Plug X^* into the budget constraint to get Y^*

- Graphical Solution for Cobb-Douglas case



Exercise. Let $U(X, Y) = XY$, $P_X = 50$, $P_Y = 200$, $I = 1000$. Find the optimal X^* and Y^*

Exercise. Let $U(X, Y) = 2X^{\frac{1}{2}}Y^{\frac{1}{2}}$, $P_X = 5$, $P_Y = 10$, $I = 100$. Find the optimal X^* and Y^*

- Case2: Solving **Perfect Substitutes** function

- Step 0: Set up the consumer's problem

$$\begin{aligned} \text{Max } U(X, Y) &= AX + BY \quad : \text{ Objective Function} \\ \text{s.t. } P_X X + P_Y Y &= I \quad : \text{ Budget Constraint} \end{aligned}$$

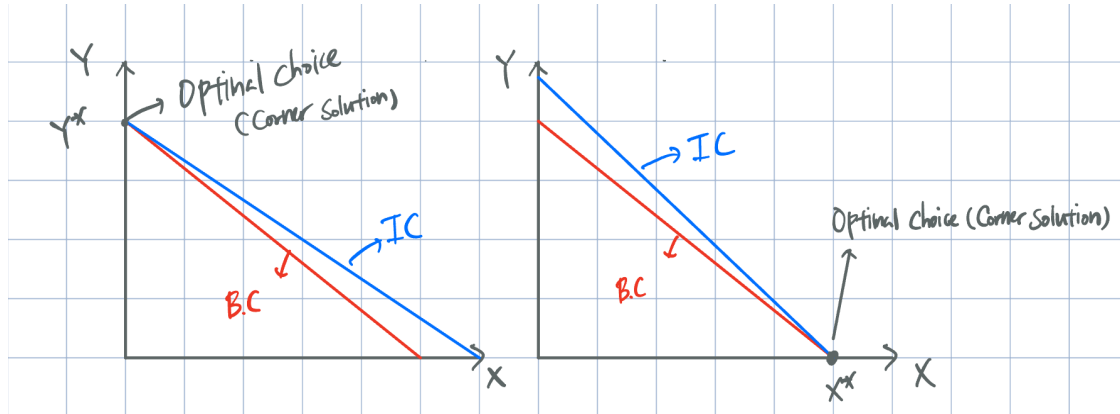
- Step 1: "Bang for Buck Solution". Compare how much more utility a consumer get from consuming X or Y if spend a dollar(\$1) more on X or Y

$$\begin{aligned} \frac{MU_X}{P_X} &> \frac{MU_Y}{P_Y} \Rightarrow \text{Consume all } X \\ \frac{MU_X}{P_X} &< \frac{MU_Y}{P_Y} \Rightarrow \text{Consume all } Y \end{aligned}$$

- Step 2: Find the optimal bundle at the corner point.

- Graphical solution for perfect substitutes case

- left: if $\frac{MU_X}{P_X} < \frac{MU_Y}{P_Y}$
- right: if $\frac{MU_X}{P_X} > \frac{MU_Y}{P_Y}$



- Case3: Solving **Perfect Complements** function

- Step 0: Set up the consumer's problem

$$\begin{aligned} \text{Max} U(X, Y) &= \min(AX, BY) : \text{Objective Function} \\ \text{s.t. } P_X X + P_Y Y &= I : \text{Budget Constraint} \end{aligned}$$

- Step 1: Rewrite the corner point equation($AX = BY$) in terms of one of the variables

$$AX = BY \Rightarrow X = \frac{B}{A}Y$$

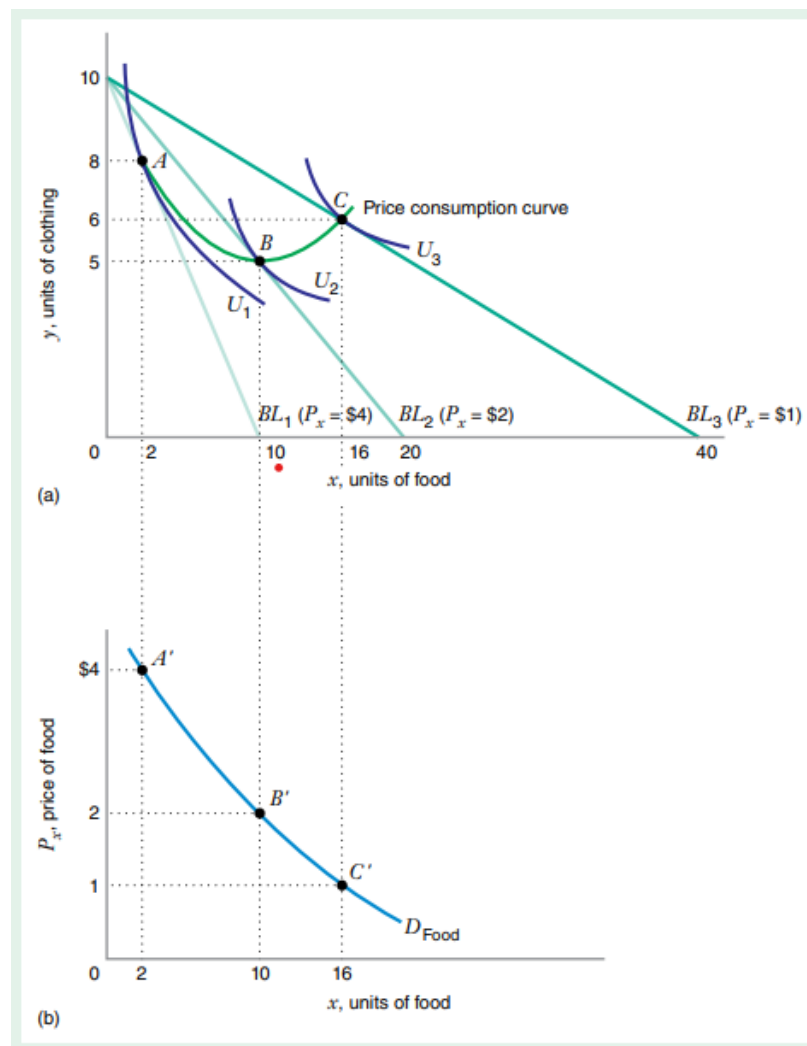
- Step 2: Plug it into the budget constraint and solve this for Y^* and substitute this Y^* into $X = \frac{B}{A}Y$ to get X^*

Exercise. Let $U(X, Y) = \min(X, 2Y)$, $P_X = 2$, $P_Y = 1$, and $I = 10$. Find the optimal choice bundle X^* and Y^* .

Ch5. The Theory of Demand

The effects of a change in price

- Price consumption curve
 - The set of utility-maximizing baskets as the price of one good varies (holding constant income and the prices of other goods.)
 - Describe how changes in the price of the good affect the consumer's purchases of both goods
 - ⇒ Draw a curve connecting all of the baskets that are optimal as the price of the good changes
 - e.g. $x = \text{food}$, $y = \text{clothing}$, $P_y = \$4$, $I = \$40$, and P_x varies.
 - * Observe that the consumer is better off (higher utility level) as the price of food falls.



- We can use the optimal choice diagram to trace out the **demand curve** for food like the above.
 - * A decrease in the price of food leads the consumer to move down and to the right *along* her demand curve for food.
- Demand curve states a "willingness to pay" curve
 - * If a consumer on A , her willingness to pay for another unit of food is \$4.
 - * This is because her $MRS_{x,y} = 1$ at A (it's from $\frac{P_x}{P_y} = 1$)
 - $\Rightarrow$ thus, at A , one more unit of food is worth the same amount to her as 1 more unit of clothing.
 - $\Rightarrow$ since the price of clothing(y) is \$4, the value of additional unit of food(x) is also be \$4.
 - * On B and C , the $MRS_{x,y} = 1/2, 1/4$ respectively
 - $\Rightarrow$ So the willingness to pay for food is \$2 and \$1 respectively.
 - $\Rightarrow$ Her willingness to pay for an additional unit of food falls as she buys more and more food (diminishing $MRS_{x,y}$)

• Case1: Cobb-Douglas Function

$MaxU(X, Y) = XY$: Objective Function

s.t. $P_X X + P_Y Y = I$: Budget Constraint

$$\Rightarrow X = \frac{I - P_Y Y}{P_X}$$

$$MaxU(X, Y) = \left[\frac{I}{P_X} - \frac{P_Y Y}{P_X} \right] Y = \frac{I}{P_X} Y - \frac{P_Y}{P_X} Y^2$$

$$(F.O.C) \Rightarrow \frac{dU}{dY} = 0 = \frac{I}{P_X} - 2 \cdot \frac{P_Y}{P_X} Y \Rightarrow Y^* = \frac{I}{2P_Y}$$

If you plug Y^* into the budget constraint, you will get $X^* = \frac{I}{2P_X}$

• Case2: Perfect substitutes (Linear utility)

$MaxU(X, Y) = 2X + 3Y$: Objective Function

s.t. $P_X X + P_Y Y = I$: Budget Constraint

$$MU_X = \frac{dU}{dX} = 2, \quad MU_Y = \frac{dU}{dY} = 3$$

("Bang for Buck") $\Rightarrow$ when $\frac{MU_X}{P_X} > \frac{MU_Y}{P_Y}$, consume only $X \Rightarrow X^* = \frac{I}{P_X}, Y^* = 0$

$\Rightarrow$ when $\frac{MU_X}{P_X} < \frac{MU_Y}{P_Y}$, consume only $Y \Rightarrow X^* = 0, Y^* = \frac{I}{P_Y}$

- Case3: Perfect complements

$MaxU(X, Y) = \min(2X, 3Y) : \text{Objective Function}$

$s.t. P_X X + P_Y Y = I : \text{Budget Constraint}$

1) On the corner of IC, $2X = 3Y \Rightarrow X = \frac{3}{2}Y$

2) On the budget constraint $P_X X + P_Y Y = I$

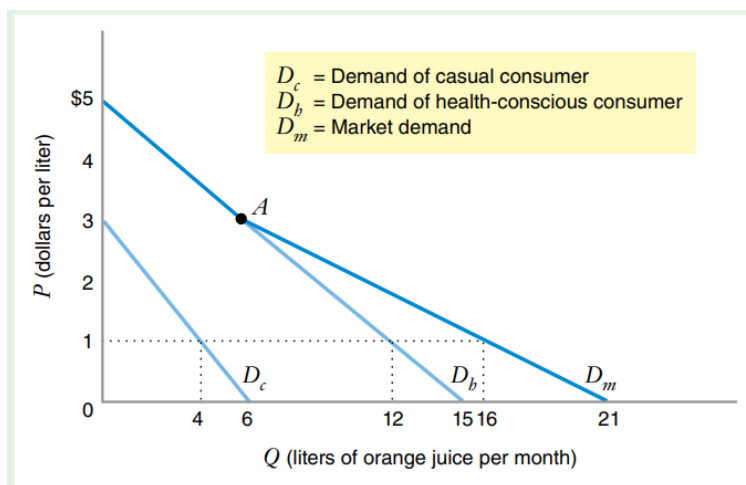
Plug X into the budget constraint $\Rightarrow P_X(\frac{3}{2}Y) + P_Y Y = I \Rightarrow Y(\frac{3}{2}P_X + P_Y) = I$

As a result, $Y^* = \frac{I}{\frac{3}{2}P_X + P_Y}$

If you solve for X, you will get $X^* = \frac{I}{P_X + \frac{2}{3}P_Y}$

- Market Demand

- The market demand curve is the horizontal sum of the demands of the individual consumers.
- Business firms and policymakers are often more concerned with the demand curve for an entire market of consumers.



- * Since the construction of a market demand curve involves adding *quantities*, you must write the demand curves in the normal form (not inverse demand) before adding them.
- * You must pay attention to how the underlying individual demands vary across the range of prices.

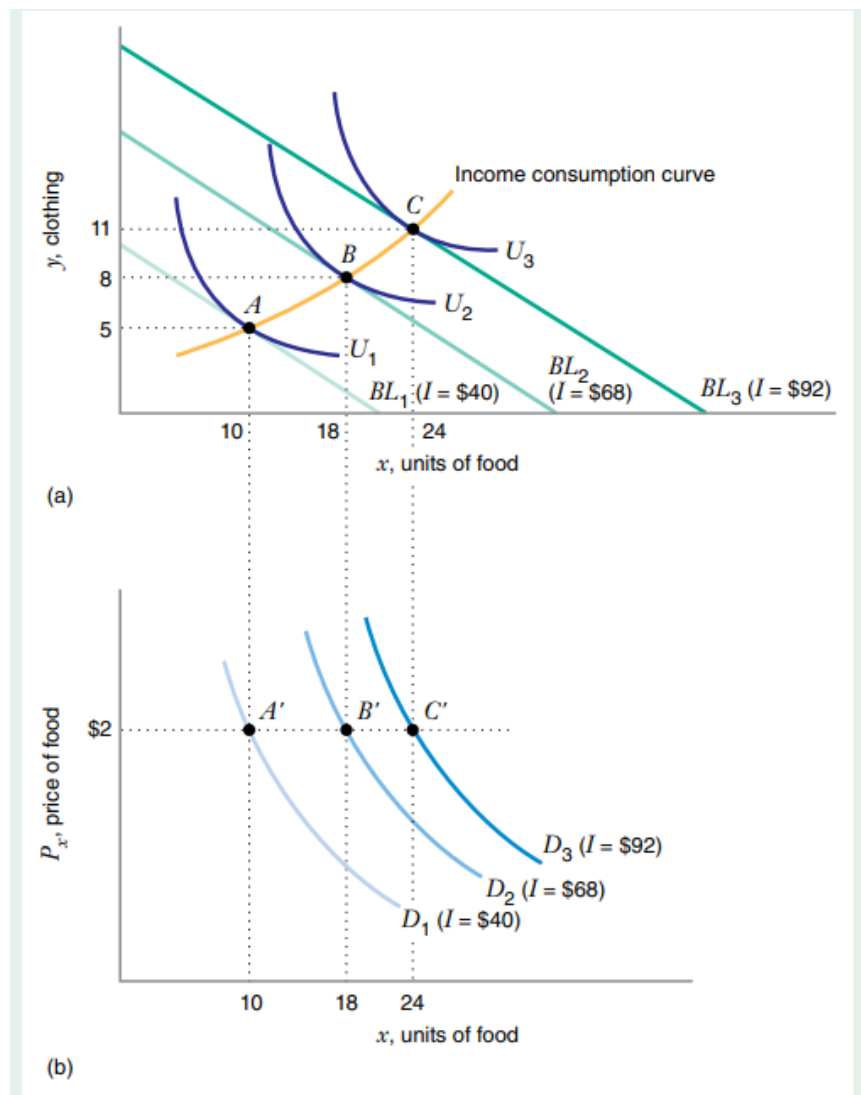
Exercise. Assume 2 people in a market for 1 good, and their demand is like the below. Find the market demand and draw the individual and market demand curves.

$$\text{Person1 : } Q_1(P) = \begin{cases} 15 - 3P & P < 5 \\ 0 & P \geq 5 \end{cases}$$

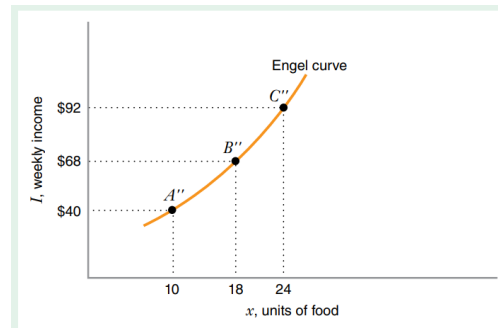
$$\text{Person2 : } Q_2(P) = \begin{cases} 6 - 2P & P < 3 \\ 0 & P \geq 3 \end{cases}$$

The effects of a change in income

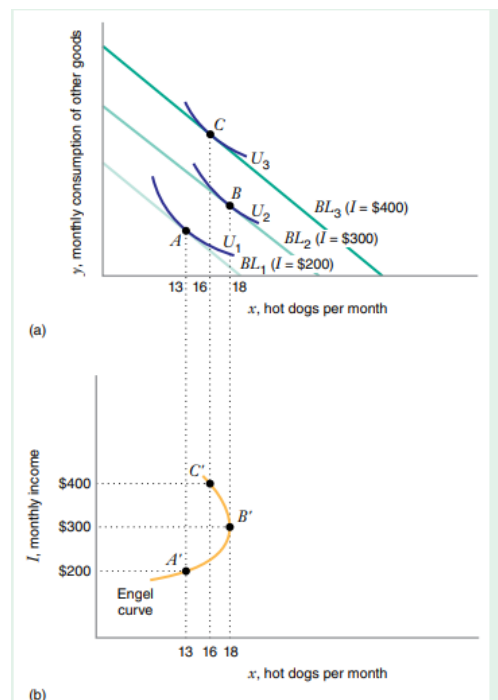
- Income consumption curve
 - The set of utility-maximizing baskets as income varies (and prices are held constant)
 - An increase in income results in an outward, parallel shift of the budget line (vice versa)
 - e.g. $x = \text{food}$, $y = \text{clothing}$, $P_x = \$2$, $P_y = \$4$, and I varies
 - * The consumer purchases more of both goods and her income rises.
 - * An increase in income results in a rightward shift in her demand curve for each good.



- Engel curve
 - A curve that relates the amount of a commodity purchased to the level of income, holding constant the prices of all goods.
 - **Normal good:** A good that a consumer purchases more of as income rises
 - * For a normal good the Engel curve will have a positive slope, as below.
 - * A good with a positive income elasticity of demand will have a positive sloped Engel curve



- **Inferior good:** A good that a consumer purchases less of as income rises
 - * Engel curve has a *positive slope* over the range of incomes for which the goods are a *normal good* and a *negative slope* over the range of incomes for which the goods are an *inferior good*



Change in the price of a good: Substitution Effect & Income Effect

- Goods and Bads

- Goods: In economics, goods(and services) are items that satisfy human wants and provide utility. It is useful to people but scarce in relation to its demand so human effort(price) is required to obtain it. Always provides positive marginal utility
 - * **Normal Goods:** A type of good that experiences an increase in demand due to an increase in income. The demand rises because of the attractiveness to consumers.
 - * **Inferior Goods:** A good whose demand decreases when consumer income rises.
 - * **Inferiority:** it is an observable fact relating to affordability rather than a statement about the quality of the good. As a rule, these goods are affordable and adequately fulfill their purpose, but as more costly substitutes that offer more pleasure become available, the use of the inferior goods diminishes.
 - * **Giffen goods:** a special type of inferior good that would disobey the "law of demand". Too much inferior, so that the income effect is larger than typical inferior good. The price elasticity of demand is positive.
- Bads: The opposite of an economic good. A 'bad' is anything with a negative value(disutility) to the consumer. Always provides negative marginal utility.

- Substitution Effect

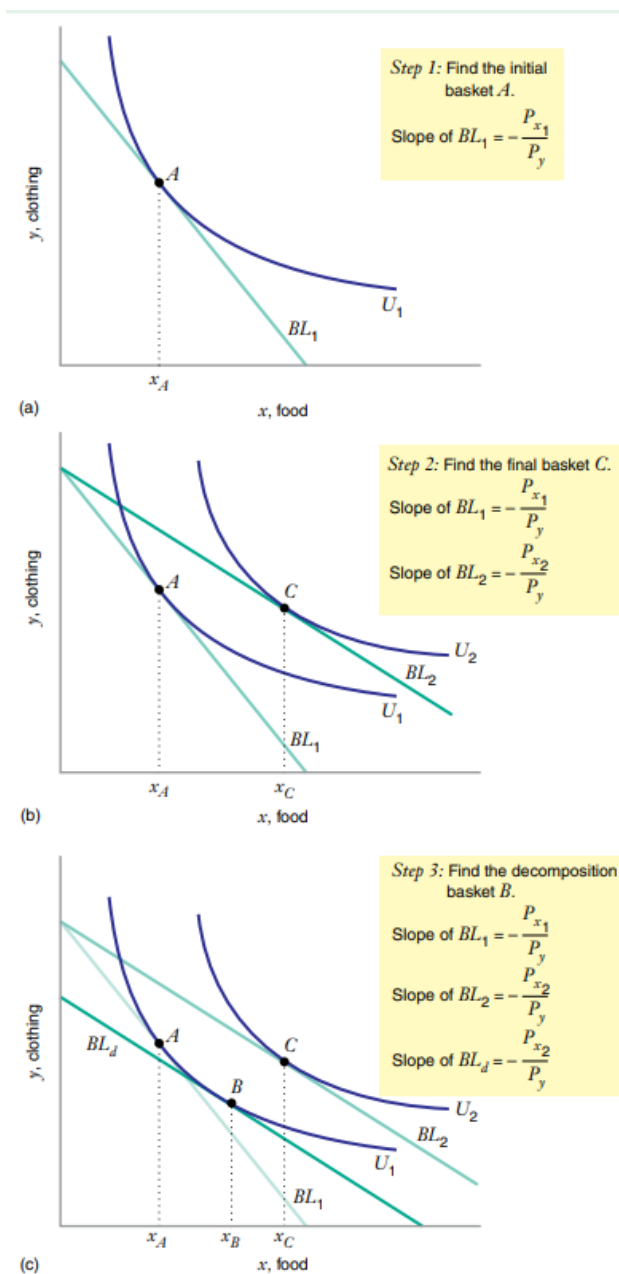
- The change in the amount of a good that would be consumed as the price of that good changes, **holding constant all other prices and the level of utility** ($\bar{U}$). Diminishing $MRS_{x,y}$ also matters.
- Either price increase or decrease changes the relative prices between x and composite goods y . As a result, there are changes in consumption for x because of the substituting activity

- Income Effect

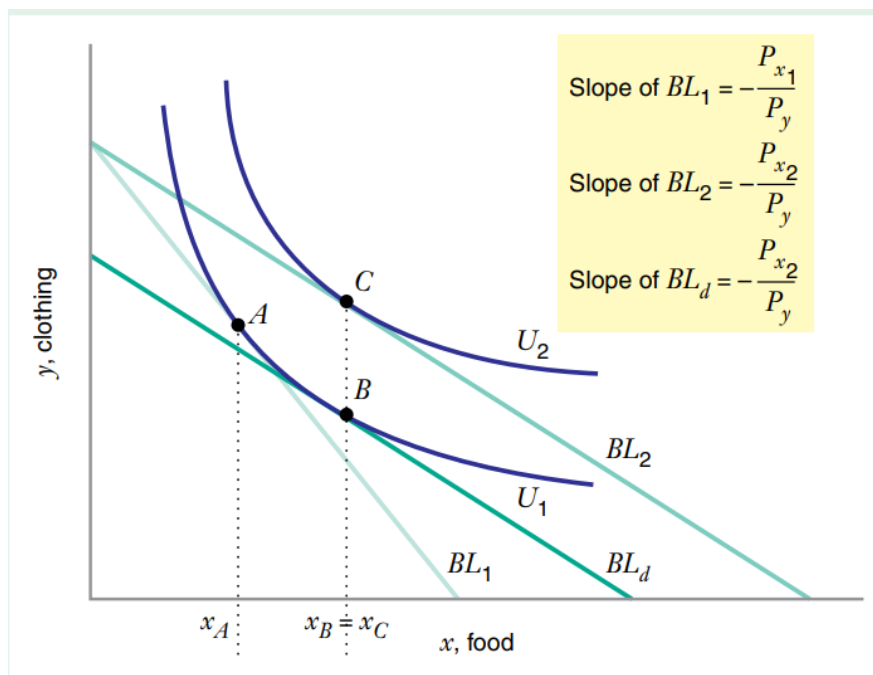
- The change in the amount of a good that a consumer would buy as **purchasing power changes**, holding all prices constant.
- When the price of a good falls, the consumer's purchasing power increases. Conversely, a rise in price decreases the consumer's purchasing power.
- Even though there is no numerical change in income I , the purchasing power(real income) has changed because of the price change in goods.
- It is called the income effect because it affects the consumer in much the same way as a change in income

- Steps of the substitution effect
 - Suppose that a consumer buys x, y that have $MU_x, MU_y > 0$.
 - 1. Find the initial basket(the basket the consumer chooses at the initial price P_x)
 - 2. Find the final basket(the basket the consumer chooses after the price falls or rises to $P_{x'}$)
 - 3. Find an intermediate decomposition basket that will enable us to identify the portion of the change in quantity due to the substitution effect.
 - First, the decomposition basket reflects the price decrease(or increase), so it must lie on a budget line that is parallel to BL_2
 - Second, the decomposition basket reflects the assumption that the consumer achieves the initial level of utility after the price decrease(or increase), so the basket must be at the point where the budget line is tangent to indifference curve U_1
- Steps of the income effect
 1. The change in the amount of a good consumed as the consumer's utility changes.
 2. In reality, the consumer's income doesn't fall or rise actually when the price increases or decreases.
 3. So we shift the budget line in parallel to the changed relative price($\frac{P_x}{P_y} \Rightarrow \frac{P_{x'}}{P_y}$)
 4. Thus, the income effects account for the consumer's movement from the decomposition basket to the final basket.
- Summary
 - Total effect: $(x_C - x_A)$
 - Substitution effect: $(x_B - x_A)$
 - Income effect: $(x_C - x_B)$
 - Total effect($x_C - x_A$)=Substitution($x_B - x_A$) + Income($x_C - x_B$)

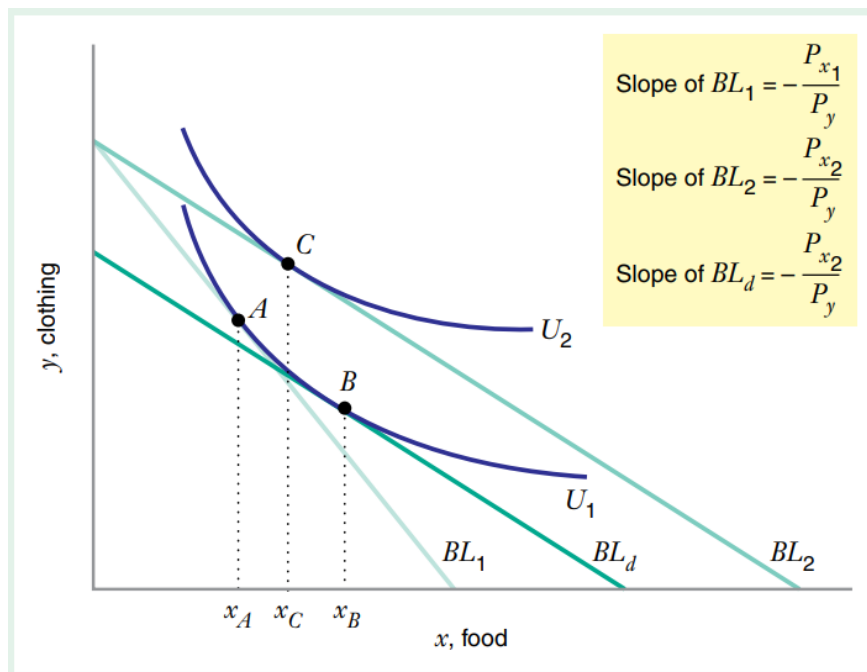
- Case: Normal good



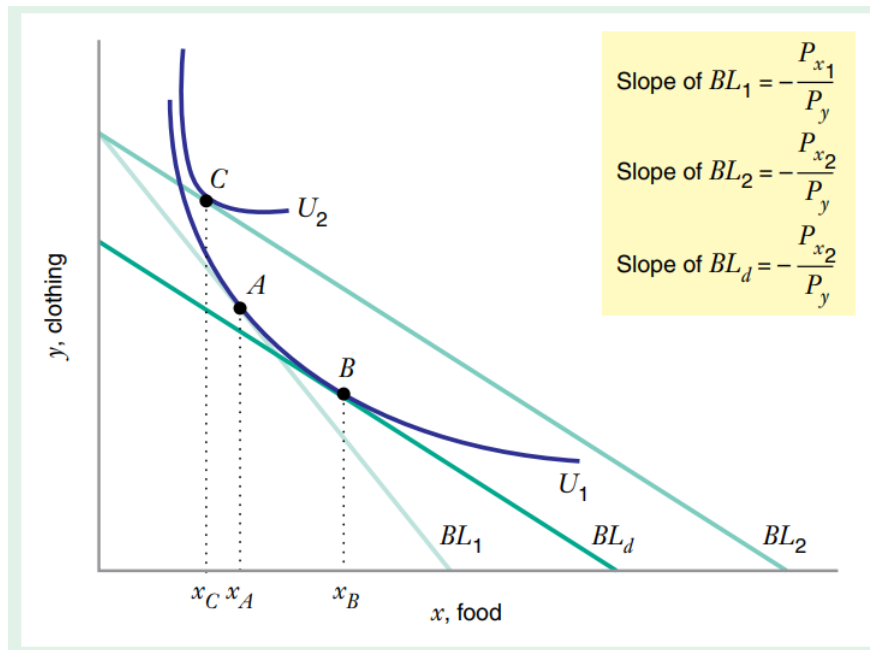
- When goods are not normal
 - For normal goods, the income and substitution effects work in the same direction.
 - The income and substitution effects do not always work in the same direction.
 - e.g. Zero income effect case. (x is neither a normal good nor an inferior good.)
 - * Because it's a good, it has a substitution effect as any other good.
(P_x decrease $\Rightarrow$ positive substitution effect)
 - * However, the magnitude of the income effect is zero in this case.
($x_C - x_B = 0 \Rightarrow$ zero income effect)



- When x is an *Inferior good*
 - The income and substitution effects might even work in opposite directions.
 - The indifference curves will show that the income effect is negative
 - However, for inferior good, **the substitution effect is larger than the income effect.**
 - As a result, the total effect is in the same direction as the substitution effect
 - Also, the demand curve for x will still be **downward sloping**



- When x is an *Giffen good*
 - Definition: A good so strongly inferior that the **income effect outweighs the substitution effect**, resulting in an **upward-sloping** demand curve over some region of prices.
 - Price elasticity of demand for Giffen goods is positive, unlike normal or inferior goods.



- However, the largest income effects are usually associated with goods that are normal rather than inferior, or Giffen goods, such as housing.

- When the price of good $X(P_X)$ decreases($\downarrow$)

	Normal goods	Inferior goods	Giffen goods
Total effect($X_C - X_A$)	$>$	$>$	$<$
Substitution effect($X_B - X_A$)	$>$	$>$	$>$
Income effect($X_C - X_B$)	$>$	$<$	$<$

- When the price of good $X(P_X)$ increases($\uparrow$)

	Normal goods	Inferior goods	Giffen goods
Total effect($X_C - X_A$)	$<$	$<$	$>$
Substitution effect($X_B - X_A$)	$<$	$<$	$<$
Income effect($X_C - X_B$)	$<$	$>$	$>$

Exercise. Draw the graphs of the normal, inferior, and Giffen goods when the price of good $X(P_X)$ increases.

Exercise. Find the demand for X if $U(X, Y) = X^{\frac{1}{4}}Y^{\frac{3}{4}}$, $P_X = 4$, $P_Y = 2$, and $I = 16$. a) Set up the consumer's maximization problem, and indicate objective function and constraint. b) Solve mathematically using the method in class. c) graph the solution

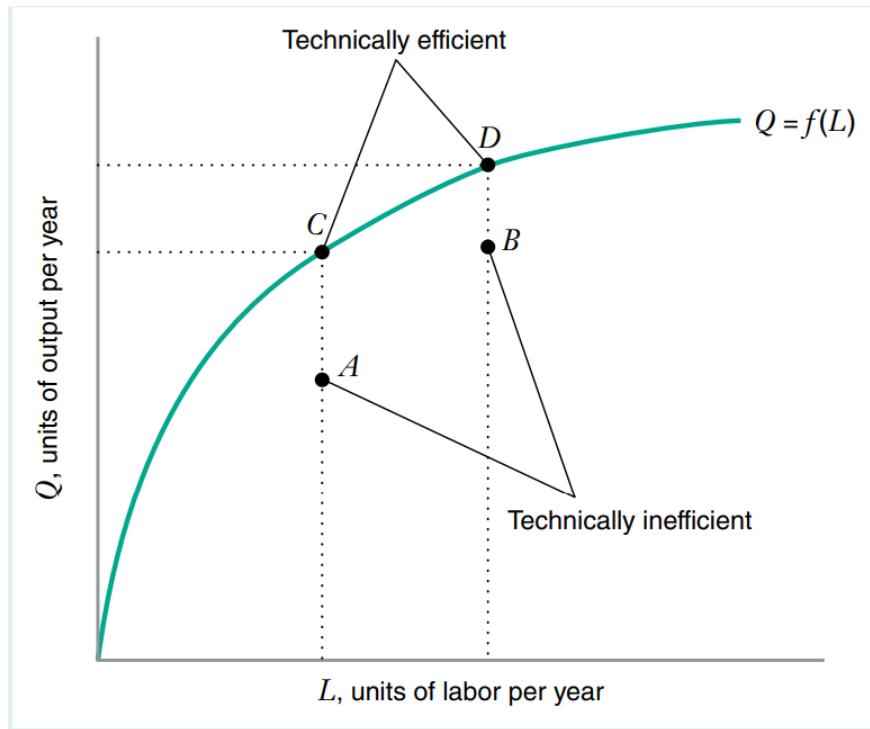
Ch6. Inputs and Production Functions

Introduction to inputs and production functions

- Production
 - Production of goods and services involves transforming resources into finished products
 - (1) *Inputs*: resources, such as labor, capital equipment, and raw materials, that are combined to produce finished goods.
 - (2) *Factors of production*: resources that are used to produce a good
 - (3) *Output*: The amount of a good or service produced by a firm
 - (4) *Production function*: a mathematical representation that shows the *maximum* quantity of output a firm can produce given the quantities of inputs that it might employ
 - Many of the important trade-offs that real firms face involve choices between labor and capital.
 - We can develop the main ideas of production theory using just these two categories of inputs

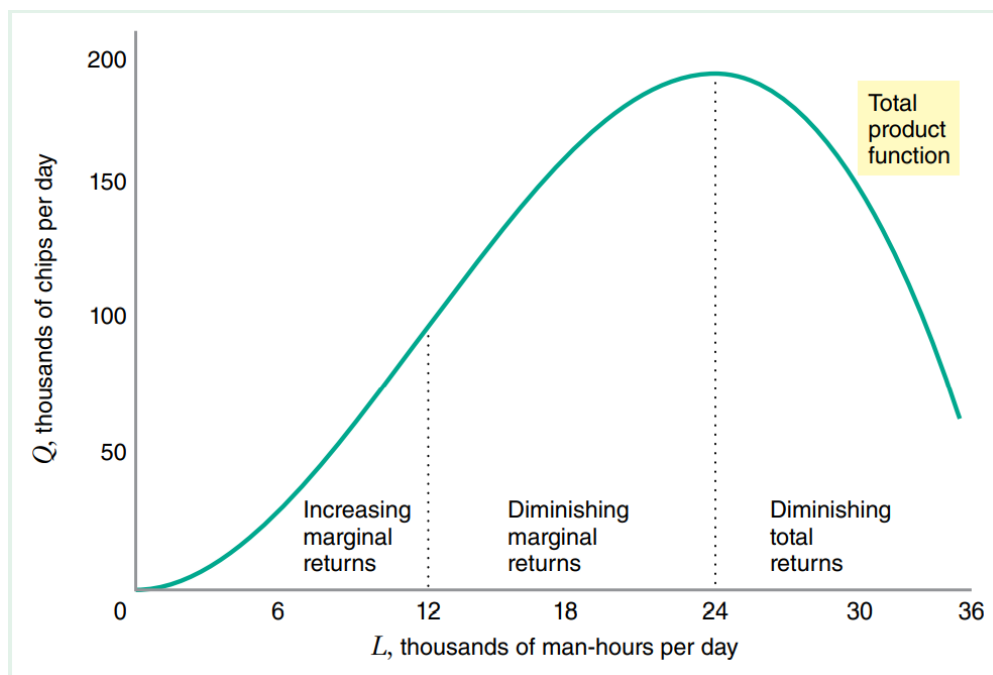
$$Q = f(L, K)$$

- Exogenous technological conditions
 - *Production set*: the set of technically feasible combinations of inputs and outputs
 - *Technically inefficient*: the set of points in the production set at which the firm is getting less output from its labor than it could
 - *Technically efficient*: the set of points in the production set at which the firm is producing as much output as it possibly can given the amount of labor it employs



- Single input production function

- *Total product function*: a production function. A total production function with a single input shows how total output depends on the level of the input
 - * How Q changes when L varies given a fixed $K \Rightarrow Q = f(L, \bar{K})$
- *Increasing marginal returns to labor*: the region along the total product function where output rises with additional labor at an increasing rate
 - * $L = 0 - 12$: convex region. Output rises with additional labor at an increasing rate
 - * Positive slope of the tangent line: $MP_L > 0$
- *Diminishing marginal returns to labor*: the region along the total product function in which output rises with additional labor but at a decreasing rate
 - * $L = 12 - 24$: concave region. Output rises with additional labor but at a decreasing rate
 - * Negative slope of the tangent line: $MP_L < 0$
- *Diminishing total returns to labor*: the region along the total product function where output decreases with additional labor



- Marginal and average product

- Average product of labor(AP_L): the average amount of output per unit of labor

$$AP_L = \frac{\text{total product}}{\text{quantity of labor}} = \frac{Q}{L} \quad (1)$$

- * The AP_L at any quantity of labor corresponds to the *slope of a ray* drawn from the origin to the point along the total product function.

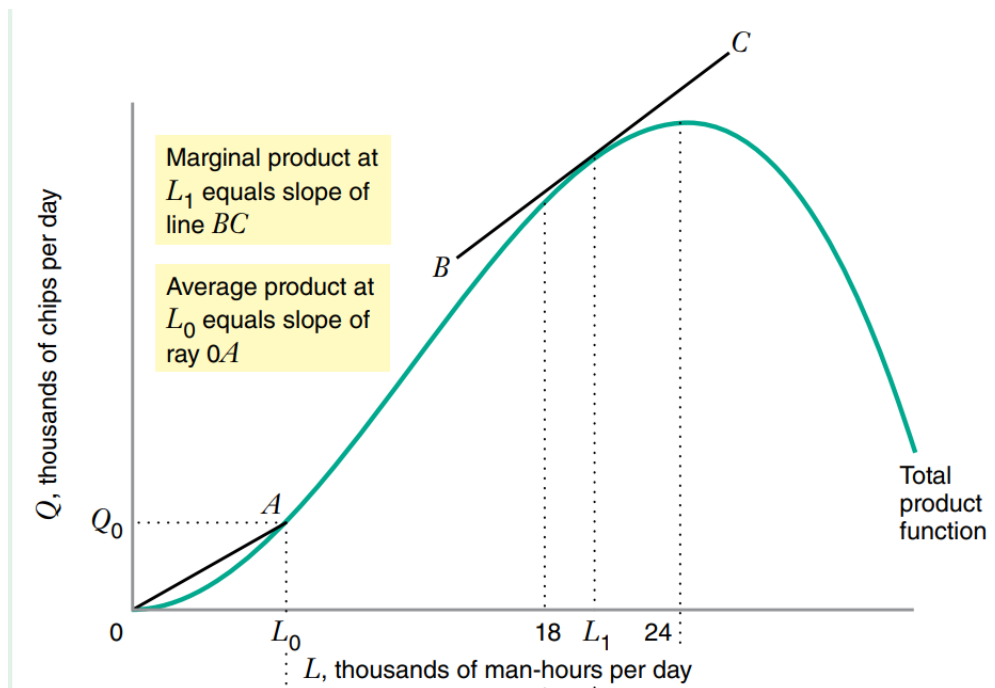
- Marginal product of labor(MP_L): the rate at which total output changes as the quantity of labor the firm uses is changed

$$MP_L = \frac{\text{change in total product}}{\text{change in quantity of labor}} = \frac{\Delta Q}{\Delta L} \quad (2)$$

- * The MP_L is the slope *tangent* of each point on the total product function graph

- * There are two regions: increasing marginal returns, diminishing marginal returns

- Law of diminishing marginal returns: As the usage of one input increases, **the quantities of other inputs being held fixed**, a point will be reached beyond which the marginal product of the variable input will decrease

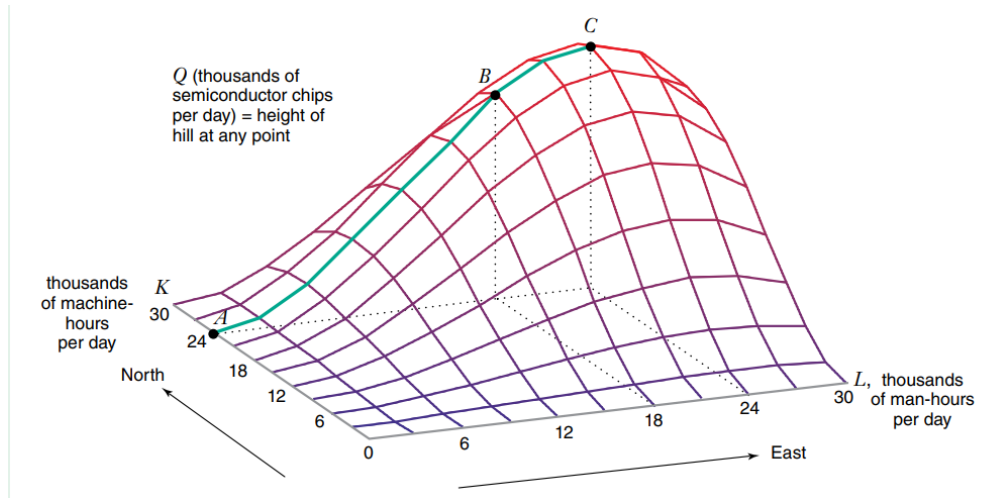


- **Relationship between marginal(MP) and average product(AP)**

- (1) *Average product is increasing in labor:* AP_L increases in L , then $MP_L > AP_L$
- (2) *Average product is decreasing in labor:* AP_L decreases in L , then $MP_L < AP_L$
- (3) *Neither increases nor decreases in labor:* When the AP_L is at a maximum, then $MP_L = AP_L$

Production function with multiple inputs

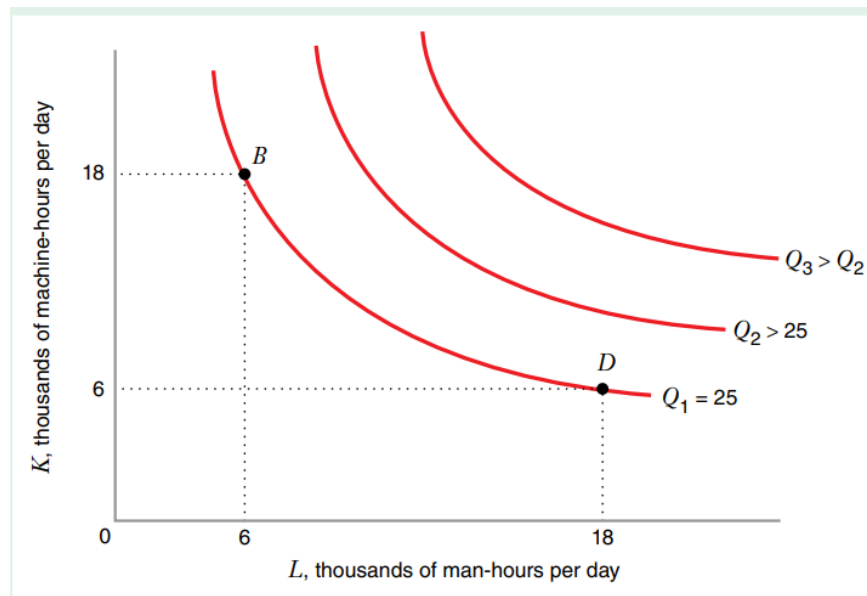
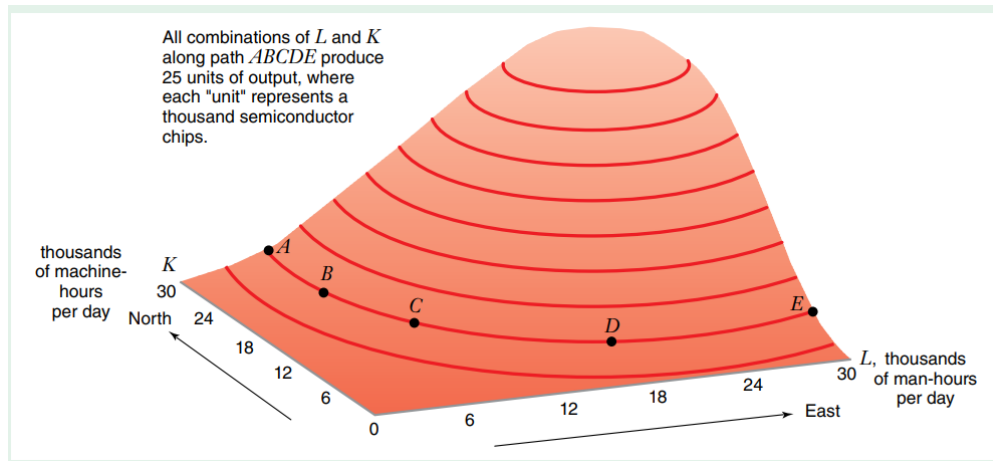
- Total product and marginal product with two inputs
 - $Q = f(L, K)$
 - * Quantity of output Q depends on the quantity of labor L and the quantity of capital K employed by a firm.
 - Total product hill: a 3-D graph of a production function



- $MP_L = \frac{\text{change in } Q}{\text{change in } L} \Big|_{K \text{ is held constant}} = \frac{\Delta Q}{\Delta L} \Big|_{K \text{ is held constant}}$
- $MP_K = \frac{\text{change in } Q}{\text{change in } K} \Big|_{L \text{ is held constant}} = \frac{\Delta Q}{\Delta L} \Big|_{L \text{ is held constant}}$
- Marginal product tells us how the steepness(slope) of the total product hill varies as we change the quantity of input, holding the quantities of all other input fixed

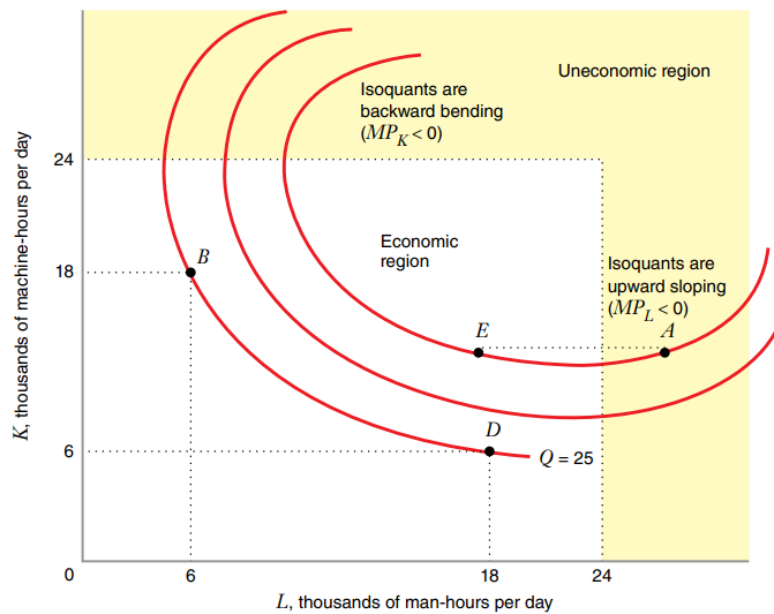
- Isoquants

- Isoquant: a curve that shows all of the combinations of labor and capital that can produce a given level of output (“same quantity”)

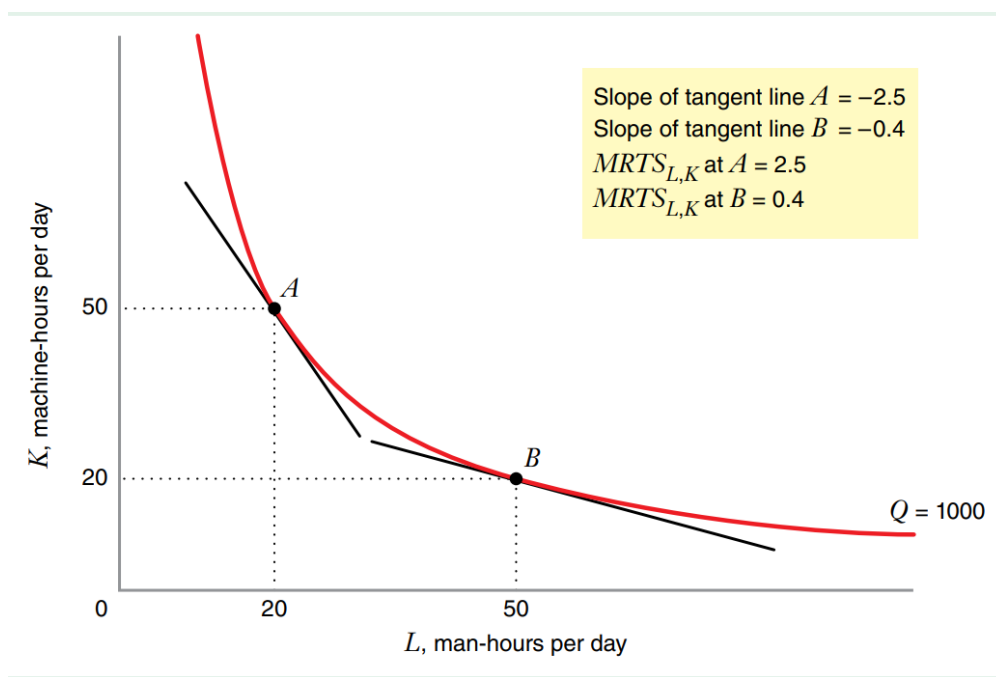


- Isoquants are downward sloping, and it illustrates an important economic **trade-off**
 - * A firm can substitute capital for labor and keep its output unchanged
 - * Such substitution is always possible whenever both labor and capital have positive marginal products
 - * Isoquants to the northeast correspond to larger quantities of outputs
 - * e.g. $Q = \sqrt{KL}$

- Economic and uneconomic regions
 - Economic region: the region of downward-sloping isoquants.
 - Uneconomic region: the region of upward-sloping or backward-bending isoquants. In the uneconomic region, at least one input has a negative marginal product.
 - * Diminishing total product: a situation in which one input has a negative marginal product ($MP_L, MP_K < 0$)
 - A firm that wants to minimize its production costs should never operate in a region of upward-sloping or backward-bending isoquants.



- $MRTS_{L,K}$: Marginal Rate of Technical Substitution of labor for capital
 - The rate at which the quantity of capital can be reduced for every 1-unit increase in the quantity of labor, holding the quantity of output constant.
 - The slope of an isoquant at a particular point is tangent to the isoquant at that point
 - Diminishing Marginal Rate of Technical Substitution: A feature of a production function in which the marginal rate of technical substitution of labor for capital diminishes as the quantity of labor increases along an isoquant
 - * The isoquants are convex to the origin



- Mathematical expression

$$\Delta Q = (\Delta K)(MP_K) + (\Delta L)(MP_L) \quad (3)$$

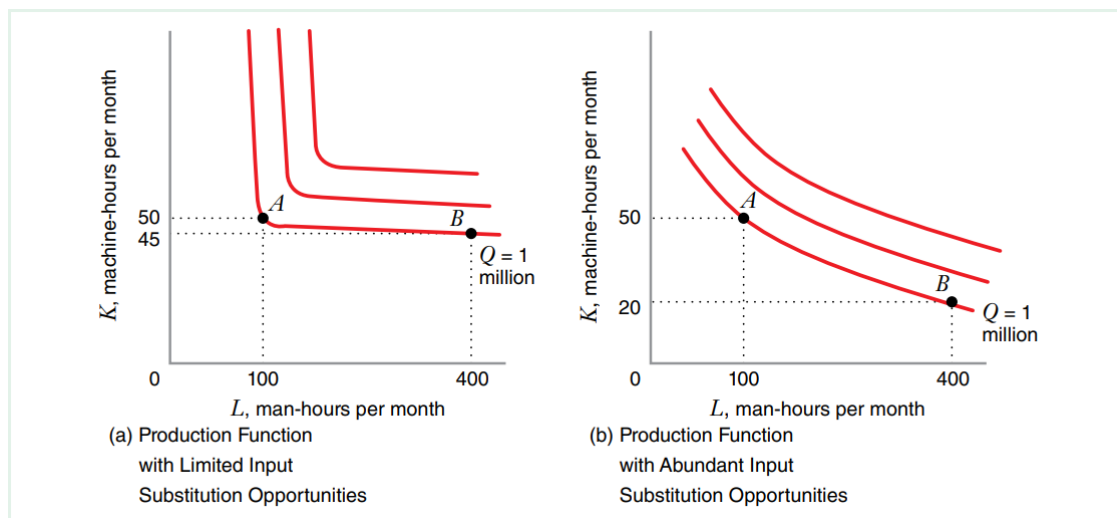
$$0 = (\Delta K)(MP_K) + (\Delta L)(MP_L) \quad (4)$$

$$-\frac{\Delta K}{\Delta L} = \frac{MP_L}{MP_K} \quad (5)$$

$$\text{As a result } MRTS_{L,K} = \frac{MP_L}{MP_K} \quad (6)$$

- If $MRTS_{L,K} = 5$ then it means that the firm can substitute 1 unit of labor for 5 units of capital without changing output quantity.

- Substitutability among inputs among inputs
 - Below the graph, the two production functions differ in terms of how easy it is for the firm to substitute between labor and capital
 - (a) At point A , it becomes *hard for the firm to substitute labor for capital*. Even if the firm quadruples its use of labor, it can reduce its quantity of capital by only a small amount to keep output at $Q = 1$
 - * It indicates that the firm would face a similar difficulty in substituting capital for labor.
 - * $MRTS_{L,K}$ changes substantially as we move along an isoquant. The isoquants are nearly L-shaped
 - (b) At point A , the firm's substitution opportunities are more abundant (linear isoquant). The firm can reduce its employment of capital significantly if it increases the quantity of labor relative to (a)
 - * It gives the firm more opportunities to substitute between labor and capital
 - * $MRTS_{L,K}$ changes gradually as we move along an isoquant. The isoquants are nearly straight lines (linear)



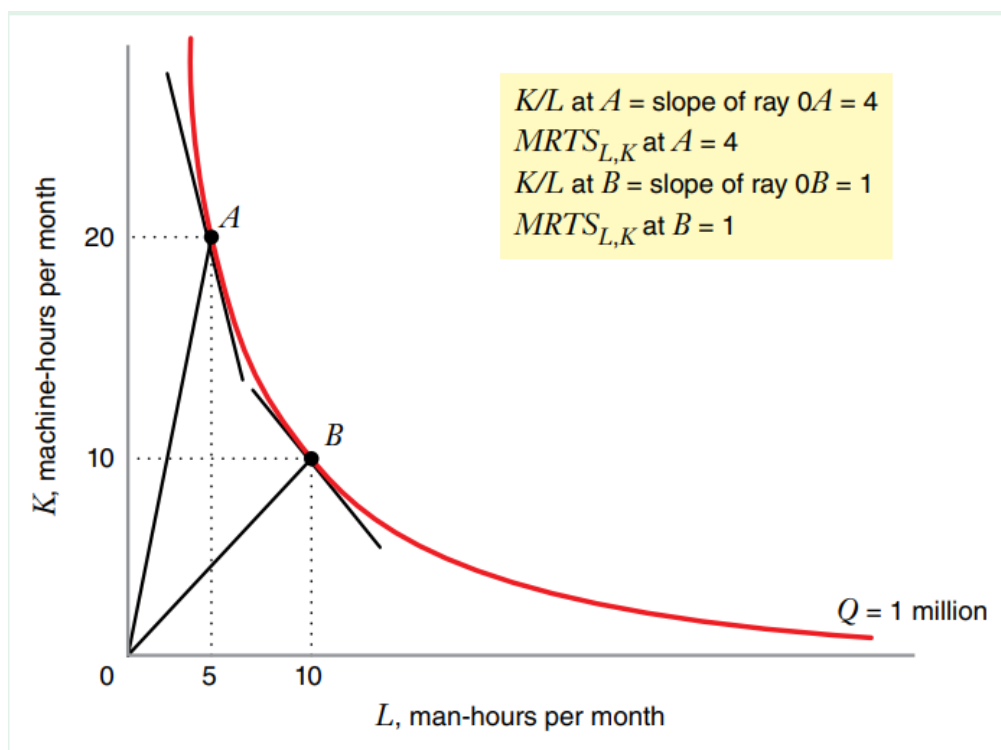
- Elasticity of substitution

- A measure of how easy it is for a firm to substitute labor for capital.
- It is equal to the percentage change in the capital-labor ratio for every 1 percent change in the marginal rate of technical substitution of labor for capital as we move along an isoquant.
- The elasticity of substitution measures how quickly the $MRTS_{L,K}$ changes as we move along an isoquant.
- Capital-labor ratio($\frac{K}{L}$): the ratio of the quantity of capital to the quantity of labor
 - * As labor(L) is substituted for capital(K), the ratio(K/L) must fall and the $MRTS_{L,K}$ also falls.

$$\sigma = \frac{\text{percentage change in } \frac{K}{L}}{\text{percentage change in } MRTS_{L,K}} \quad (7)$$

$$= \frac{\% \Delta \left(\frac{K}{L} \right)}{\% \Delta MRTS_{L,K}} \quad (8)$$

- If σ close to 0, there is little opportunity to substitute between inputs
 - * When $\% \Delta MRTS_{L,K}$ is large
- If σ is large, there is a substantial opportunity to substitute between inputs
 - * When $\% \Delta MRTS_{L,K}$ is small

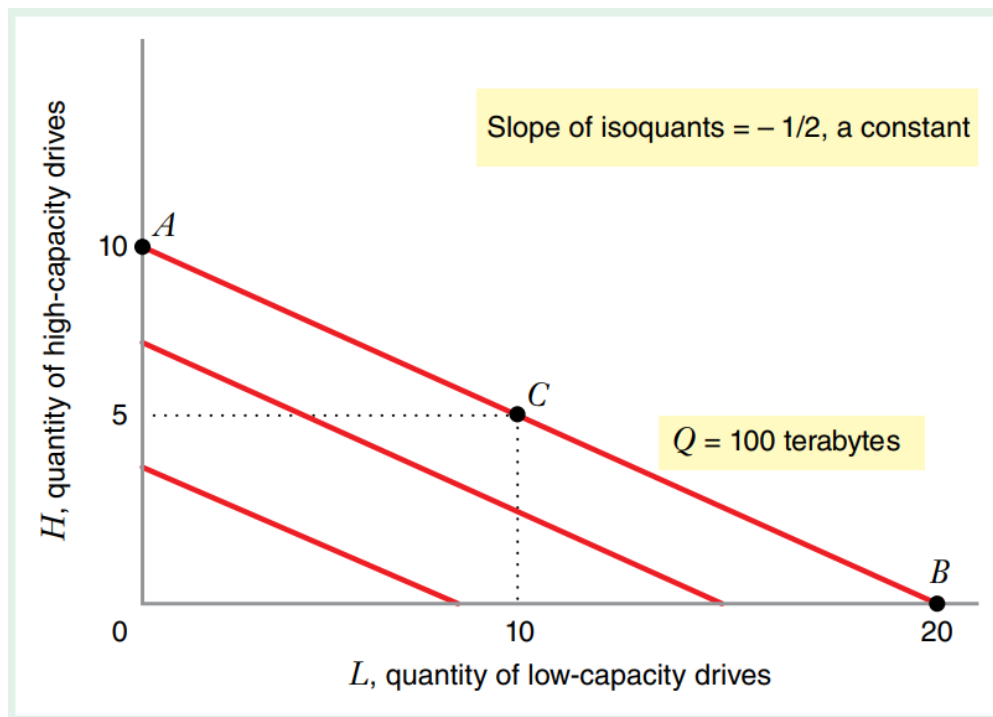


Exercise. Consider $Q = \sqrt{KL}$. Show that the elasticity of substitution for this production is exactly equal to 1, no matter what the values of K and L are.

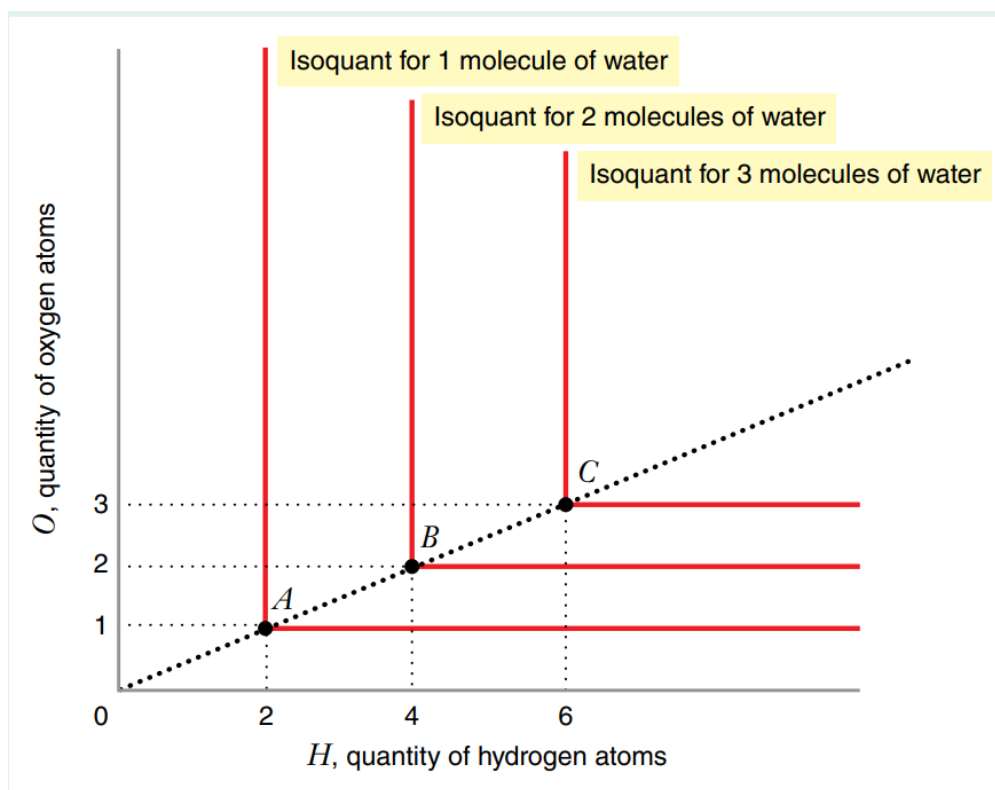
$$MRTS_{L,K} = \frac{MP_L}{MP_K} = \frac{\frac{\sqrt{K}}{2\sqrt{L}}}{\frac{\sqrt{L}}{2\sqrt{K}}} = \frac{K}{L}$$
$$\sigma = \frac{\% \Delta \left(\frac{K}{L} \right)}{\% \Delta MRTS_{L,K}} = \frac{\% \Delta \left(\frac{K}{L} \right)}{\% \Delta \left(\frac{K}{L} \right)} = 1$$

Special Production Functions

- Linear Production Function(Perfect Substitutes)
 - Perfect substitutes: (in production) inputs in a production function with a constant marginal rate of technical substitution.
 - Linear production function: $Q = aL + bK$, where a and b are positive constants.
 - $MRTS$ of one input for another can be constant
 - * e.g. A manufacturing process may require energy in the form of natural gas or fuel oil. A given amount of natural gas can always be substituted for each liter of fuel oil
 - * e.g. $Q = 10H + 5L$
 - In this case $\% \Delta MRTS_{L,H} = 0$
 - As a result, $\sigma = \infty$
 - The inputs in a linear production function are infinitely(perfectly) substitutable for each other



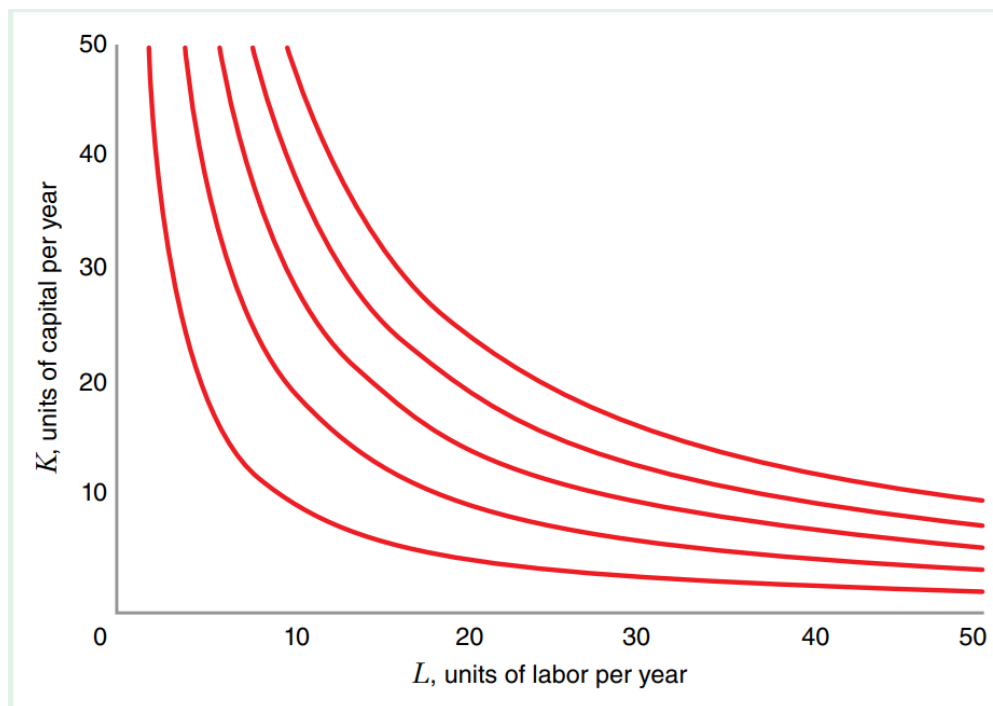
- Fixed-Proportions Production Function(Perfect Complements)
 - Fixed-proportions production function: a production function where the inputs must be combined in a constant ratio to one another
 - Perfect complements: (in production) inputs in a fixed-proportions production function
 - $Q = \min(AK, BL)$
 - e.g. Production of water, where inputs are hydrogen(H), and oxygen(O)
 - * Fixed-proportion: two H and one O
 - * $Q = \min(\frac{H}{2}, O)$



- $\sigma = 0$, because the $MRTS$ changes from ∞ to 0 when we pass through the corner($AK = BL$) of an isoquant
 - * A firm has no flexibility in its ability to substitute among inputs

- Cobb-Douglas Production Function

- $Q = AL^\alpha K^\beta$: where Q is the quantity of output, from L units of labor and K units of capital, and all others are positive
- With the Cobb-Douglas production function, capital and labor can be substituted for each other
 - * Unlike perfect complements, capital and labor, can be used in variable proportions
 - * Unlike perfect substitutes, the rate at which labor can be substituted for capital is not constant as you move along an isoquant
- σ for a Cobb-Douglas falls somewhere between 0 and ∞ .
 - * In fact, it turns out that the σ along a Cobb-Douglas production function is **always equal to 1**



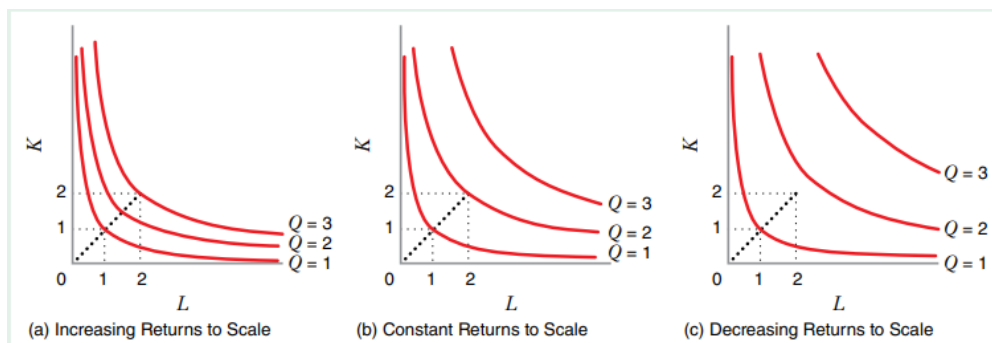
Returns to Scale

- Returns to scale

- The concept that tells us the percentage by which output will increase when all inputs are increased by a given percentage
- A firm's scale of operations increases, we might want to know by how much output will increase.
 - * e.g. When $MP_L, MP_K > 0$, a firm's total output must increase when the quantities of all inputs are increased simultaneously

$$\text{Returns to scale} = \frac{\% \Delta Q}{\% \Delta \text{all inputs}} \quad (9)$$

- L, K, Q , and suppose all inputs are "scaled up" by the same proportionate amount $\lambda > 1$. Let ψ represent the resulting proportionate increase in the quantity of output $Q(\psi Q)$
 - (1) Increasing returns to scale ($\psi > \lambda$)
 - * A proportionate increase in all input quantities results in a greater than proportionate increase in output.
 - (2) Constant returns to scale ($\psi = \lambda$)
 - * A proportionate increase in all input quantities results in the same proportionate increase in output
 - (3) Decreasing returns to scale ($\psi < \lambda$)
 - * A proportionate increase in all input quantities results in a less than proportionate increase in output
- The cost advantage of large-scale operation has been the traditional justification for allowing firms to operate as regulated monopolists in markets such as electric power and oil pipeline transportation
 - * e.g. When a large firm has a cost advantage over smaller firms, a market is most efficiently served by one large firm rather than several smaller firms.



- Cobb-Douglas and returns to scale

- $Q = AL^\alpha K^\beta$

- * $A(\lambda L)^\alpha (\lambda K)^\beta = \lambda^{\alpha+\beta} AL^\alpha K^\beta$

- * $\alpha + \beta > 1$: increasing returns to scale

- * $\alpha + \beta = 1$: constant returns to scale

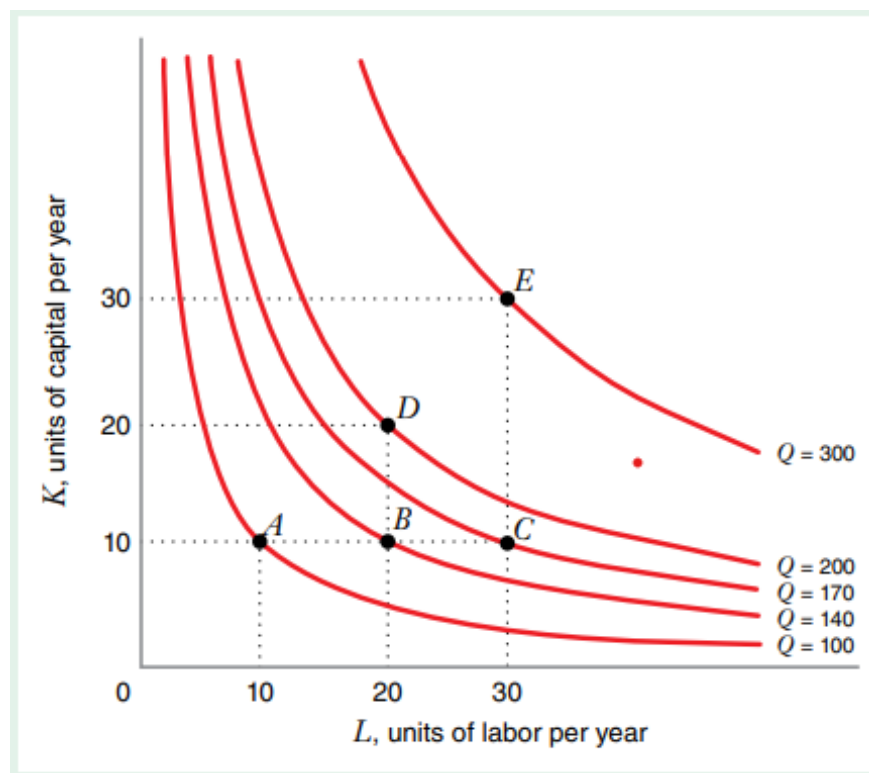
- * $\alpha + \beta < 1$: decreasing returns to scale

- Returns to scale vs. Diminishing marginal returns

- Returns to scale pertains to the impact of an increase in all input quantities simultaneously

- Marginal returns pertain to the impact of an increase in the quantity of a single input

- * Such as labor, holding the quantities of all other inputs fixed.

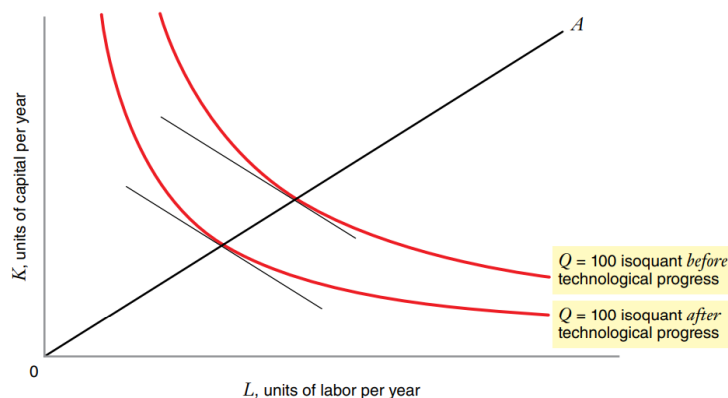


- Technological progress

- A change in a production process that enables a firm to achieve more output from a given combination of inputs or the same amount of output from fewer inputs.

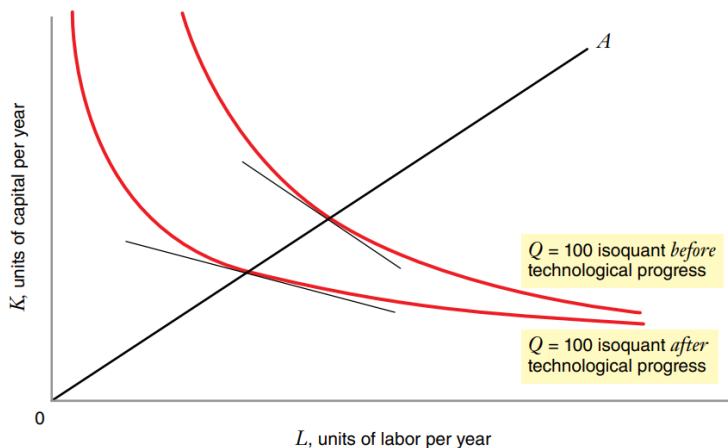
(1) *Neutral technological progress*

- Progress that decreases the amounts of labor and capital needed to produce a given output, without affecting the $MRTS_{L,K}$
- Under neutral progress, each isoquant corresponds to a higher level of output than before, but the isoquants themselves retain the same shape



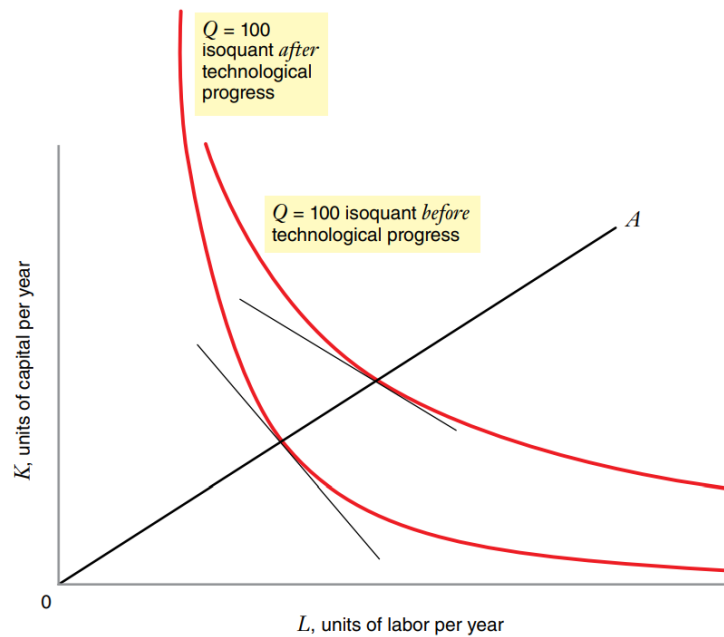
(2) *Labor-saving technological progress*

- Progress that causes the marginal product of capital to increase relative to the marginal product of labor
- Isoquant corresponding to a given level of output shifts inward, the isoquant becomes flatter, indicating that the $MRTS_{L,K}$ is less than it was before
 - * It implies that MP_K increase more rapidly than the MP_L
 - * e.g. Advance in equipment, robotics, or computers



(3) *Capital-saving technological progress*

- Progress that causes the marginal product of labor to increase relative to the marginal product of capital
- An isoquant shifts inward, and $MRTS_{L,K}$ increases, indicating that the MP_L increases more rapidly than the MP_K .
 - * e.g. Educational or skill level of the firm's actual workforce rises.



Ch7. Costs and Cost Minimization

- Costs

- Costs are not necessarily synonymous with monetary outlays.
- Explicit costs
 - * Costs that involve a direct monetary outlay
 - * e.g. Airline's expenditures on fuel and salaries
- Implicit costs
 - * Costs that do not involve outlays of cash
 - * e.g. Income it forgoes by not leasing airline's jets

- Opportunity Cost

- Definition: The value of the next best alternative that is forgone when another alternative is chosen.
- The opportunity cost of an alternative includes all of the *explicit* and *implicit* costs associated with that alternative.
- To assess opportunity cost we ask: "What does the decision maker give up at the time the decision is being made?"
- Opportunity costs depend on the decision being made
 - * Because there are different opportunity costs for different decisions under different circumstances.
- Opportunity costs and market prices
 - * From the firm's perspective, the opportunity cost of using the input is the current *market price* of the input.

- **Economic vs. Accounting costs**

- Economic costs
 - * The sum of the firm's explicit costs and implicit costs
 - * All decision-relevant costs
- Accounting costs
 - * The total of explicit costs that have been incurred in the past
 - * Historical expenses only
 - * It does not include implicit costs because such costs are often hard to measure in an objectively verifiable way.

- **Sunk(unavoidable) vs. Nonsunk(avoidable) costs**

- Sunk costs
 - * Costs that have already been incurred and cannot be recovered
- Nonsunk costs
 - * Costs that are incurred only if a particular decision is made
- When evaluating alternative decisions, the decision maker should ignore sunk costs and consider only nonsunk costs.
- Whether a cost is sunk or nonsunk depends on the decision that is being contemplated.

The Cost Minimization Problem

- The cost minimization problem
 - Cost-minimization problem
 - * The problem of finding the input combination that **minimizes** a firm's total cost of producing a particular level of output
 - Cost-minimizing firm
 - * A firm that seeks to minimize the cost of producing a given amount of output
- Long run vs. Short run
 - Long run
 - * The period of time that is long enough for the firm to vary the quantities of all of its inputs as much as it desires
 - * A firm that makes a long-run decision faces a blank slate(i.e., no constraints)
 - * The costs associated with this long-run decision are necessarily nonsunk
 - Short run
 - * The period of time in which at least one of the firm's input quantities cannot be changed
 - * A firm facing a short-run decision is subject to constraints
 - * It will not be able to adjust the quantities of some of its inputs and/or reverse the consequences of past decisions that it has made regarding those inputs

- The **long-run** cost-minimization problem
 - We want to minimize the firm's total costs(TC), subject to the requirement that the firm produce a given amount of output(Q_0) using production function $f(K, L)$
 - * Total cost: $TC(L, K) = wL + rK$
 - * For now, the target quantity Q_0 is exogenously decided
 - Long-run cost-minimization problem for a firm that uses two inputs: K, L
 - * w : *Wage rate* - the price of a unit of labor services
 - The price of labor could be either an explicit cost or an implicit cost
 - explicit cost: if the firm hires workers in the open market
 - implicit cost: if the firm's owner provides her own labor to run the firm, sacrifices outside employment opportunities
 - * r : *Rental rate* - the price of a unit of capital services
 - The price of capital could either be an explicit cost or an implicit cost
 - explicit cost: if the firm leased capital services from another firm
 - implicit cost: if the firm owned the physical capital and, by using it in its own business, sacrificed the opportunity to sell capital services to other firms
 - **The manager(producer) must choose a quantity of K, L that minimize the total cost $TC = wL + rK$ of producing Q_0 .**

$$\begin{aligned}
 \text{Min}TC(L, K) &= rK + wL && : \text{Objective Function} \\
 \text{s.t. } Q_0(K, L) &= f(K, L) && : \text{Constraint}
 \end{aligned}$$

- Isocost lines

- Definition: The set of combinations of labor and capital that yield the same total cost for the firm.

- $TC = wL + rK \Rightarrow K = \frac{TC}{r} - \frac{w}{r}L$

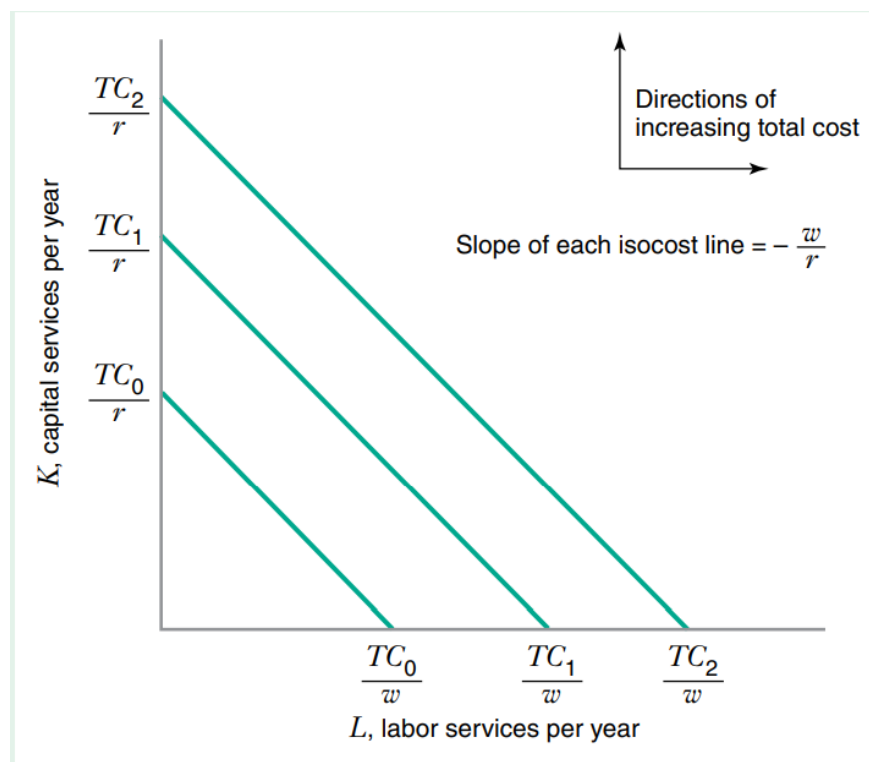
- * e.g. $w = 10, r = 20, TC = \$1m$

- * Isocost line: $1,000,000 = 10L + 20K \Rightarrow K = (1,000,000/20) - (10/20)L$

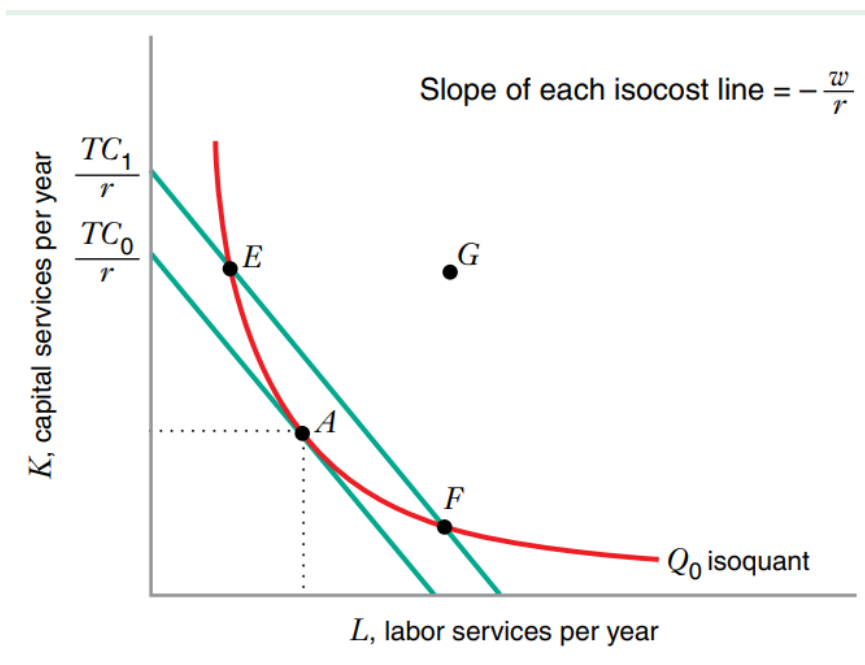
- The graphs of isocost lines where $TC_0 < TC_1 < TC_2$

- * The slope of isocost lines: $-\frac{w}{r}$

- * Intercepts: $\frac{TC}{r}, \frac{TC}{w}$



- Graphical solution to the Long-run cost-minimization problem
 - The solution to the firm's cost-minimization problem occurs at point A , where the isoquant is just tangent to an isocost line
 - * Point A : Optimal point(solution). It provides the firm with the lowest level of cost given the Q
 - * Point G : Technically inefficient. This input combination could produce Q_0 of output, in using it the firm would be wasting inputs
 - Cannot be optimal because input combination A also produces Q_0 but uses fewer units of L, K
 - * Point E, F : Technically efficient. But they are not cost-minimizing because they are on an isocost line that is higher than the isocost line passing through the cost-minimizing point A



- The slope of the isoquant at the cost-minimizing point A is equal to the slope of the isocost line.

slope of isoquant = $-(\text{slope of isocost line})$

$$\begin{aligned}
 MRTS_{L,K} &= -\left(-\frac{w}{r}\right) \\
 &= \frac{MP_L}{MP_K} = \frac{w}{r} \\
 &= \frac{MP_L}{w} = \frac{MP_K}{r}
 \end{aligned}$$

- This condition tells us that at a cost-minimizing input combination, the additional output per dollar spent on labor services equals the additional output per dollar spent on capital services

Exercise. (Cobb-Douglas production function case) $Q = 50\sqrt{LK}$, $w = 5$, $r = 20$, and $Q_0 = 1000$. What is the cost-minimizing amount of L and K ? a) Set up the firm's minimization problem. b) solve for K^* and L^* .

$$\begin{aligned} \text{a) } \textit{MinTC}(L, K) &= 5L + 20K : \textit{Objective Function} \\ \text{s.t. } 50\sqrt{LK} &= 1000 : \textit{Constraint} \end{aligned}$$

$$\text{b) } \sqrt{LK} = 20 \Rightarrow LK = 400 \Rightarrow L = \frac{400}{K}$$

$$\text{plug into the TC} \Rightarrow TC = 5\left(\frac{400}{K}\right) + 20K = 2000K^{-1} + 20K$$

$$\text{F.O.C : } \frac{\partial TC}{\partial K} = -2000K^{-2} + 20 = 0 \Rightarrow 2000 = 20K^2 \Rightarrow K^2 = 100 \Rightarrow K^* = 10$$

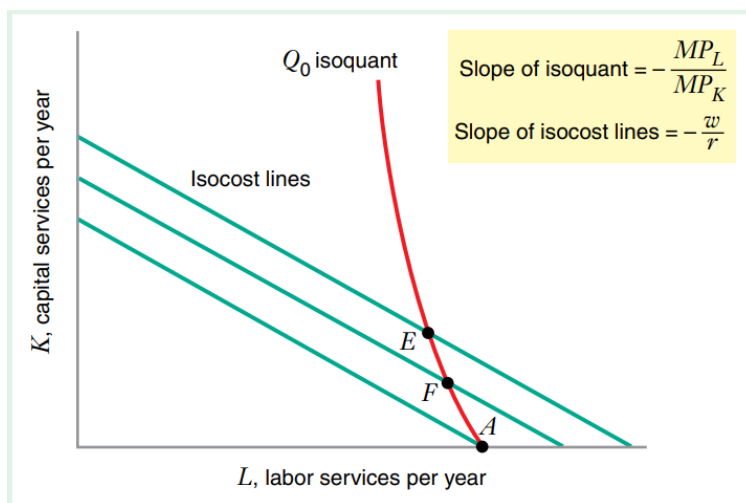
$$\text{plug } K^* \text{ into the constraint} \Rightarrow L^* = \frac{400}{10} = 40$$

- Long-run perfect substitutes total cost minimization(corner point solutions)

$$\text{MinTC}(L, K) = wL + rK : \text{Objective Function}$$

$$\text{s.t. } Q_0(L, K) = \mathbf{aL} + \mathbf{bK} : \text{Constraint}$$

- Corner point solutions happens when $\frac{MP_L}{w} \neq \frac{MP_K}{r}$. Solve by using the "Bang for Buck" method similar to a maximizing utility under perfect substitute utility function.
- Suppose $\frac{MP_L}{w} > \frac{MP_K}{r}$
 - * In this case, there is no reason to use K for producing Q_0 amount. Every additional dollar spent on labor yields more output than every additional dollar spent on capital.
 - * The firm should substitute labor for capital until it uses no capital at all. So the point A in the below graph is the corner point solution if $\frac{MP_L}{w} > \frac{MP_K}{r}$



Exercise. (Perfect Substitute production function case) Let $Q = 10L + 2K$. $w = \$5, r = \2 . Find the optimal solution at $Q = 200$. a) Set up the firm's minimization problem. b) solve for K^* and L^* using the "Bang for Buck" method

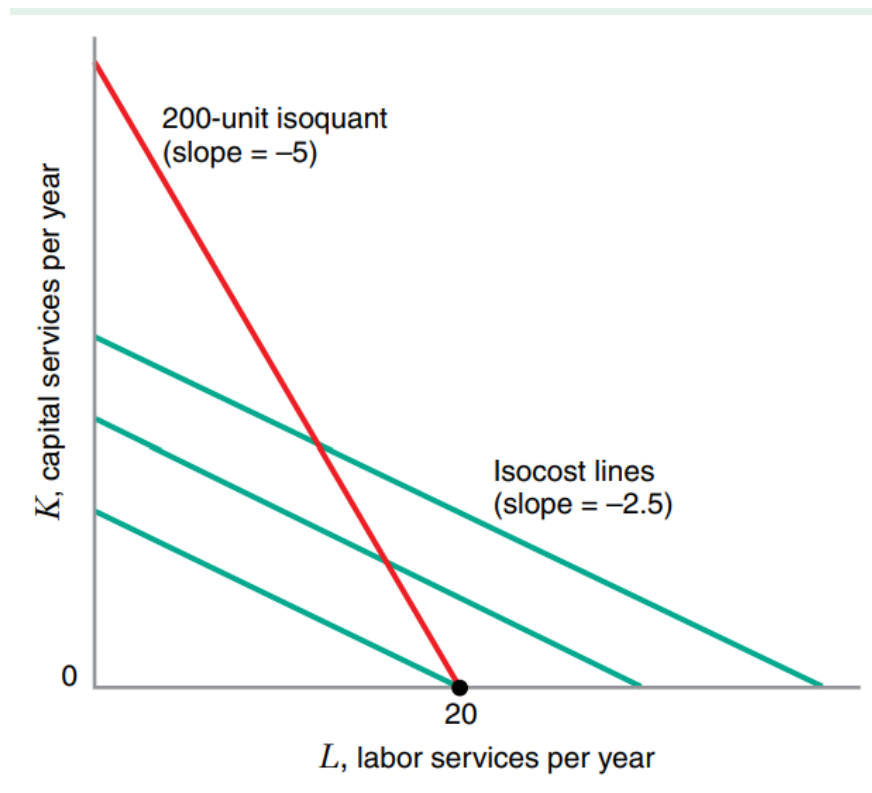
$$\begin{aligned} a) \text{ MinTC}(L, K) &= 5L + 2K : \text{Objective Function} \\ \text{s.t. } 10L + 2K &= 200 : \text{Constraint} \end{aligned}$$

$$b) \text{ Bang for buck : } \frac{MP_L}{w} = \frac{10}{5} > \frac{2}{2} = \frac{MP_K}{r}$$

The producer should only use labor to minimize TC to produce $Q = 200$

$$10L^* + 2 \cdot 0 = 200 \Rightarrow 10L^* = 200$$

As a result, $L^* = 20, K^* = 0$

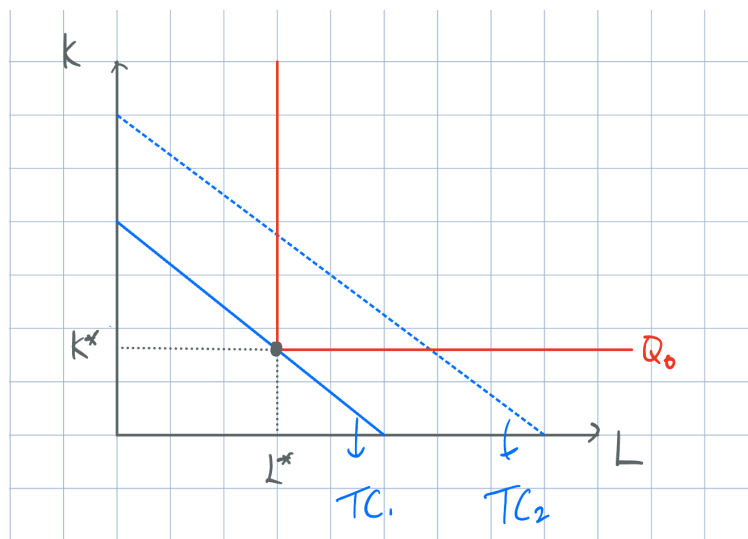


- Long-run perfect complements total cost minimization

$$\text{Min}TC(L, K) = wL + rK : \text{Objective Function}$$

$$\text{s.t. } Q_0(L, K) = \min(AL, BK) : \text{Constraint}$$

- To find the solution, we need to find the point where the isocost line and the isoquant line intersect.
 - * The intersection happens where $AL = BK = Q_0$
- Below is the graphical understanding of the solution under the perfect complement production function.



Exercise. (Perfect complement production function case) Let $Q = \min(2L, 4K)$ and $Q_0 = 20$. a) Set up the firm's minimization problem. b) solve for K^* and L^* .

$$a) \text{Min}TC(L, K) = wL + rK : \text{Objective Function}$$

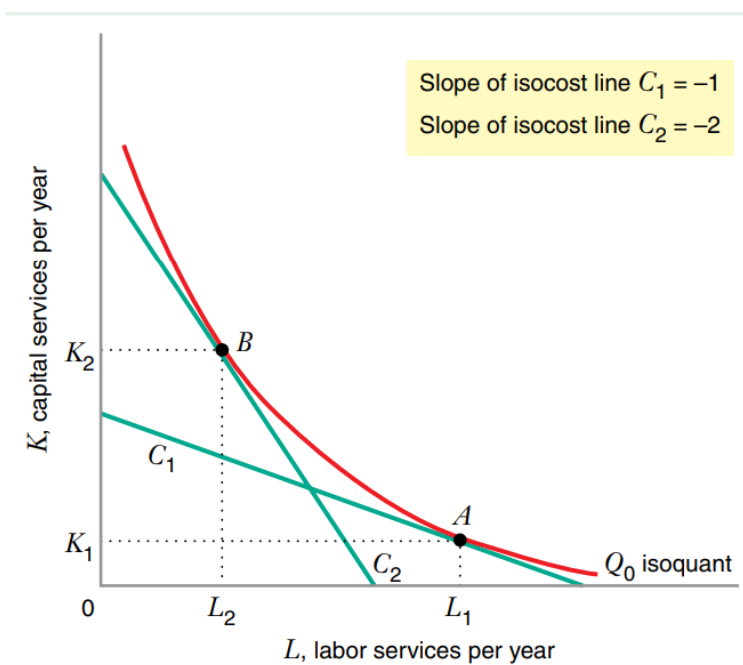
$$\text{s.t. } \min(2L, 4K) = 20 : \text{Constraint}$$

b) Find the intersection by setting : $2L = 4K = 20$

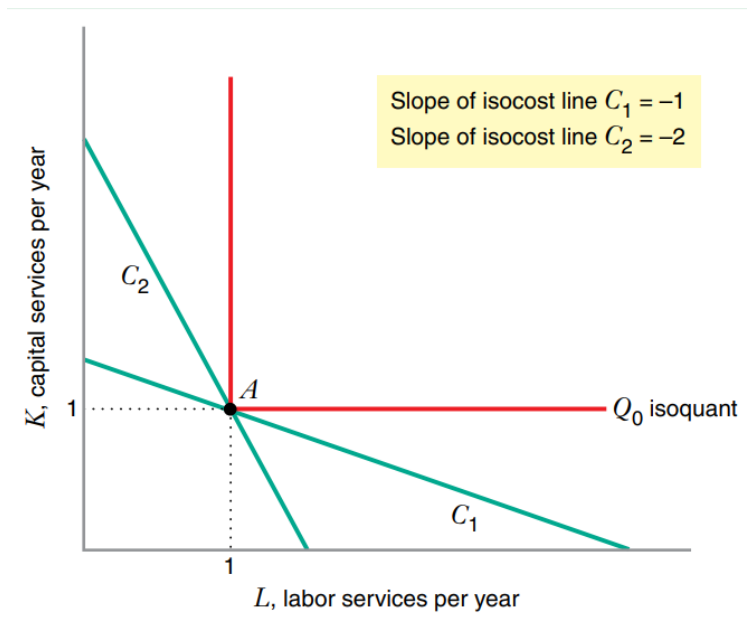
As a result, $L^* = 10$, and $K^* = 5$

Prices of the inputs do not matter in this case.

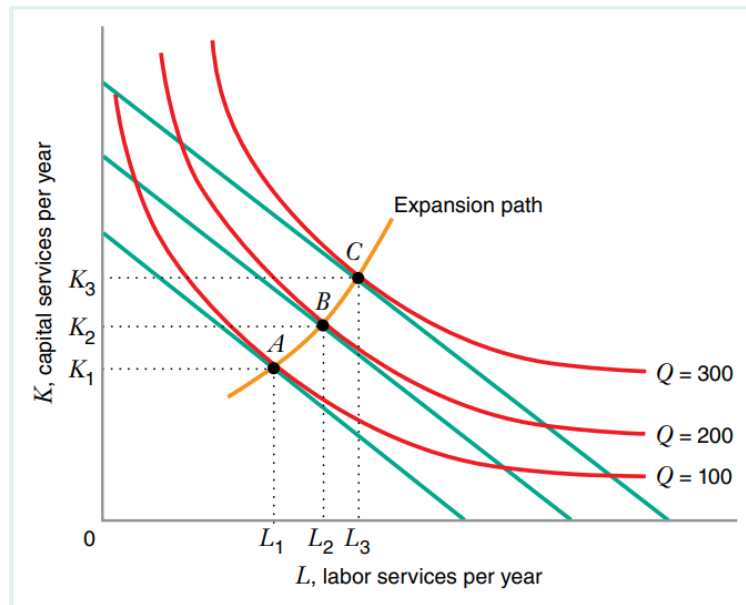
- Comparative statics analysis of changes in input prices(w or r)
 - Let's suppose the price of labor w changes, with the price of capital r held constant and the quantity of output held constant at Q_0
 - * w increases: the cost-minimizing quantity of labor goes down, while the cost-minimizing quantity capital goes up.
 - * **The increase in w causes the firm to substitute capital for labor($A \Rightarrow B$).**
 - This is because the labor has become more expensive relative to capital(the slope of the isocost line has changed: $C_1 \Rightarrow C_2$)



- Summary
 - * Smooth isoquants with a diminishing $MRTS_{L,K}$, and is initially using positive quantities of input, then an increase in the price of that input will cause the cost-minimizing quantity of that input to go down
 - * When the firm is initially using a *zero* quantity of the input or the firm has a *fixed-proportions(perfect complements)*, an increase in the price of the input will leave the cost-minimizing input quantity unchanged(see the graph on the next page)

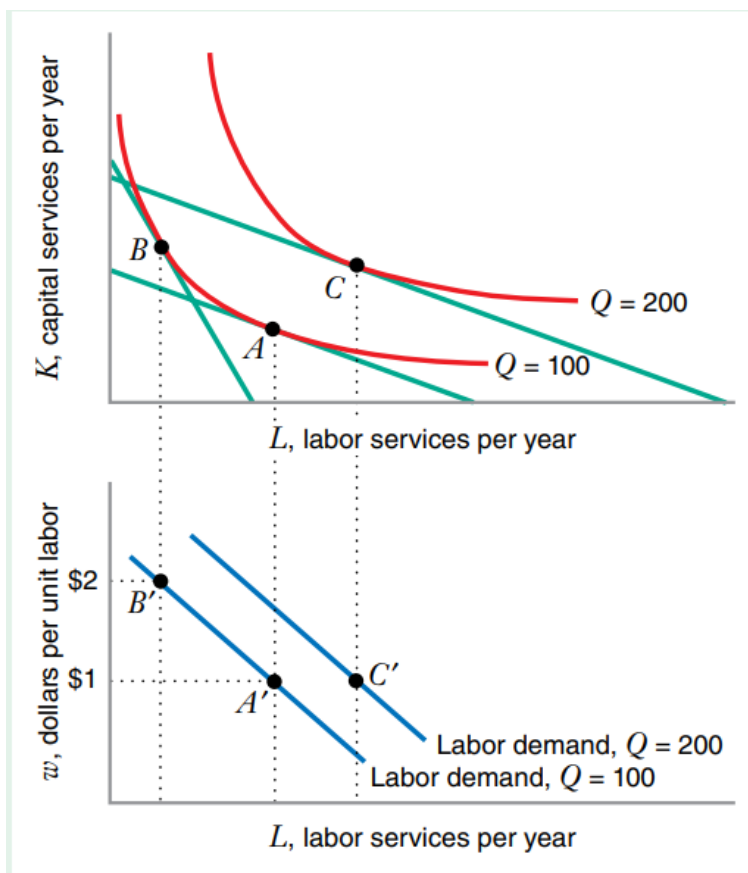


- Comparative statics analysis of changes in output(Q)
 - Let's suppose the Q changes with the prices of inputs(w, r) held constant
 - * As Q increases, the cost-minimizing combination of inputs moves to the north-east. from point A to C .
 - Expansion path
 - * A line that connects the cost-minimizing input combinations as the Q varies, holding input prices constant.



- Normal & Inferior input
 - (1) Normal input: An input whose cost-minimizing Q *increases* as the firm produces more output
 - (2) Inferior input: An input whose cost-minimizing Q *decreases* as the firm produces more output

- The input Demand curves
 - Labor and Capital Demand curve
 - * Labor demand curve: A curve that shows how the firm's cost-minimizing quantity of L varies with the w
 - * Capital demand curve: A curve that shows how the firm's cost-minimizing quantity of K varies with the r
 - e.g. Initially, $Q = 100, w = \$1, \bar{r} = \1
 - (1) $A \Rightarrow B$: when w increase from \$1 to \$2, holding $Q = 100$
 - * From this, we can see that the labor demand curve is (generally) downward sloping (movement along the same labor demand curve)
 - (2) $A \Rightarrow C$: when w, r are constant, Q increases from 100 to 200
 - * A change in the level of Q leads to a *shift* in the labor demand curve
 - * If output increase and an input is *normal*, the demand for that input will shift to the *right*
 - * If output increases and an input is *inferior*, the demand for that input will shift to the *left*



- Deriving the input demand from a production function
 - (Cobb-Douglas production function). $Q = 50\sqrt{KL}$. What are the demand curves for labor and capital?

$$\min(TC) = wL + rK$$

$$\text{subjected to } Q = 50\sqrt{KL}$$

$$\Rightarrow Q^2 = 2500KL \Rightarrow K = \frac{Q^2}{2500L}$$

$$(\text{plug-in}) \Rightarrow wL + r\left(\frac{Q^2}{2500L}\right)$$

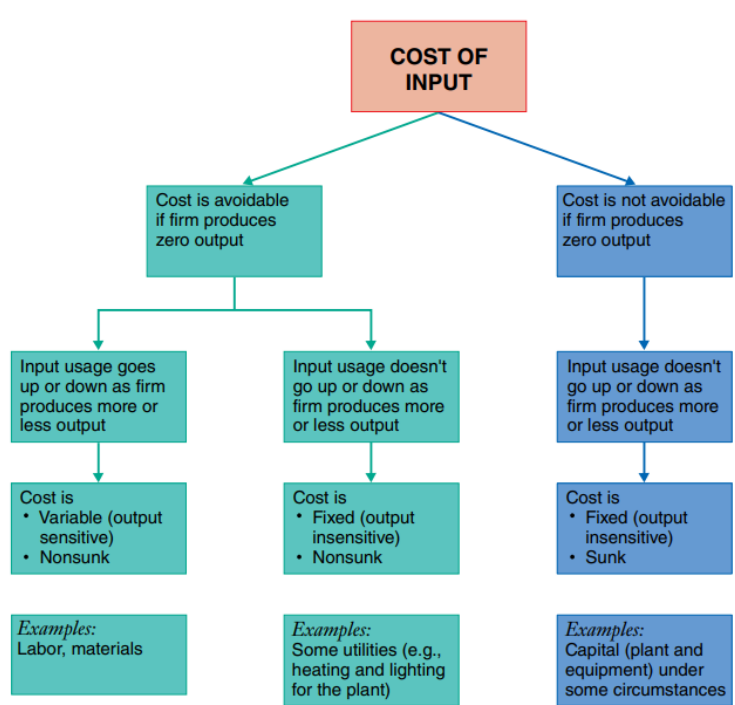
$$(\text{F.O.C}) \Rightarrow w + \frac{rQ^2}{2500} \cdot (-1L^{-2}) = 0$$

$$\Rightarrow L^2 = \frac{rQ^2}{w \cdot 2500} \Rightarrow L(w, r, Q) = \sqrt{\frac{r}{w}} \cdot \frac{Q}{50}$$

$$\text{Labor demand function : } L(w, r, Q) = \sqrt{\frac{r}{w}} \cdot \frac{Q}{50}$$

$$\text{Capital demand function : } K(w, r, Q) = \sqrt{\frac{w}{r}} \cdot \frac{Q}{50}$$

- The **short-run** cost-minimization problem
 - The firm's cost-minimization problem in the short-run
 - * When the firm faces the constraint that one or more of the firm's inputs cannot be changed(held constant)
 - * Suppose that the firm is unable to alter its quantity of $\bar{K}$, even if it produces zero output, but can alter its quantity of L ($TC = wL + r\bar{K}$)
 - Costs
 1. ***TVC(total variable cost)***
 - * The sum of expenditures on variable inputs, such as labor and materials, at the short-run cost-minimizing input combination
 - * wL : output-sensitive component of the TC . Also a non-sunk cost
 2. ***TFC(total fixed cost)***
 - * The cost of fixed inputs; it does not vary with output
 - * $r\bar{K}$: output-insensitive component of the TC .
 - Are fixed costs and sunk costs the same?
 - * Variable costs are always non-sunk costs
 - * However fixed costs are not necessarily sunk costs
 - e.g. The rental proceeds will cover the mortgage payment, the firm can avoid all of the fixed costs by renting its plant.



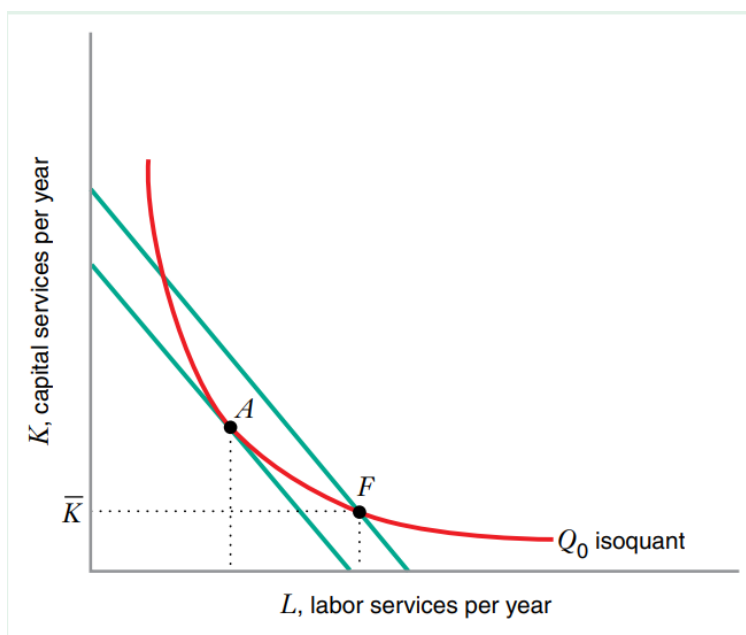
- Cost minimization in the short-run

$$\text{MinTC}(L, K) = wL + r\bar{K} : \text{Objective Function}$$

$$\text{s.t. } aL^\alpha \bar{K}^\beta = Q_0 : \text{Constraint}$$

– Firm's problem: seeks to produce Q_0 under $(\bar{K}, L)$.

- * The firm's only technically efficient combination of inputs occurs at point F , where the firm uses the minimum quantity of labor that, in conjunction with the fixed quantity of $\bar{K}$
- * Point A is unable because of the $\bar{K}$ (feasible in the long-run)



Exercise. (Cobb-Douglas production function case) Let $Q = 2L^{\frac{1}{2}}K^{\frac{1}{2}}$, $w = 4$, $r = 1$, $Q_0 = 16$, and $K = \bar{K}$. Find the cost minimization L^* , and K^* in the short-run.

$$\text{MinTC}(L, \bar{K}) = 4L + 1\bar{K} : \text{Objective Function}$$

$$\text{s.t. } 2L^{\frac{1}{2}}\bar{K}^{\frac{1}{2}} = 16 : \text{Constraint}$$

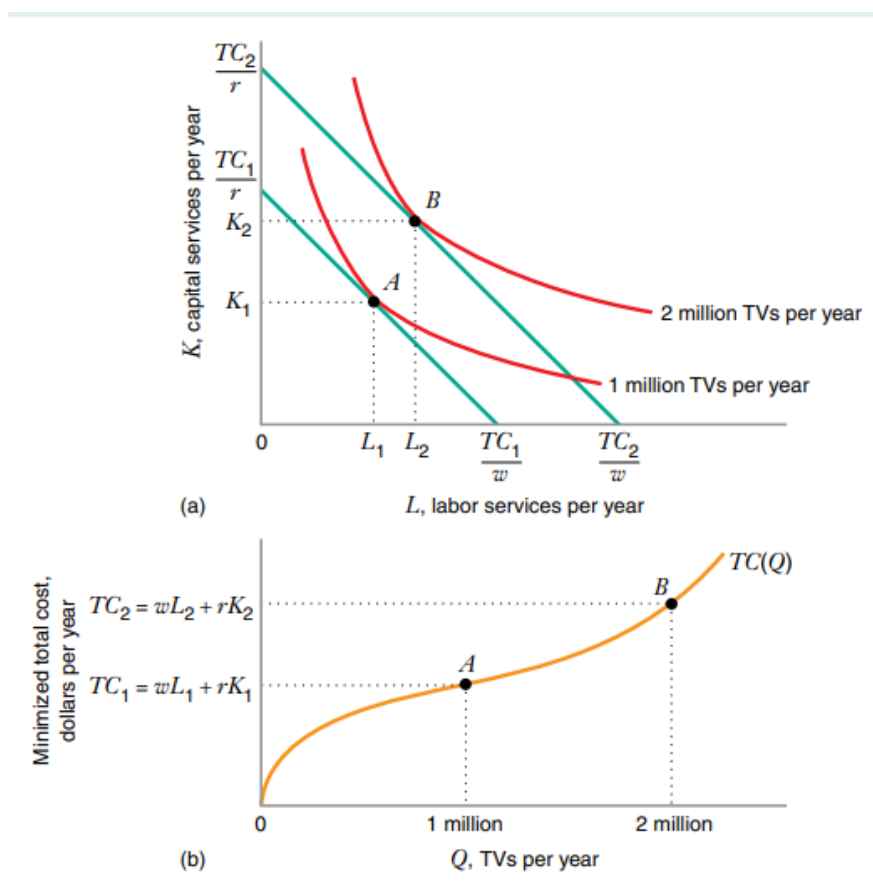
$$\Rightarrow (8)^2 = (L^{\frac{1}{2}}\bar{K}^{\frac{1}{2}})^2 \Rightarrow 64 = L\bar{K}$$

$$\text{As a result, } L^* = \frac{64}{\bar{K}}$$

Ch8. Cost Curves

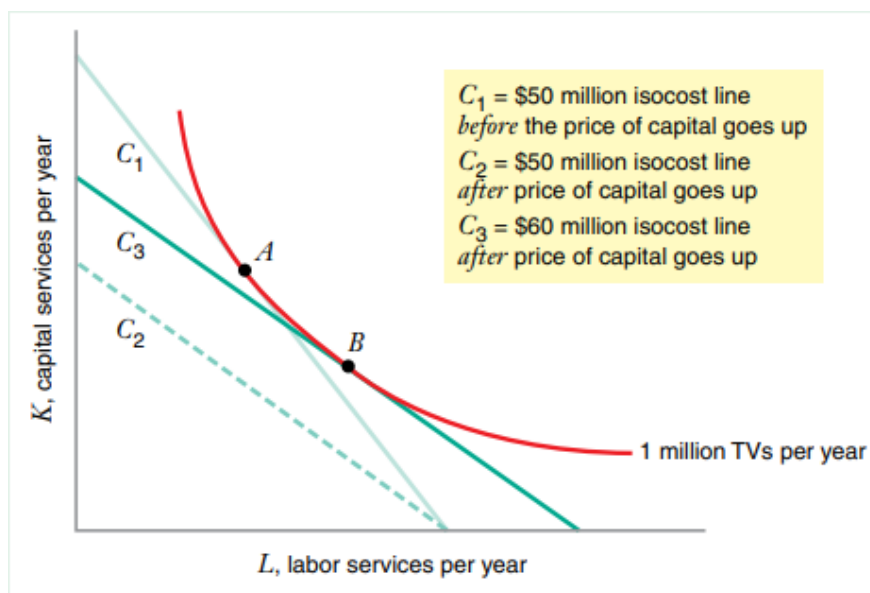
Long-run Total Cost Curve

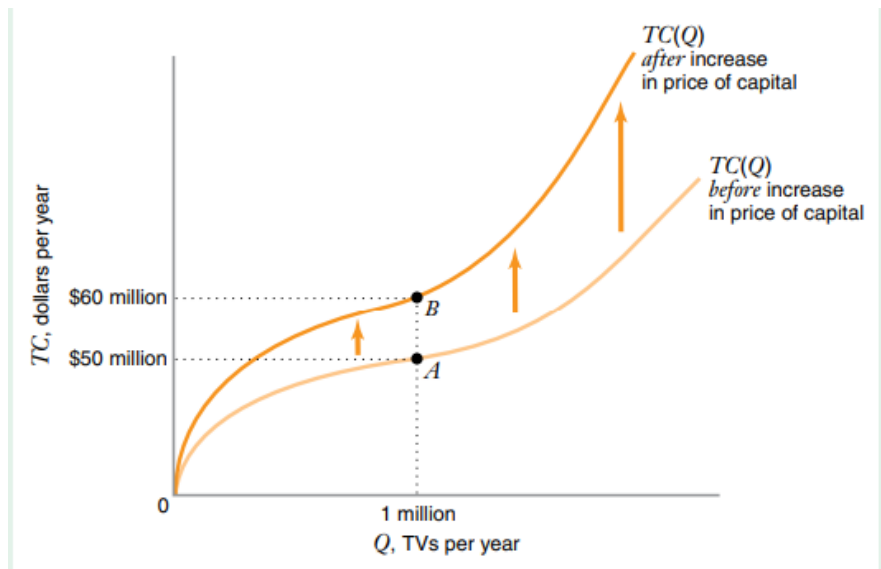
- Long-run Total Cost Curves
 - Definition: A curve that shows how total cost varies with output(Q) holding input prices fixed($\bar{r}, \bar{w}$) and choosing all inputs to minimize cost.
 - Because the cost-minimizing input combination moves us to higher isocost lines, the long-run total cost curve must be increasing in Q .
 - Comparative statics analysis of the cost-minimization problem implies that the long-run total cost curve must be increasing in Q and must equal 0 when $Q = 0$
 - * This is because, in the long run, the firm is free to vary all its inputs.



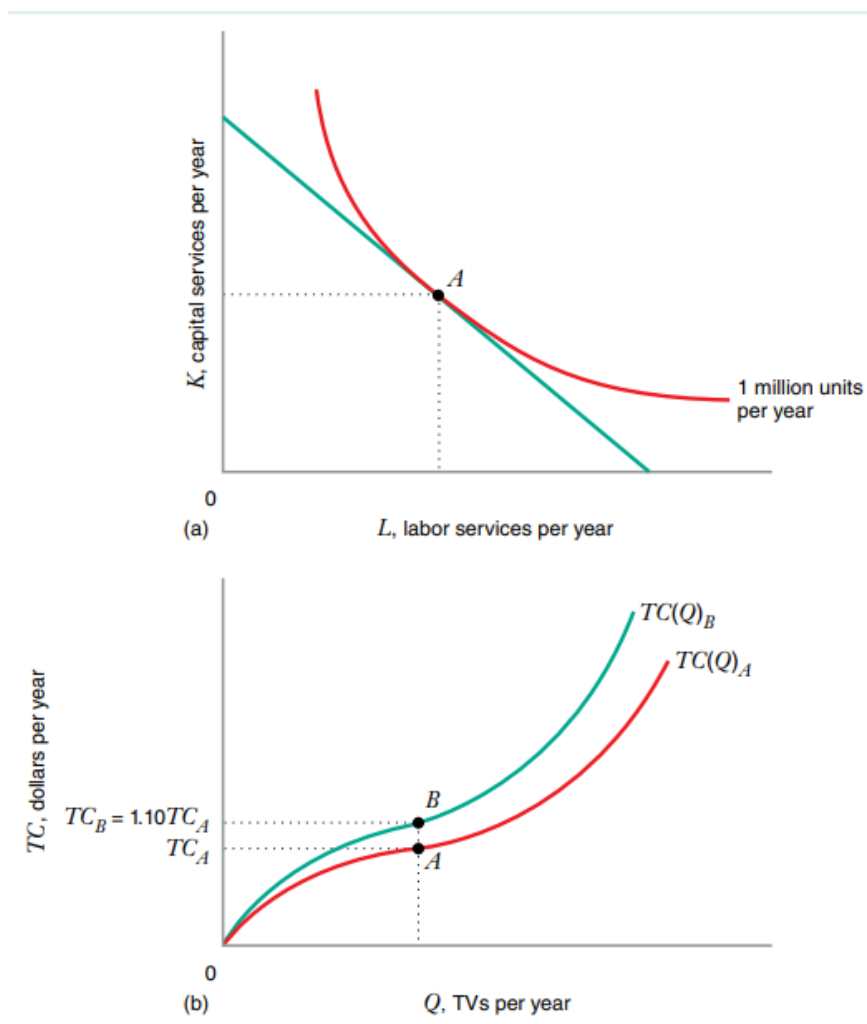
How does the long-run TC shift when input prices change?

- What happens when just one input price changes?
 - e.g When the price of capital increases $r \Rightarrow r' (r < r')$ holding output and the price of labor(w) constant
 - (1) Optimal input combination for $\bar{Q}$ occurs at A on isocost line C_1 (\$50million)
 - (2) After the r increases, the \$50million isocost line at the new input prices(C_2) intersects the horizontal axis in the sample place as the \$50million isocost line at the old input prices
 - (3) Thus, the firm could not operate on cost line C_2 because it would be unable to produce $\bar{Q}$
 - (4) Instead, the firm must operate on an isocost C_3 and thus corresponds to a higher level of cost(\$60million)
 - Thus, holding output fixed, the minimized total cost goes up when the price of an input goes up(vice versa)
 - This implies that an increase in the price of an input results in a new TC that lies above the original TC at every $Q > 0$.
 - * *An increase in an input price rotates the long-run total cost curve upward(graph on the next page)*



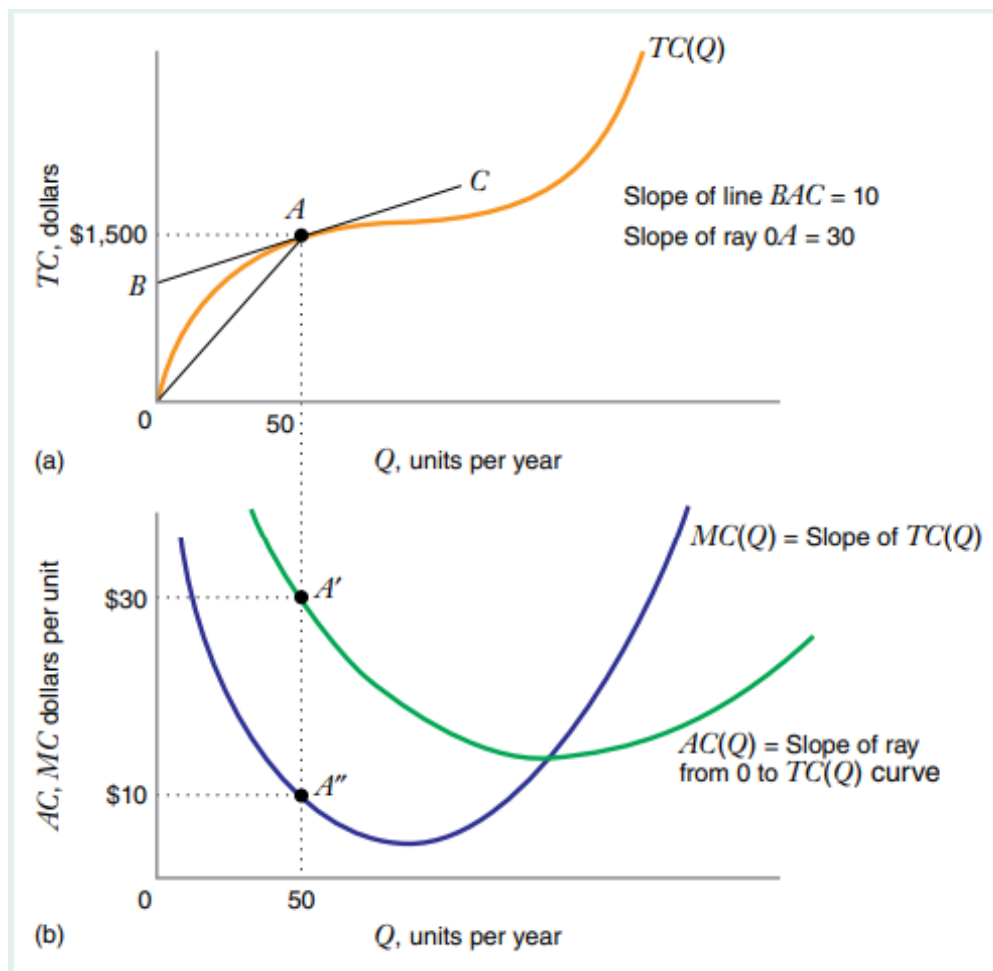


- When input prices change *proportionately*
 - A given percentage increase in both input prices leaves the cost-minimizing input combination unchanged, while the TC curve shifts up by exactly the same percentage.
 - e.g. At the initial prices of labor w and capital r , the cost-minimizing input combination is at point A
 - * If both input prices increase by 10 percent, the ideal combination is still at point A
 - * This is because the slope is still the same, so the point of tangency between the isocost line and the isoquant is also unchanged.
 - * $TC_A = wL + rK \Rightarrow TC_B = 1.10wL + 1.10rK = 1.10TC_A$ (Just TC changes)



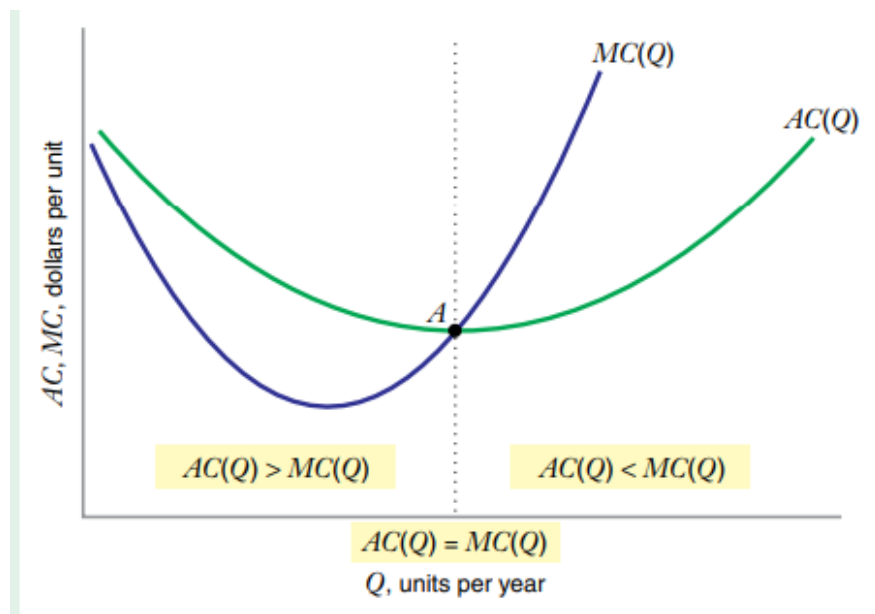
- Long-run average and marginal cost curves

- **Long-run average cost:** The firm's total cost per unit of output. It equals the long-run total cost divided by the total quantity ($AC(Q) = \frac{TC(Q)}{Q}$)
 - * Long-run average cost is equal to the slope of a ray from the origin
- **Long-run marginal cost:** The rate at which long-run total cost changes with respect to change in output ($MC(Q) = \frac{\Delta TC}{\Delta Q} = \text{slope of } TC(Q)$)
 - * Long-run marginal cost is equal to the slope of the long-run total cost curve itself at that point
- e.g. $TC(Q)$, $AC(Q)$, $MC(Q)$



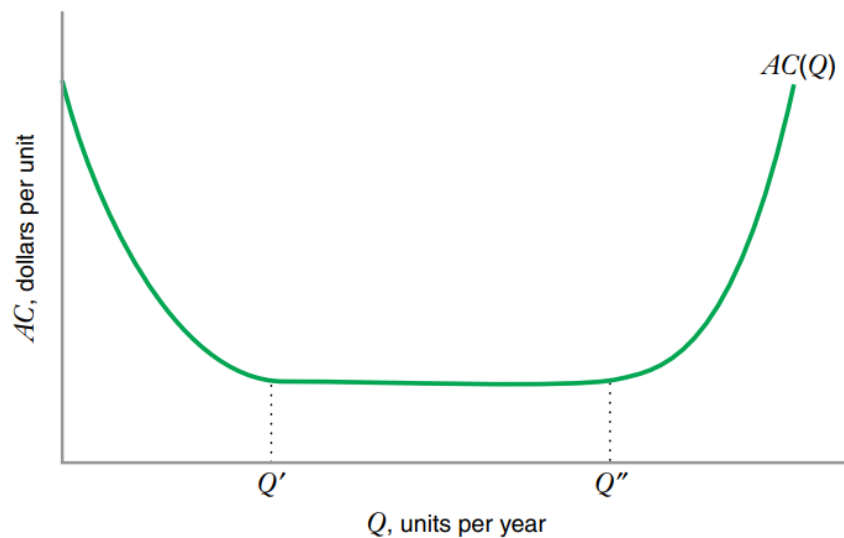
- Relationship between Long-run Average and Marginal Cost curves
 - (1) If AC is decreasing as quantity is increasing, then AC is greater than MC:
 $AC(Q) > MC(Q)$
 - (2) If AC is increasing as quantity is increasing, then AC cost is less than MC:
 $AC(Q) < MC(Q)$
 - (3) If AC is neither increasing nor decreasing as quantity is increasing, then AC is equal to MC: $AC(Q) = MC(Q)$

– $MC(Q)$ cuts the $AC(Q)$ curve at its minimum(Point A)



- Economies and Diseconomies of Scale

- **Economies of scale:** A characteristic of production in which AC decreases as Q goes up
- **Diseconomies of scale:** A characteristic of production in which AC increases as Q goes up
- Economies of scale can also explain why some firms are more profitable than others in the same industry.
 - * Claims of economies of scale are often used to justify mergers between two firms producing the same product
- e.g. Below is an average cost curve
 - * $Q < Q'$: a range of economies of scale
 - * $Q > Q''$: a range of diseconomies of scale



- Various causes of the economies of scale
 - * Specialization of labor
 - * Indivisible input: An input that is available only in a certain minimum size. Its quantity cannot be scaled down as the firm's output goes to zero. ex) high-speed packaging line for breakfast cereal
 - Indivisible inputs lead to decreasing AC because when a firm purchases the services of an indivisible input, it can *spread* the cost of the input over more units of output goes up

- Various causes of the diseconomies of scale
 - * Managerial diseconomies: A situation in which a given percentage increase in output forces the firm to increase its spending on the services of managers by more than this percentage.
- **Minimum Efficient Scale(MES)**
 - * The smallest quantity at which the long-run AC curve attains its minimum point
 - * The size of MES relative to the size of the market often indicates the significance of economies of scale in particular industries.

- Economies of scale and returns to scale
 - The returns to scale of the production function determine how long-run AC varies with output.
 - (1) If AC decreases as Q increases, we have *economies of scale* and *increasing returns to scale*
 - (2) If AC increases as Q increases, we have *diseconomies of scale* and *decreasing returns to scale*
 - (3) If AC stays the same as Q increases, we have *neither economies nor diseconomies of scale* and *constant returns to scale*

	Production Function		
	$Q = L^2$	$Q = \sqrt{L}$	$Q = L$
Labor requirements function	$L = \sqrt{Q}$	$L = Q^2$	$L = Q$
Long-run total cost	$TC = w\sqrt{Q}$	$TC = wQ^2$	$TC = wQ$
Long-run average cost	$AC = \frac{w}{\sqrt{Q}}$	$AC = wQ$	$AC = w$
How does long-run average cost vary with Q ?	Decreasing	Increasing	Constant
Economies/diseconomies of scale?	Economies of scale	Diseconomies of scale	Neither
Returns to scale	Increasing	Decreasing	Constant

- The output elasticity of total cost
 - Definition: The percentage change in total cost per 1 percent change in output
 - * We use the elasticity to tell how sensitive total cost is to the factors that influence it.

$$\epsilon_{TC,Q} = \frac{\frac{\Delta TC}{TC}}{\frac{\Delta Q}{Q}} = \frac{\frac{\Delta TC}{\Delta Q}}{\frac{TC}{Q}}$$

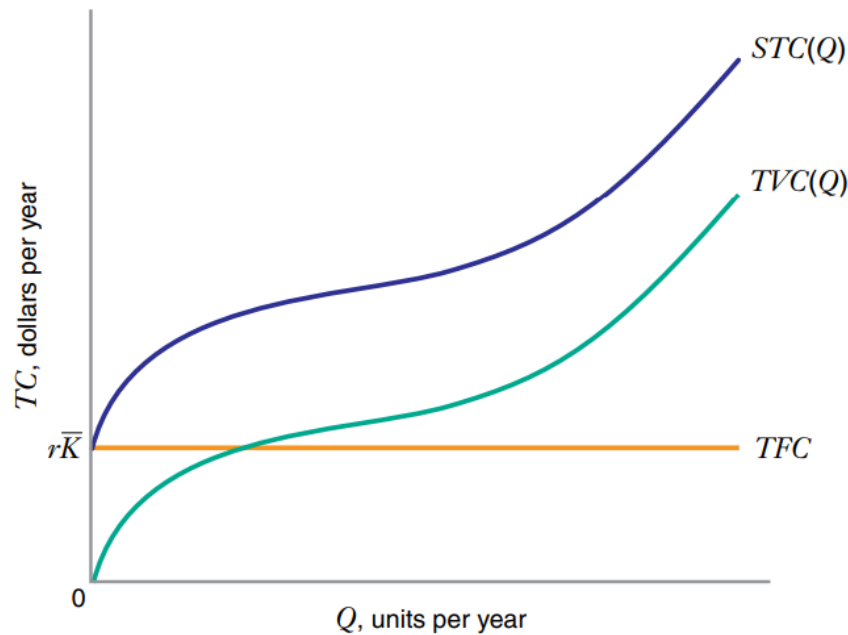
$$= \frac{MC}{AC}$$

- The output elasticity of total cost is equal to the ratio of marginal to average cost.

Value of $\epsilon_{TC,Q}$	<i>MC</i> versus <i>AC</i>	How <i>AC</i> Varies as <i>Q</i> Increases	Economies/ Diseconomies of Scale
$\epsilon_{TC,Q} < 1$	$MC < AC$	Decreases	Economies of scale
$\epsilon_{TC,Q} > 1$	$MC > AC$	Increases	Diseconomies of scale
$\epsilon_{TC,Q} = 1$	$MC = AC$	Constant	Neither

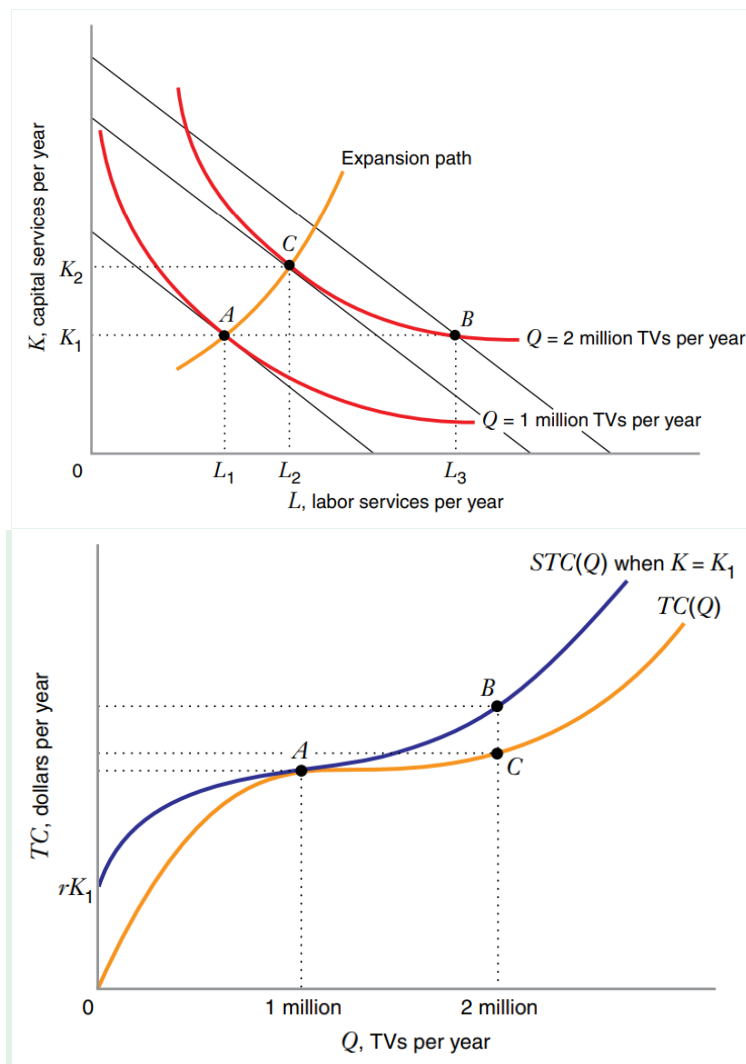
Short-run Total Cost Curve

- **Short-run total cost curve: $STC(Q)$**
 - Definition: A curve that shows the minimized total cost of producing a given quantity of output when at least one input is fixed
 - We assume that the amount of capital(K) used by the firm is fixed at $\bar{K}$
- $STC(Q) = TVC(Q) + TFC = wL + r\bar{K}$
 - Total variable cost curve: $TVC(Q)$
 - * A curve that shows the sum of expenditures on variable inputs, such as labor and materials, at the short-run cost-minimizing input combination
 - Total fixed cost curve: TFC
 - * A curve that shows the cost of fixed inputs and does not vary with output.
 - $STC(Q) = TVC(Q) + r\bar{K}$
 - * The vertical distance between $STC(Q)$ and $TVC(Q)$ is equal to the $TFC = r\bar{K}$ at every quantity Q



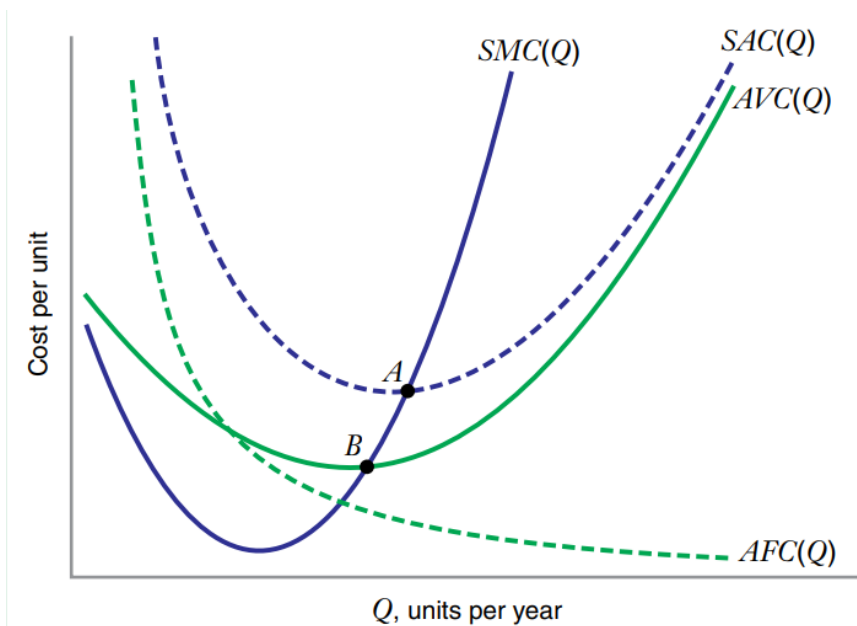
- Relationship between the **long-run** and the **short-run TC**

- In the long-run, the firm can freely vary the quantity of both inputs, but in the short-run the quantity of capital is fixed ($K = \bar{K}$).
 - * Thus, the firm is more constrained in the short-run, so it makes sense that it will be able to achieve *lower* total costs in the long-run
- e.g. Graphical analysis of the cost-minimization between long and short
 - * *A*: When the firm wants to produce 1million TV in the long run
 - * *B*: In the short-run, K is fixed but want to produce 2million
 - * *C*: In the long-run, when the firm wants to produce 2million
 - * *B* is on a higher isocost line than *C*, given the same prices of the inputs, the short-run total cost is higher than the long-run total cost

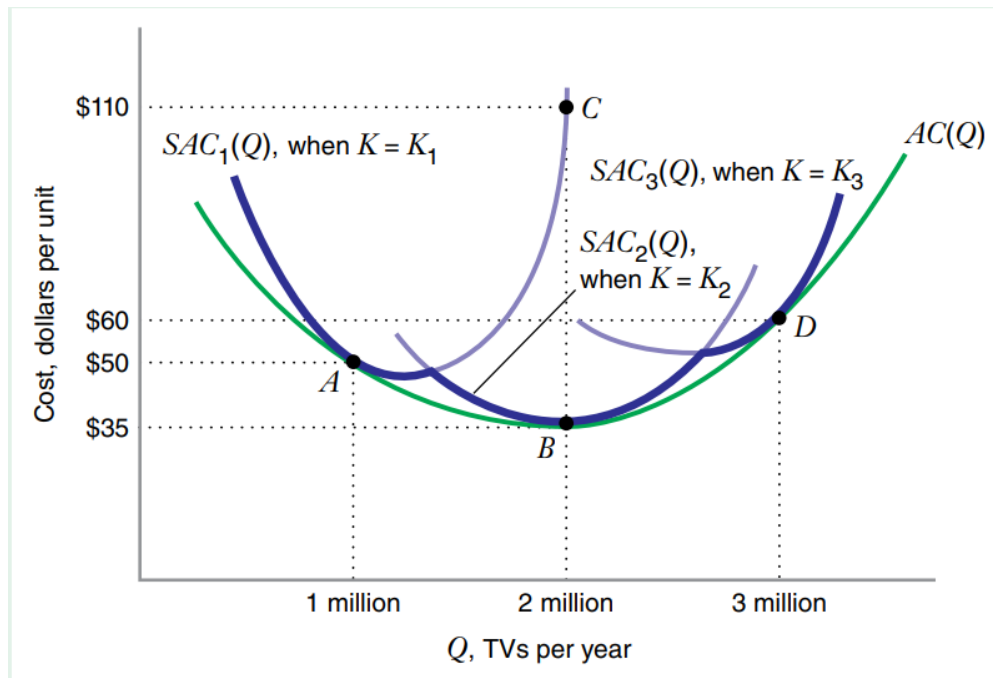


• Short-run average(SAC) and marginal(SMC) cost curves

- Short-run average cost($SAC(Q) = \frac{STC(Q)}{Q}$): The firm's total cost per unit of output when it has one or more fixed inputs
- Short-run marginal cost($SMC(Q) = \frac{\Delta STC}{\Delta Q}$): The slope of the short-run total cost curve
- $SAC = AVC + AFC$
 - * Average variable cost($AVC = \frac{TVC}{Q}$): Total variable cost per unit of output
 - * Average fixed cost($AFC = \frac{TFC}{Q}$): Total fixed cost per unit of output. It decreases everywhere and approaches the horizontal axes as Q becomes very large
- e.g. Typical graphs of the short-run marginal cost
 - * The $AVC(Q)$ and $SAC(Q)$ curves get closer and closer together as Q increases.
 - * $SMC(Q)$ intersects the $SAC(Q)$ and $AVC(Q)$ both at the minimum point of each curve(point A and point B)



- The long-run average cost curve as an envelope curve
 - The long-run average cost curve forms a boundary(or envelope) around the set of short-run average cost curves corresponding to different levels of output and fixed input
 - $AC(Q)$ is U-shaped, as are its short-run average cost curves $SAC_1(Q)$, $SAC_2(Q)$, $SAC_3(Q)$, which correspond to different levels of fixed capital K_1 , K_2 , K_3



- Economies of scope
 - Definition: A production characteristic in which the total cost of producing given quantities of two goods in the same firm is less than the total cost of producing those quantities in two single-product firms
 - * $TC(Q_1, Q_2)$: denote how the firm's costs vary with Q_1 and Q_2
 - * $TC(Q_1, Q_2) < TC(Q_1, 0) + TC(0, Q_2)$
 - Stand-alone cost: The cost of producing a good in a single-product firm
 - Intuitively, the existence of economies of scope tells us that "variety" is more efficient than "specialization".
 - * An important reason is a firm's ability to use a common input to make and sell more than one product.

Ch9. Perfectly Competitive Markets

What is perfect competition?

- Characteristics of perfectly competitive markets
 - (1) **Fragmented industry:** An industry that consists of many small buyers and sellers; one of the characteristics of a perfectly competitive industry
 - Each seller's output is so small in comparison to market demand that it has an *imperceptible impact* on the input prices and market prices.
 - (2) **Undifferentiated products:** Products that consumers perceive as being *identical*; one of the characteristics of a perfectly competitive industry
 - *Homogeneous* goods are produced by each producer
 - (3) **Perfect information about prices:** Full awareness by consumers of the prices charged by all sellers in the market; one of the characteristics of a perfectly competitive industry.
 - (4) **Equal access to resources:** A condition in which all firms—those currently in the industry, as well as prospective entrants have access to the same technology and inputs; one of the characteristics of a perfectly competitive industry
- Three implications of perfectly competitive markets
 - (1) **Price taker:** A seller or a buyer that takes the price of the product as given when making an output decision(seller) or a purchase decision(buyer)
 - (2) **Law of one price:** In a perfectly competitive industry, the occurrence of all transactions between buyers and sellers at a single, common market price
 - Because the products of all firms are perceived to be identical and the prices of all sellers are known, a consumer will purchase at the lowest price available in the market
 - (3) **Free entry:** Characteristic of an industry in which any potential entrant has access to the same technology and inputs that existing firms have
 - If it is profitable for new firms to enter the industry, they will eventually do so.
 - It does not mean that a new firm incurs no cost when it enters the industry, but rather that it has access to the same technology and inputs that existing firms have

Profit maximization by a price-taking firm

- Economic profit versus accounting profit
 - Economic profit: The difference between a firm's sales revenue and the totality of its economic costs, including all relevant opportunity costs.

$$\text{Economic profit} = \text{Sales revenue} - \text{economic costs}$$

$$\text{Accounting profit} = \text{Sales revenue} - \text{accounting costs}$$

- The profit-maximizing output choice for a price-taking firm
 - Economic value added: A widely used measure of economic profit, equal to the company's accounting profit minus the minimum return on invested capital demanded by the firm's investors.

$$\pi = TR(Q) - TC(Q)$$

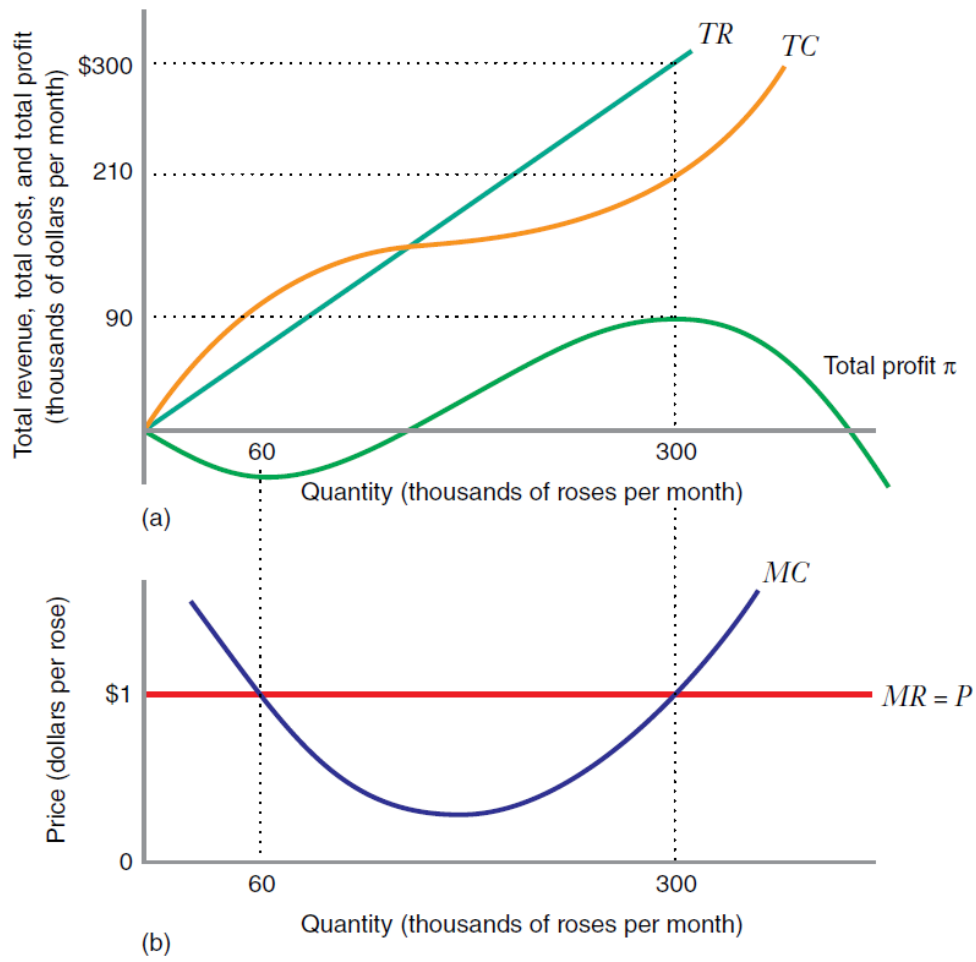
$$TR(Q) = P \times Q$$

$$TC(Q) : \text{total cost curve}$$

- * Because the firm is a price-taker, it perceives that its volume decision has a negligible impact on the market price
- * **Thus, it takes the market price P as given. Its goal is to choose Q to maximize its total profit.**
- Marginal revenue: The rate at which total revenue changes with respect to Q .

$$MR = \frac{\Delta TR}{\Delta Q}$$

- * For a price-taking firm, each additional unit sold increases $TR(Q)$ by an amount of equal to the market price; $\Delta TR / \Delta Q = P$
- * Thus, for a price-taking firm, $MR = P$
- $P = MC$
 - * A price-taking firm maximizes its profit when it produces a quantity Q^* at which the $MC = P = MR$



How the market price is determined: Short-run equilibrium

- The price-taking firm's short-run cost structure
 - **Total variable costs**($TVC(Q)$): These are output-sensitive costs that they go up or down as the firm increases or decreases its output(Q)
 - **Sunk fixed costs**(SFC): A fixed cost that the firm cannot avoid if it shuts down and produces zero output. Also called as *unavoidable costs*
 - **Nonsunk fixed cost**($NSFC$): A fixed cost that must be incurred for a firm to produce any output but that does not have to be incurred if the firm produces no output

$$\begin{aligned} TFC(\text{Total Fixed Costs}) = \\ NSFC(\text{Nonsunk Fixed Costs}) + SFC(\text{Sunk Fixed Costs}) \end{aligned}$$

- Short-run Total Costs

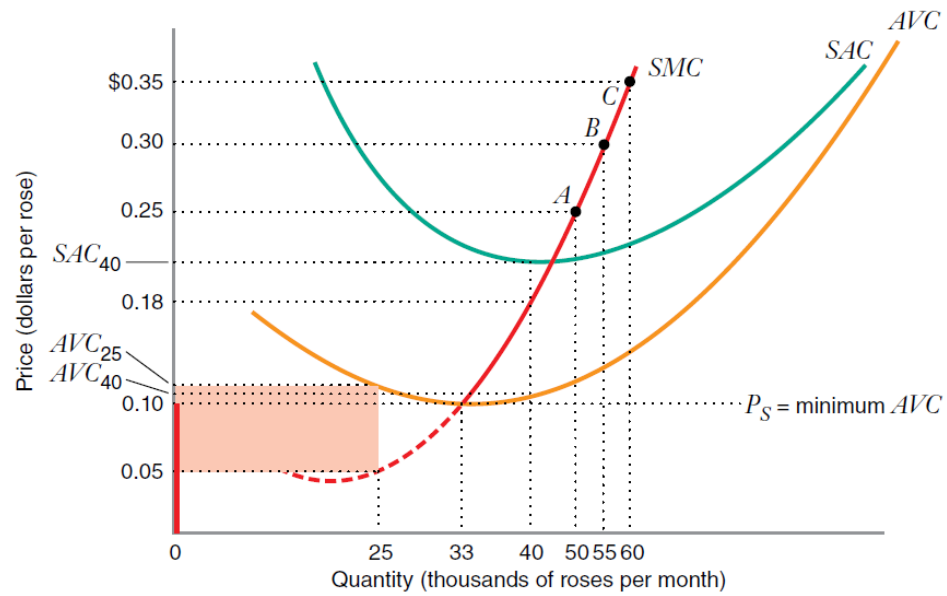
$$STC(Q) = \begin{cases} TFC(NSFC + SFC) + TVC(Q), & \text{when } Q > 0 \\ SFC, & \text{when } Q = 0 \end{cases}$$

- Case1) Short-run supply curve for a price-taking firm **when all fixed costs are sunk**

- Short-run supply curve: The supply curve that shows how the firm's profit-maximizing output decision changes as the market price changes, assuming that the firm cannot adjust all of its inputs($\bar{K}$)
- If $NSFC = 0$ then $TFC = SFC$
- The firm's short-run marginal cost(SMC) curve and the firm's short-run supply curve do not necessarily coincide at all possible prices
 - * The firm is better off cutting its losses by temporarily shutting down if the market price P is less than the average variable cost $AVC(Q^*)$ at the Q^* at which P equals short-run marginal cost, or $P < AVC(Q^*)$
- Draw the firm's short-run supply curve

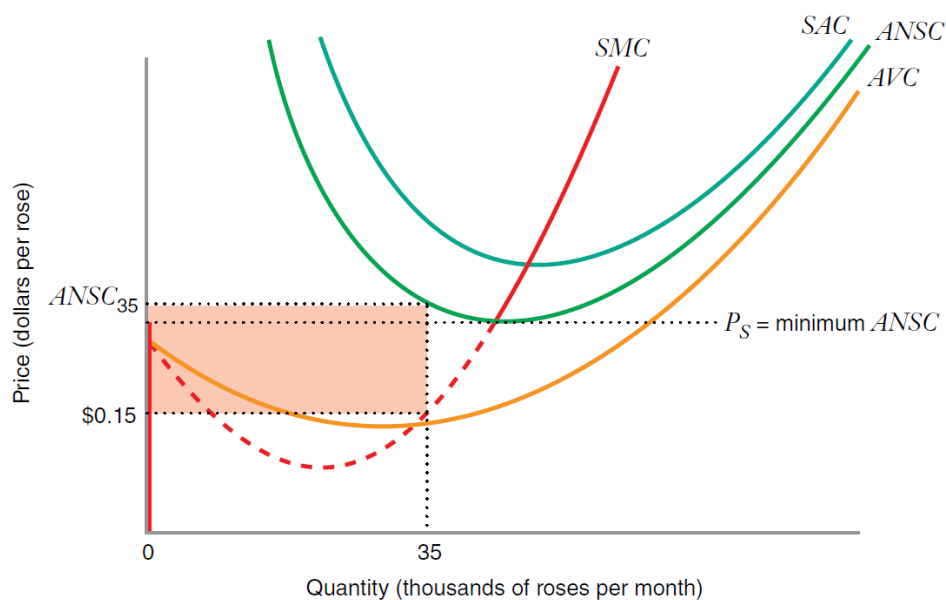
$$\text{Supply curve} = \begin{cases} \text{if } Q > 0, \text{ produces where } P = SMC \text{ and } SMC \text{ slopes upward} \\ \text{firm never produces where } P < AVC \end{cases}$$

- A firm's supply curve has two parts(next page):
 - * If market price(P) is less than P_{AVCmin} , the firm will supply zero output(i.e., $Q=0$)
 - * **Shutdown price:** The price below which a firm supplies zero output in the short run
 - * If the market price(P) is greater than P_{AVCmin} , the firm will produce a positive amount of output, and its short-run supply curve will coincide with its short-run marginal cost(SMC) curve



- Case2) Short-run supply curve for a price-taking firm **when some fixed costs are sunk and some are nonsunk**

- $TFC = SFC + NSFC$, $NSFC > 0$
- Average nonsunk cost($ANSC$): The sum of average variable cost(AVC) and average nonsunk fixed cost($ANFC$)
 - * $ANSC = AVC + \frac{NSFC}{Q}$
 - * $ANSC$ is U-shaped and lies between the short-run average cost curve(SAC) and the average variable cost curve(AVC).
 - * At the minimum point, $SMC = ANSC$.
- If the firm did not produce, its loss would only be its sunk fixed cost(SFC). That is, by temporarily shutting down, the firm would avoid both its variable costs and its nonsunk fixed costs.
 - * By doing so, it avoids an additional loss.
- The firm is better off cutting its short-run losses by not producing if the market price(P) is less than the average nonsunk cost($ANSC(Q^*)$) at the output Q^* at which P equals short-run marginal cost, $P < ANSC(Q^*)$
- Short-run supply curve with sunk and nonsunk cost
 - * A profit-maximizing price-taking firm, if it produces positive output, produces where $P = SMC$ and SMC slopes upward
 - * A profit-maximizing price-taking firm with nonsunk fixed costs would never produce where $P < ANSC$



- Three special cases of the firm's supply curve and shutdown price
 - (1) All fixed costs are sunk: When all fixed costs are sunk($NSFC = 0$), $ANSC = AVC$, and our shutdown rule, $P < ANSC$, becomes $P < AVC$. The firm's short-run supply curve is thus the portion of SMC above the minimum point of the average variable cost curve(AVC)
 - (2) All fixed costs are nonsunk: In this case, $ANSC = SAC(AVC + ANSFC)$. Our shutdown rule, $P < ANSC$, now becomes $P < SAC$. When all fixed costs are nonsunk, the firm's short-run supply curve is the portion of SMC above the minimum point of the short-run average costs curve(SAC)
 - (3) Some fixed costs are sunk and some are nonsunk: The firm's short-run supply curve is the portion of SMC above the minimum point of the average nonsunk cost curve($ANSC$).

Exercise. Suppose that a firm's short-run total cost curve is $STC = 100 + 20Q + Q^2$.

(a) Suppose that $SFC=36$, while $NSFC = 64$. What is the firm's average nonsunk cost curve?

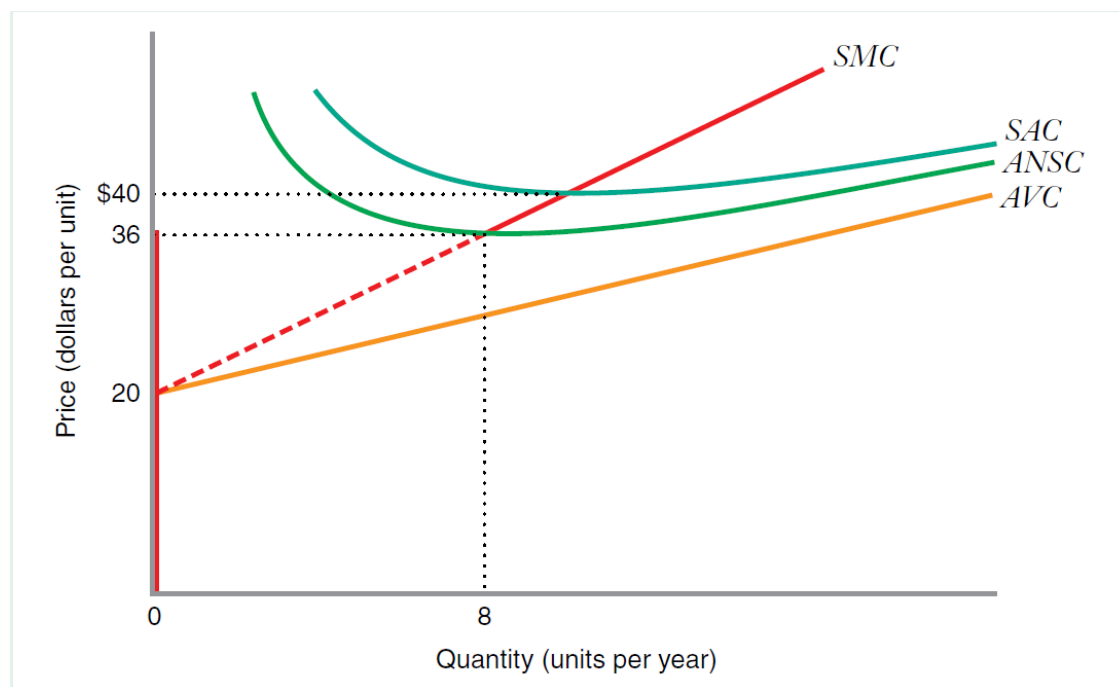
- $ANSC = AVC + ANSFC$. Thus, $ANSC = 20 + Q + \frac{64}{Q}$

(b) What is the minimum level of average nonsunk cost?

- It's when $ANSC = SMC(\frac{\partial STC}{\partial Q})$. Thus, $ANSC = 20 + Q + \frac{64}{Q} = 20 + 2Q = SMC$
- The $Q_{ANSCmin} = 8$. If we plug $Q_{ANSCmin}$ in SMC then we can get $P_{ANSCmin} = 36$

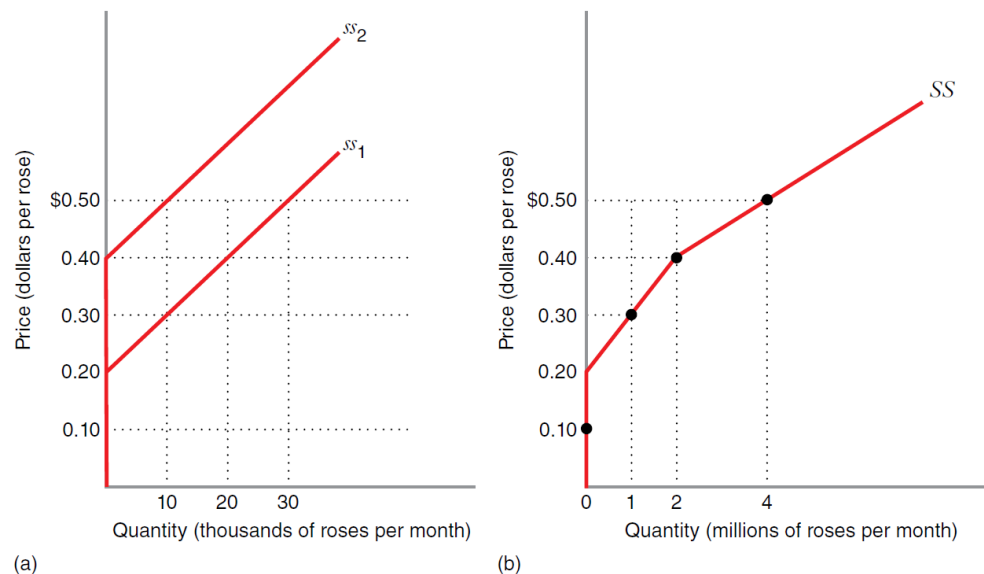
(c) What is the firm's short-run supply curve?

- If $P < 36 = P_{ANSCmin}$, then $s(p) = 0$
- If $P \geq 36 = P_{ANSCmin}$, then $s(p) = -10 + \frac{1}{2}P$

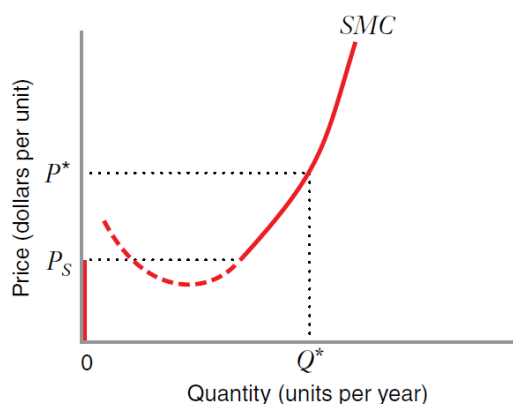


- Short-run Market Supply Curve

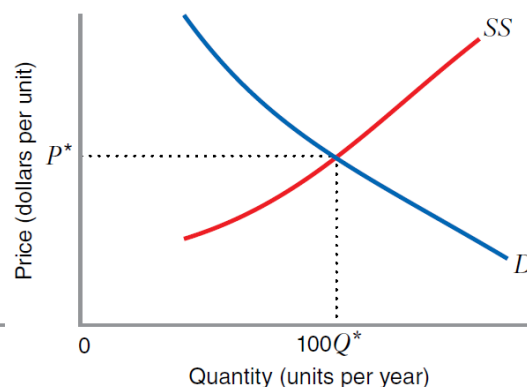
- Having derived the short-run supply curve for an individual price-taking firm, next we need to see how to go from the firm's supply curve to the supply curve for the entire industry.
- Because the number of producers in the industry is fixed in the short-run, market supply at any price is equal to the sum of the quantities that each established firm supplies at that price.
- **Short-run market supply curve:** The supply curve that shows the quantity supplied in the aggregate by all firms in the market for each possible market price when the number of firms in the industry is fixed
- The process of obtaining the market supply curve by summing the individual firm supply curves is subject to one important qualification



- Short-run perfectly competitive equilibrium
 - Short-run perfectly competitive equilibrium
 - * The market price and quantity at which quantity demanded equals quantity supplied in the short-run
 - * The equilibrium price is P^* , where the quantity supplied is equal to the quantity demanded.
 - * Figure (a) shows that a typical firm will produce output Q^* , at which its marginal cost equals the market price P^*



(a) Typical firm

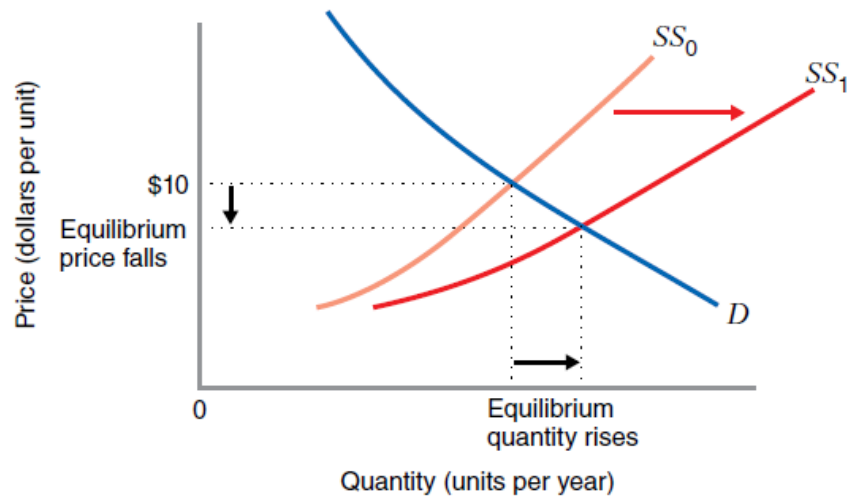


(b) Market

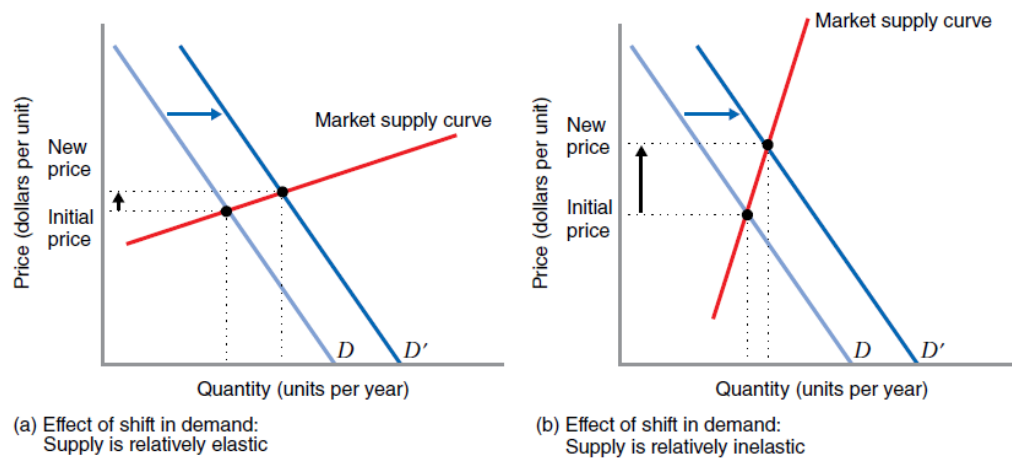
Exercise. A market consists of 300 identical firms, and $D(P) = 60 - P$. $STC = 0.1 + 150Q^2$, and all fixed costs are sunk. What is the short-run equilibrium price in this market?

- $ANSC = AVC = 150Q$, $SMC = 300Q$, thus, the $Q_{ANSCmin} = 0$ and $P_{ANSCmin} = 0$. Therefore, the supply curve $s(p)$ of an individual firm is $s(p) = \frac{P}{300}$
- Since the 300 firms in this market are all identical, the market supply is $300s(P)=P$. Therefore, the short-run equilibrium occurs where market supply equals market demand, or $P = 60 - P$. So the equilibrium price is $P^* = \$30$, and the equilibrium output $Q^* = 30$

- Comparative statics analysis of the short-run equilibrium
 - e.g. What happens when the number of firms in the market goes up.
 - * Adding more firms $\Rightarrow$ shift short-run market supply curve to the right
 - * P^* falls, Q^* rises



- e.g. What happens when the market demand increases from $D \Rightarrow D'$
 - * P^* rises, Q^* rises
 - * Given the shift in demand in a market



How the market prices are determined: Long-run Equilibrium

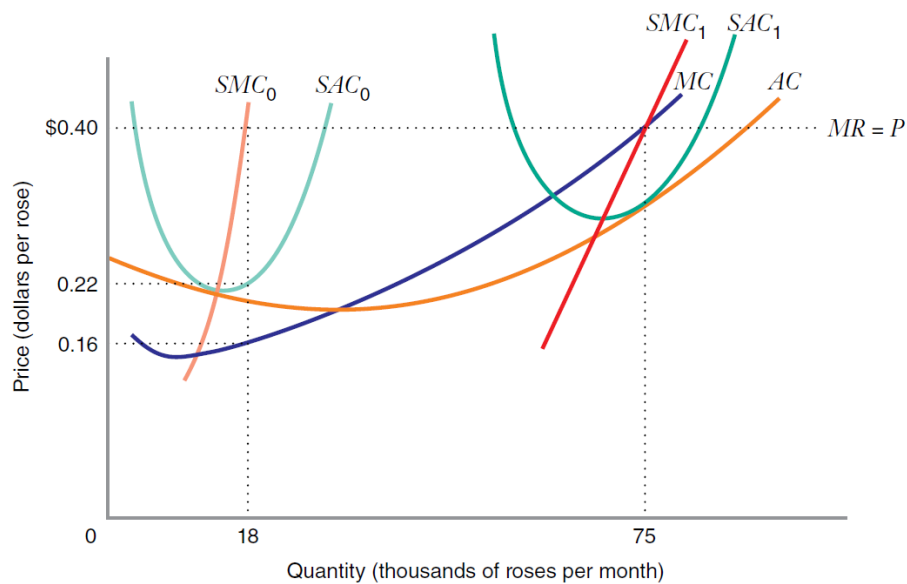
- In the **short-run**

- Firms operate within a given plant size, and the number of firms in the industry does not change.
 - * capital($\overline{K}$) is fixed
 - * number of firms($\overline{n}$) in the market is fixed
- At the short-run perfectly competitive equilibrium, firms might earn *positive* or *negative* economic profits

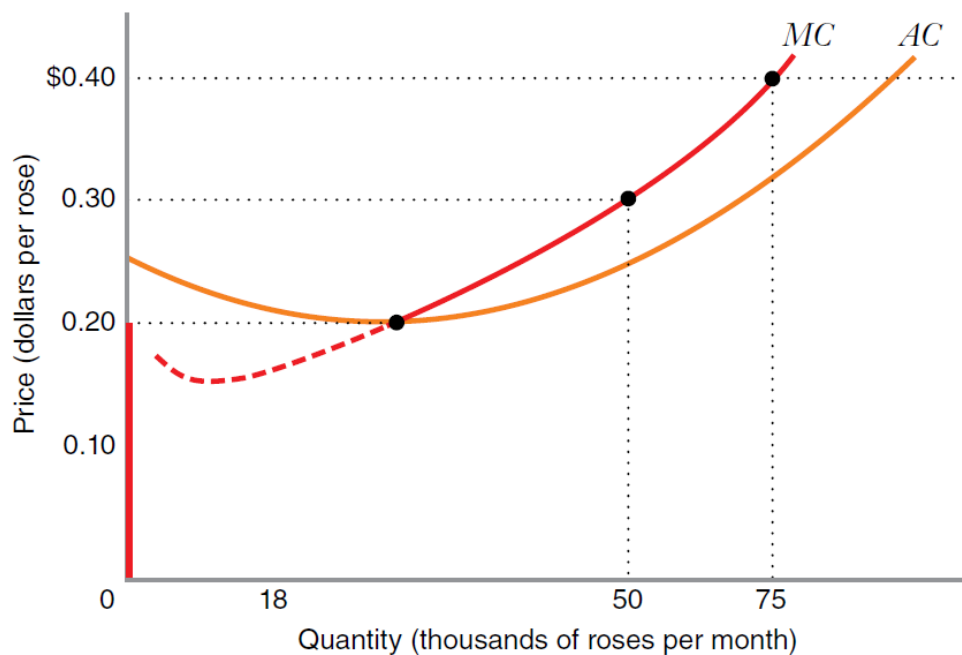
- In the **long-run**

- Established firms can adjust their plant sizes and can leave the industry altogether
 - * capital(K) is not fixed
 - * number of firms(n) in the market if not fixed
- These forces drive a *firm's economic profits to zero*(Zero Economic Profit condition)

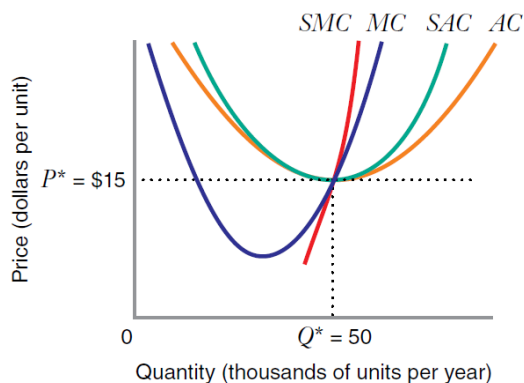
- Long-run output(Q) and plant-size(K) adjustments by established firms
 - The firm looks out over the long-run horizon, so it should evaluate the cost of those outputs(Q) using its long-run cost functions.
 - (figure) In the long-run, the producer can increase its profits by expanding its plant size and producing more output(Q) within this expanded plant size(*initial*(0) $\Rightarrow$ *afterward*(1))
 - * *initial*(0): 18,000 was the profit maximization Q because of the $P = MC = MR$
 - * *afterward*(1): 75,000 is the profit maximization Q because of the $P = MC = MR$
 - * As a result, in the long-run, the producer can increase its profit by choosing greater amount of quantity with expanded K



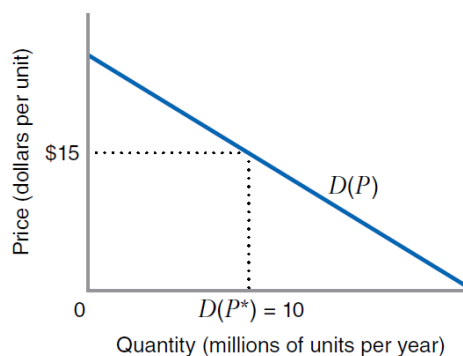
- The firm's long-run supply curve
 - A firm's long-run supply curve is its long-run marginal cost curve(MC)
 - For prices below the minimum long-run average cost, a firm would produce no output($Q = 0$), and its long-run supply curve would be a vertical spike that coincides with the vertical axis $\Rightarrow$ *shutting down*
 - * This is because the firm would earn a negative economic profit in the range between $p = 0$ to $p = \text{shut down price}$
 - * If the firm anticipated that the market price(p) would remain at such a level, its best course of action would be to exit the industry
 - The logic underlying the construction of the firm's long-run supply curve is analogous to the firm's short-run supply curve
 - * The difference in the long run is that all costs are avoidable(i.e., they are *nonsunk*)
 - * Whereas in the short run. some cost might not be avoidable(i.e., they are *sunk*)



- Free entry and long-run perfectly competitive equilibrium
 - Long-run perfectly competitive equilibrium
 - * The market price(P^*) and quantity(Q^*) at which supply equals demand, established firms(N^*) have no incentive to exit the industry, and prospective firms have no incentive to enter the industry
 - **Three conditions**
 - (1) Each firm maximizes its long-run profit with respect to output and plant size: $P^* = MC(Q^*)$
 - * Given the price P^* , each active firm chooses Q^* that maximizes its profit and selects a plant size that minimizes the cost of producing that output
 - (2) Each firm's economic profit is zero: $P^* = AC(Q^*)$
 - * Given the price P^* , a prospective entrant cannot earn positive economic profit by entering this industry. Moreover, an active firm cannot earn negative economic profit by participating in this industry.
 - (3) Market demand equals market supply: $D(P^*) = n^*Q^*$, $n^* = D(P^*)/Q^*$
 - * At the price P^* , market demand equals market supply, given the number of firms n^* and individual firm supply decisions Q^*



(a) Typical firm



(b) Market

Exercise. Suppose $TC(Q) = 40Q - Q^2 + 0.01Q^3$, and $D(P) = 25,000 - 1,000P$. Find the long-run equilibrium quantity per firm, price, and number of firms

$$1) AC = \frac{TC(Q)}{Q} = 40 - Q + 0.01Q^2$$

$$2) MC = 40 - 2Q + 0.03Q^2$$

$$3) \text{ Find } Q_f^* \text{ by setting } AC = MC$$

$$\Rightarrow 40 - Q + 0.01Q^2 = 40 - 2Q + 0.03Q^2 \Rightarrow Q_f^* = \frac{1}{0.02} = 50 (\text{for individual firm})$$

$$4) \text{ Plug } Q_f^* \text{ into } AC \text{ or } MC \text{ to find } P^*$$

$$\Rightarrow MC(Q_f^*) = 40 - 2 \cdot (50) + 0.03 \cdot (50)^2 = 15 = P^*$$

$$5) \text{ Find market demand quantity } (Q_m^*) \text{ by plugging } p^* \text{ into } D(P)$$

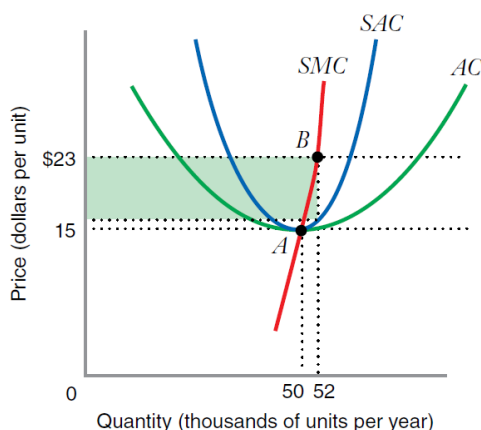
$$\Rightarrow D(P^*) = 25,000 - 1000 \cdot (15) = 10,000 = Q_m^*$$

$$6) n^* (\text{number of firms}) = \frac{Q_m^*}{Q_f^*} = \frac{10,000}{50} = 200$$

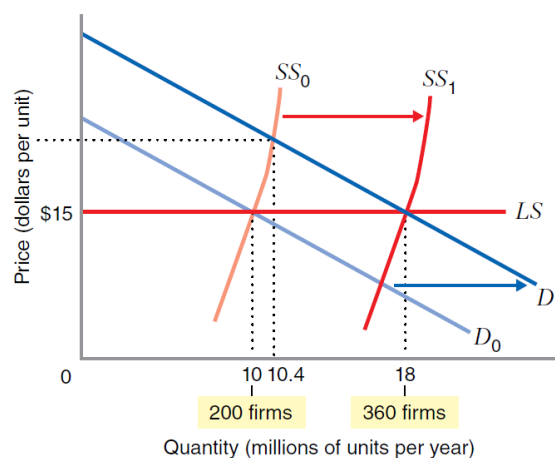
Exercise. Suppose $STC(Q) = 100 + 20Q + Q^2$, and all sunk costs are fixed. Find the shutdown price and the firm's short-run supply curve.

- Long-run market supply curve

- A curve that shows the total quantity of output that will be supplied in the market at various prices, assuming that all long-run adjustments take place
- We assume that changes in industry output do not affect input prices(w, r)
- **Long-run market supply curve is not like we obtained the short-run curve(horizontal summation of the individual supply curves)**
 - * In the long-run, market supply can vary as firms enter or exit the market; there is no fixed set of individual firm supply curves that we can sum together
- e.g. How to construct a long-run market supply curve
 - (1) Initially, the market is in long-run equilibrium at $P = \$15$. Point A represents the position of a typical firm at this long-run equilibrium
 - (2) Suppose that market demand shifts from $D_0 \rightarrow D_1$. Also, this demand shift is expected to persist, so the market will reach a new long-run equilibrium
 - (3) In the short run, equilibrium occurs at a price of \$23, with each firm maximizing profit by $Q = 52$. (Point B , intersection of SS_0 and D_1)
 - (4) At \$23, each of the firms in the market earns a positive economic profit in the short run. It attracts new firms into the market, new firms enter, and shifts the short-run supply curve rightward($SS_0 \rightarrow SS_1$)
 - (5) At this point, the price falls to \$15 again, and each firm maximizes its profit by producing $Q = 50$
 - (6) There is no incentive for additional entry or exit because each firm earns zero economic profit.
 - (7) Thus, the long-run market supply curve will be a horizontal line corresponding to the long-run equilibrium price P



(a) Typical firm



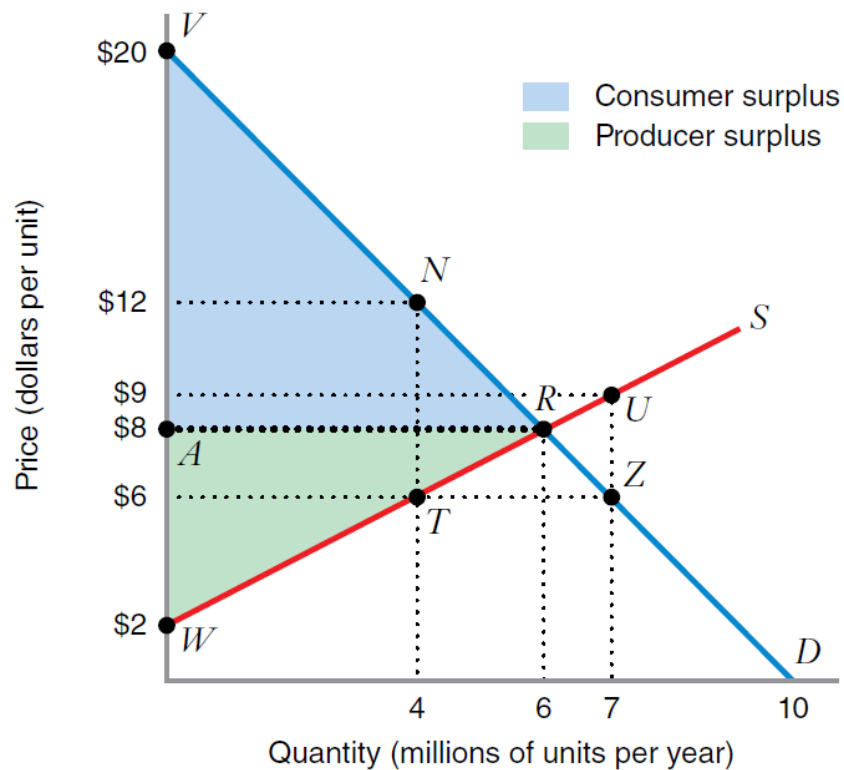
(b) Market

Ch10. Competitive Markets: Applications

The invisible hand, excise taxes, and subsidies

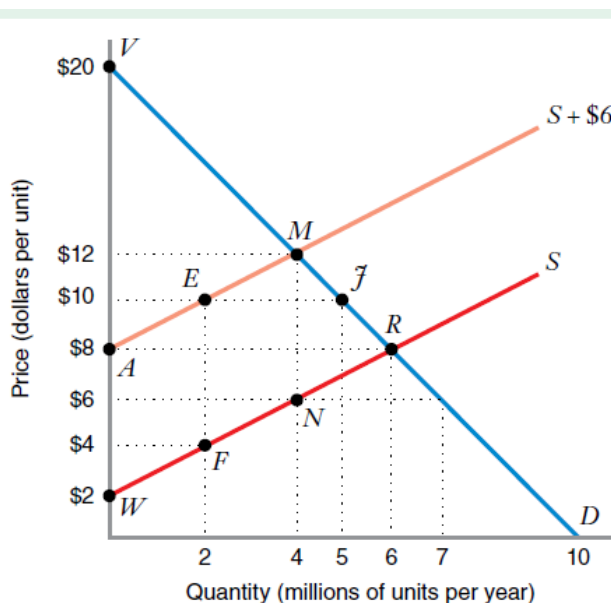
- Equilibrium
 - Partial Equilibrium
 - * An analysis that studies the determination of equilibrium price and output in a single market, taking as given the prices in all other markets
 - * e.g. the effect of rent controls on the market for housing
 - General Equilibrium
 - * An analysis that determines the equilibrium prices and quantities in more than one market simultaneously
- Government intervention
 - Sometimes governments intervene in markets because consumers are unable to gather enough information about the products in the market.
 - * e.g. Health care products, including medication and medical procedures, can be so complex that the average consumer finds it difficult to make informed choices.
- Externalities
 - Definition: The effect that an action of any decision maker has on the well-being of other consumers or producers, beyond the effects transmitted by changes in prices.
 - Externalities are present in a market if the actions of either consumers or producers lead to costs or benefits that are not reflected in the price of the product in that market.

- The Invisible Hand
 - Remarkable features of a perfectly competitive market:
 - * In equilibrium, a competitive market allocates resources efficiently
 - * The sum of consumer and producer surplus will be the triangle like the figure below
 - * In a perfectly competitive market, each producer and consumer acts in its own self-interest. There is no omniscient social planner telling producers and consumers how to behave so that an efficient level of output is produced.



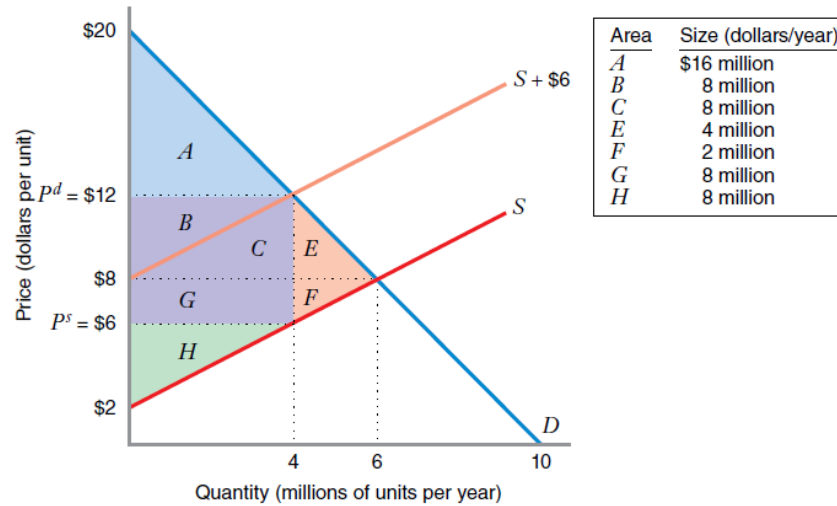
- Excise Taxes

- An excise is a tax on a specific commodity, such as gasoline, alcohol, tobacco, or airline tickets.
 - * e.g. How a gasoline tax will affect the price consumers pay for gasoline, as well as the price producers receive.
- If there is no tax
 - * $Q^s = Q^d = 6$ when $P^s = P^d = \$8$
- If the government imposes an excise tax of \$6 per unit
 - * A seller has the "administrative responsibility" to collect the tax
 - * If tax(T) is \$6, then $P^d = P^s + T \rightarrow P^d = P^s + \6
 - In general, the price P^s that a seller receives will be \$ T less than the price P^d that a buyer pays
- The effects of an excise tax
 - * The market will underproduce relative to the efficient level
 - * Consumer surplus will be lower than with no tax
 - * Producer surplus will be lower than with no tax
 - * The impact on the government budget will be positive because tax receipts are collected. The tax receipts are part of the net benefit to society because they will be distributed to people in the economy
 - * The tax receipts will be less than the decrease in consumer and producer surplus. Thus, the tax will cause a reduction in net economic benefits(DWL)



- Deadweight loss

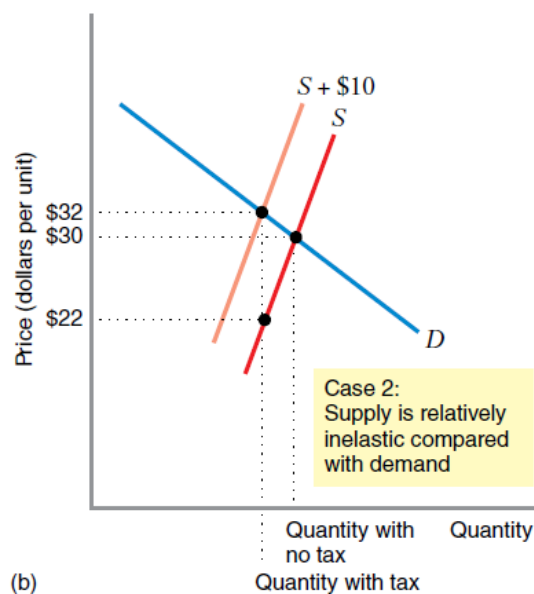
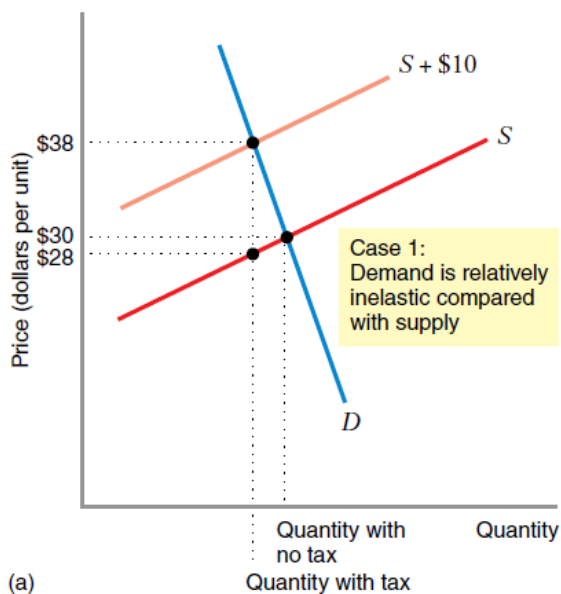
- Definition: a reduction in net economic benefits resulting from an inefficient allocation of resources



	With No Tax	With Tax	Impact of Tax
Consumer surplus	$A + B + C + E$ (\$36 million)	A (\$16 million)	$-B - C - E$ (-\$20 million)
Producer surplus	$F + G + H$ (\$18 million)	H (\$8 million)	$-F - G$ (-\$10 million)
Government receipts from tax	zero	$B + C + G$ (\$24 million)	$B + C + G$ (\$24 million)
Net benefits (consumer surplus + producer surplus + government receipts)	$A + B + C + E + F + G + H$ (\$54 million)	$A + B + C + G + H$ (\$48 million)	$-E - F$ (-\$6 million)
Deadweight loss	zero	$E + F$ (\$6 million)	$E + F$ (\$6 million)

- Incidence of a tax

- Definition: A measure of the effect of a tax on the prices consumers pay and sellers receive in a market.
- An excise tax will increase the market price that consumers pay but will decrease the net after-tax price that sellers receive.
- The incidence, or burden, of the tax is shared by both consumers and producers
- The incidence of a tax depends on the shapes of the supply and demand curves
 - * **Case 1)** The demand curve is relatively inelastic, while the supply curve is quite elastic
 - The price change resulting from the tax is larger for consumers because demand is comparatively inelastic
 - * **Case 2)** The supply curve is relatively inelastic, while the demand curve is comparatively elastic
 - The tax has a larger impact on producers, decreasing the price they receive by \$8, while increasing the price consumers pay by only \$2



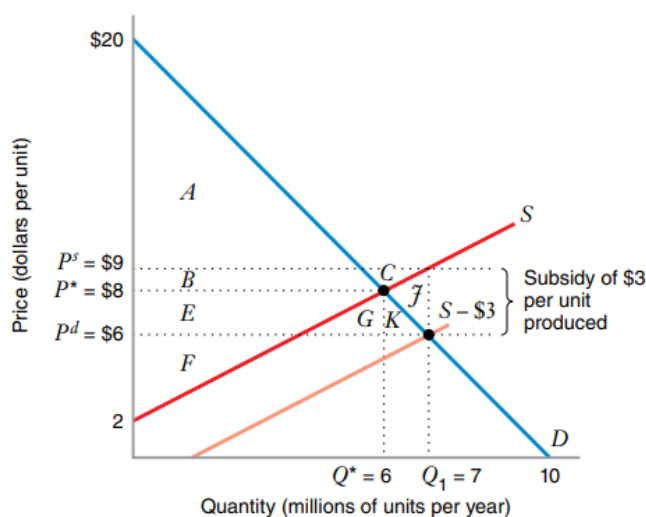
- Quantitative relationship between the incidence of a tax and the price elasticities of supply and demand

$$\frac{\Delta P^d}{\Delta P^s} = \frac{\epsilon_{Q^s,P}}{\epsilon_{Q^d,P}}$$

- The equation tells us that the impact of the price change on consumers and producers will be equal when the absolute values of the price elasticities are the same
- e.g. If $\epsilon_{Q^d,P} = -0.5$, $\epsilon_{Q^s,P} = 0.5$, then $\frac{\Delta P^d}{\Delta P^s} = -1$
 - * In other words, if a tax of \$1 were imposed, the price consumers pay would rise by \$0.5, while the price producers receive would fall by \$0.5
- e.g. If $\epsilon_{Q^d,P} = -0.5$, $\epsilon_{Q^s,P} = 2.0$, then $\frac{\Delta P^d}{\Delta P^s} = -4$.
 - * The increase in the price consumers pay will be four times as much as the decrease in the price producers receive
 - * If an excise tax of \$1 were imposed, the price consumers pay would rise by \$0.80, while the price producers receive would fall by \$0.20.

- Subsidies

- We can think of a subsidy as a *negative tax*
 - * Buyers pay the market price P^d
 - * Sellers get paid P^s which is the summation of the market price(P^d) and the government subsidy of $\$T$ per unit: $P^s = P^d + T$
- The effects of a subsidy
 - * The market will overproduce relative to the efficient level(i.e., the amount that would be supplied with no subsidy).
 - * Consumer surplus will be higher than with no subsidy
 - * Producer surplus will be higher than with no subsidy
 - * The impact on the government budget will be negative. Government expenditures on the subsidy constitute a negative net economic benefit
 - * Government expenditures on the subsidy will be larger than the increase in consumer and producer surplus $\Rightarrow$ Deadweight loss from over-production



	With No Subsidy	With Subsidy	Impact of Subsidy
Consumer surplus	$A + B$ (\$36 million)	$A + B + E + G + K$ (\$49 million)	$E + G + K$ (\$13 million)
Producer surplus	$E + F$ (\$18 million)	$B + C + E + F$ (\$24.5 million)	$B + C$ (\$6.5 million)
Impact on government budget	zero	$-B - C - E - G - K - J$ (-\$21 million)	$-B - C - E - G - K - J$ (-\$21 million)
Net benefits (consumer surplus + producer surplus – government expenditures)	$A + B + E + F$ (\$54 million)	$A + B + E + F - J$ (\$52.5 million)	$-J$ (-\$1.5 million)
Deadweight loss	zero	J (\$1.5 million)	

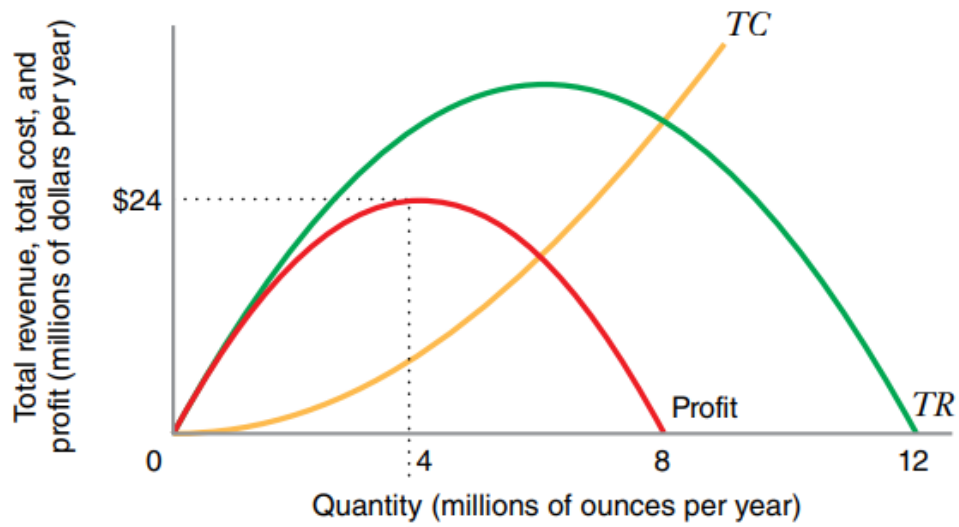
Ch11. Monopoly and Monopsony

Profit Maximization by a Monopolist

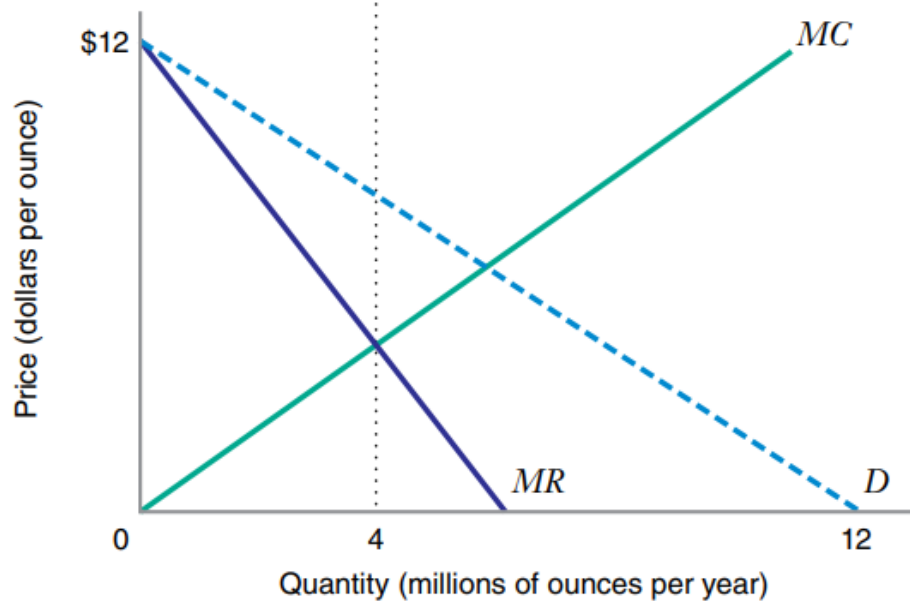
- A monopolist
 - A monopolist sets the market price for its product.
 - The monopolist must take account of the market demand curve
 - * The higher the price it sets, the fewer units of its product it will sell
 - * The lower the price it sets, the more units it will sell
 - Thus, the monopolist's market demand curve is downward sloping.
- The profit-maximization condition
 - Definition: The condition that says that a monopolist maximizes profit(π) by producing a quantity(Q^*) at which $MR(Q^*) = MC(Q^*)$
 - The profit-maximizing monopolist's problem is finding the optimal trade-off between volume(the number of units it sells) and margin(the differential between price and marginal cost on the units it sells).
 - **Monopolist's total revenue:** $TR(Q) = P(Q) \times Q$
 - **Monopolist's profit:** $\pi = TR(Q) - TC(Q)$
 - Summary
 - (1) If the firm produces a Q at which $MR > MC$, the firm cannot be maximizing its π because it could increase its output and its profit would go up
 - (2) If the firm produces a Q at which $MR < MC$, the firm cannot be maximizing its π because it could decrease its output and its profit would go up
 - (3) Thus, the only situation at which the monopolist cannot improve its profit by increasing or decreasing output is where $MR(Q^*) = MC(Q^*)$

- e.g. Suppose a monopolist faces the market demand curve $P(Q) = 12 - Q$, and $TC(Q) = \frac{1}{2}Q^2$ like the below.
 - To sell 2 million ounces $\Rightarrow$ the monopolist would charge \$10 per ounce
 - To sell 5 million ounces $\Rightarrow$ the monopolist would charge \$7 per ounce
 - As we move along the monopolist's demand curve, different Q and their associated P generate different amounts of TR for the monopolist.
 - For less than $Q = 4\text{million}$, increasing the output increases TR more than it increases TC , which moves the firm up its profit hill.
 - * Over this range, the monopolist's MR exceeds its MC : $MR > MC$
 - For greater than $Q = 4\text{million}$, producing less output increases profit. Over this range, decreasing Q decreases TC faster than it decreases TR , which also moves the firm up its profit hill.
 - * Over this range, the monopolist's MR is less than its MC : $MR < MC$
 - The profit-maximization condition for a monopolist is the quantity at which $MR = MC$ where MR and MC curves cross.
 - * In this example, $Q = 4$ satisfies the profit-maximization condition

Q (million ounces)	P (\$/oz.)	TR (\$ million)	TC (\$ million)	Profit (\$ million)
0	12	0	0	0
1	11	11.00	0.50	10.50
2	10	20.00	2.00	18.00
3	9	27.00	4.50	22.50
4	8	32.00	8.00	24.00
5	7	35.00	12.50	22.50
6	6	36.00	18.00	18.00
7	5	35.00	24.50	10.50
8	4	32.00	32.00	0
9	3	27.00	40.50	-13.50
10	2	20.00	50.00	-30.00



(a)



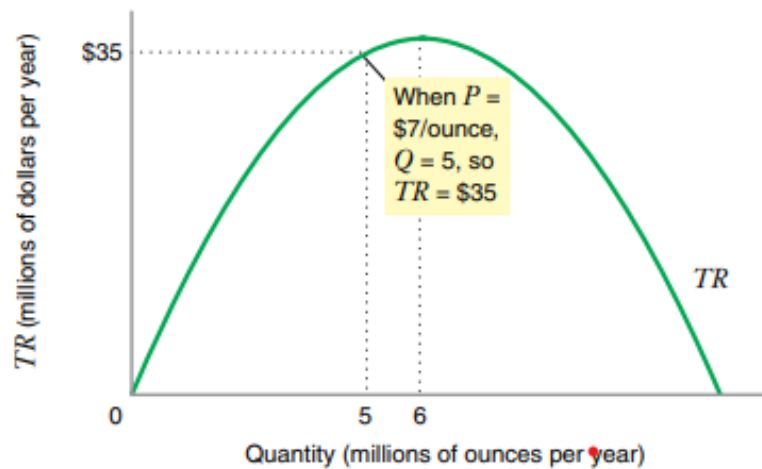
(b)

- A closer look at marginal revenue
 - In Chapter 9, for a price-taking firm, marginal revenue(MR) equals to the market price (P)
 - **However, for a monopolist, marginal revenue(MR) is not equal to market price(P)**

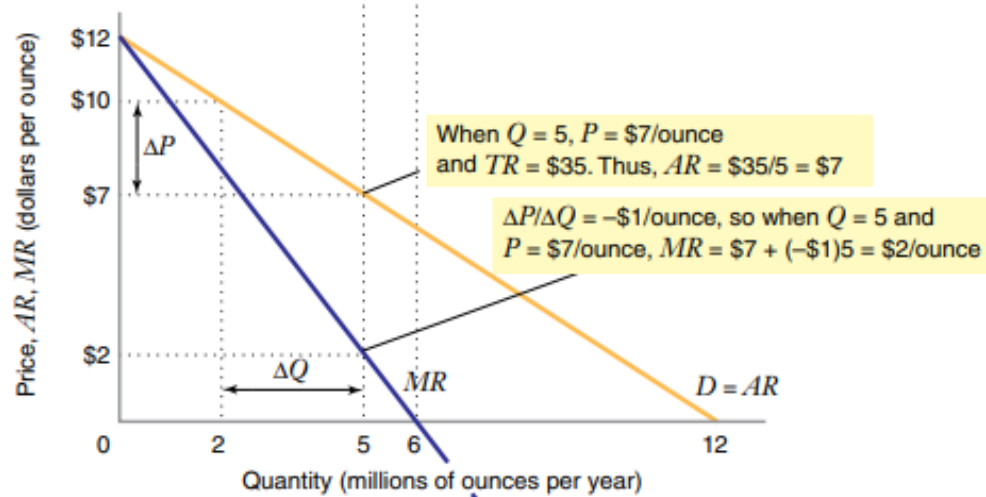
$$MR = \frac{\Delta TR}{\Delta Q} = \frac{\Delta(P(Q) \times Q)}{\Delta Q} = \frac{P\Delta Q + Q\Delta P}{\Delta Q} = P + Q \cdot \frac{\Delta P}{\Delta Q}$$

- * The first part of MR is P
- * The second part, $Q \cdot \frac{\Delta P}{\Delta Q}$ is negative since ΔP is negative.
- * Thus, $MR = P + Q \cdot \frac{\Delta P}{\Delta Q} (< 0) < P$: that is, the marginal revenue is less than the price the monopolist can charge to sell that quantity, for any quantity greater than 0
- The monopolist's Average Revenue(AR) and Marginal Revenue(MR)
 - Average revenue(AR): Total revenue per unit of output(i.e., the ratio of total revenue to quantity)

$$AR(Q) = \frac{TR}{Q} = \frac{(P(Q) \times Q)}{Q} = P(Q)$$
 - * Thus, average revenue is equal to price.
 - * The monopolist's average revenue coincides with the market demand curve($P(Q)$)
- If the monopolist's output is positive($Q > 0$)
 - (1) Marginal revenue is less than price($MR < P$)
 - (2) Because average revenue is equal to price, marginal revenue is less than average revenue($MR < AR$)
 - (3) Since the average revenue curve coincides with the demand curve, the marginal revenue curve must lie below the demand curve

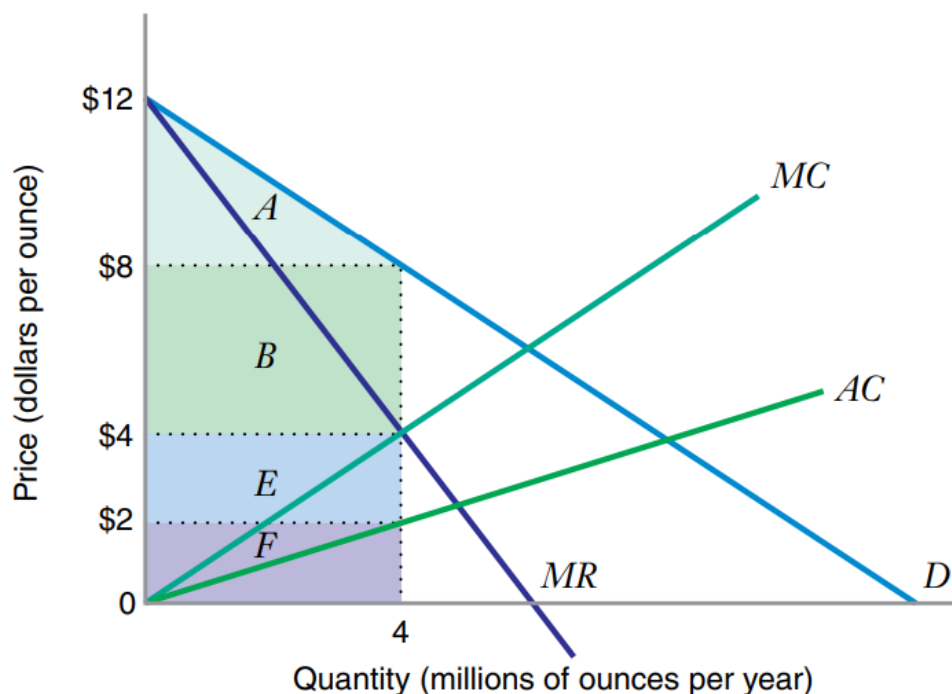


(a)



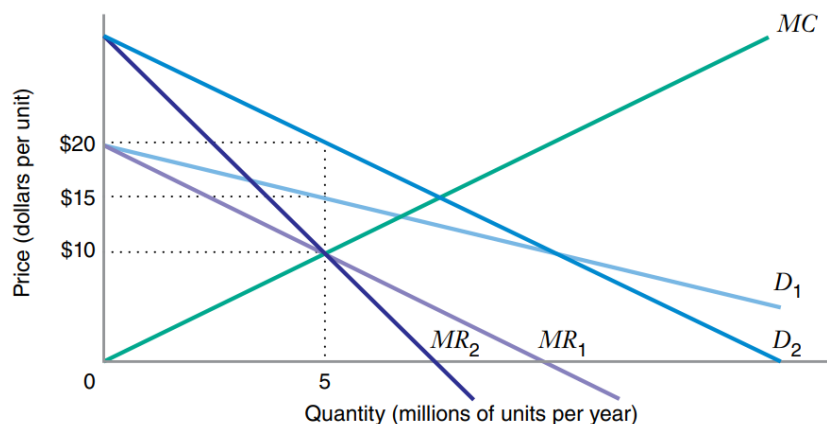
(b)

- The profit-maximization condition shown graphically



- The marginal revenue curve MR is decreasing and lies below the demand curve (which is also the average revenue curve)
- The marginal cost curve MC is a straight line from the origin, as is the average cost curve AC . For all positive output levels, the marginal cost curve lies above the average cost curve.
- The profit-maximizing quantity is the quantity at the point where MR and MC intersect: $Q = 4$
- The profit-maximizing price is the price at which that quantity meets the demand curve: \$8 per ounce
 - * At that price, the quantity demanded is 4 million ounces per year
- At this profit-maximization condition, profit(π) equals total revenue($P(Q) \times Q$) minus total cost($AC(Q) \times Q$)
 - * $TR(B + E + F) - TC(F) = profit(B + E) = \24 millions

- Properties of the equilibrium in a monopoly market:
 - (1) The monopolist's profit-maximization price(\$8) exceeds the marginal cost of the last unit supplied(\$4).
 - This differs from the outcome in a perfectly competitive market, in which price equals the marginal cost of the last unit supplied.
 - (2) The monopolist's economic profits can be positive.
 - This is in contrast to a perfectly competitive firm in a long-run equilibrium because the monopolist does not face the threat of free entry that drives economic profits to zero in competitive markets.
 - (3) Even though the monopolist raises price above marginal cost and earns positive economic profits, consumers still enjoy some benefits at the monopoly equilibrium.
 - The consumer surplus at the equilibrium: area A
 - The monopolist's producer surplus: area $B + E$
 - The total economic benefit at the monopoly equilibrium: area $A + B + E$
- A monopolist does not have a supply curve
 - For the monopolist, price is endogenous, not exogenous.
 - That is, the monopolist determines both quantity(Q) and price(P).
 - Depending on the shape of the demand curve, the monopolist might supply the same quantity at two different prices or different quantities at the same price.
 - **Thus, a monopolist does not have a supply curve**



Exercise. Consider a monopolist facing the following linear demand curve: $Q = 60 - 2P$ where Q is the quantity demanded and P is the price. The monopolist has a total cost (TC) function given by: $TC(Q) = 2Q^2$

- (1) Derive the monopolist's marginal cost (MC) curve.
- (2) Derive the monopolist's marginal revenue (MR) curve.
- (3) Determine the profit-maximizing quantity (Q^*) and price (P^*).
- (4) Calculate the monopolist's profit at equilibrium.

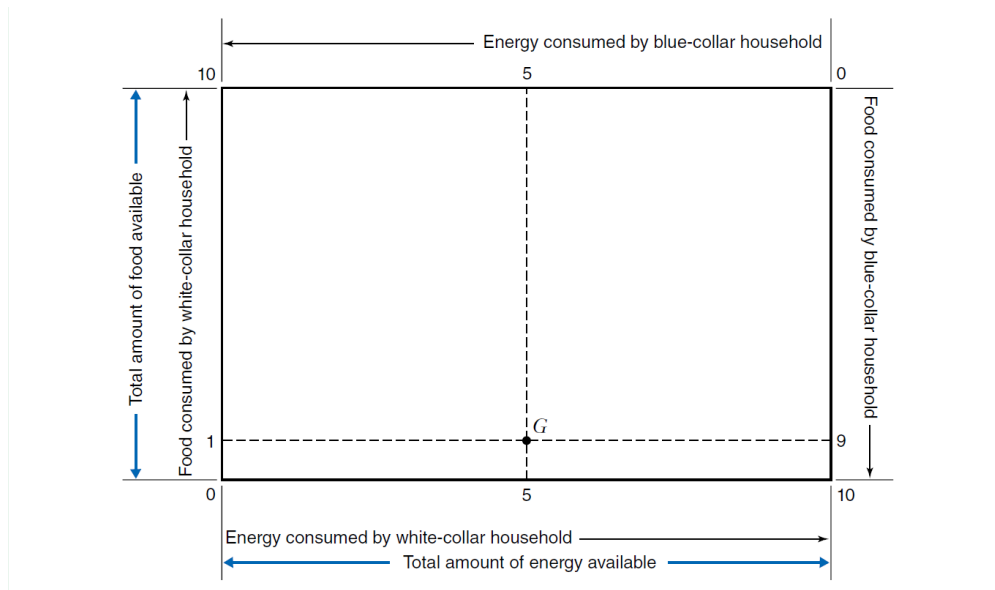
Ch16. General Equilibrium Theory

The efficiency of competitive markets

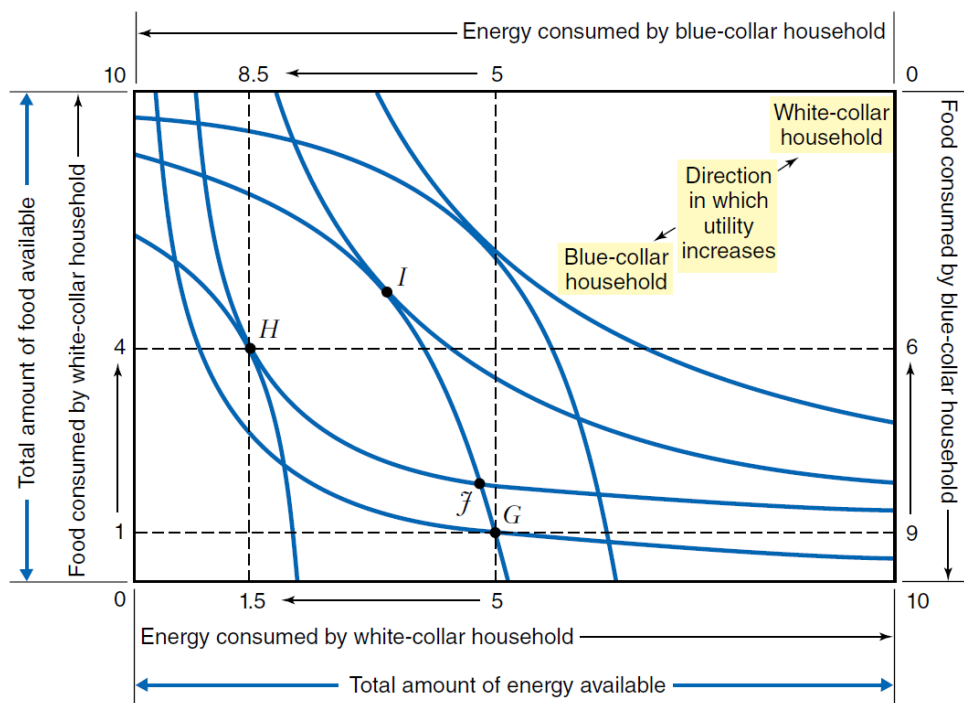
- What is Economic Efficiency?
 - Allocation of goods and inputs
 - * A pattern of consumption and input usage that might arise in a general equilibrium in an economy.
 - * e.g. energy and food consumed by households, labor and capital used by industries
 - Economically efficient
 - * (**Pareto efficient**) Characteristic of an allocation of goods and inputs in an economy if there is no other feasible allocation of goods and inputs that would make some consumers better off without hurting other consumers.
 - Economically inefficient
 - * (**Pareto inefficient**) Characteristic of an allocation of goods and inputs if there is an alternative feasible allocation of goods and inputs that would make all consumers better off as compared with the initial allocation
 - * We can always find at least one efficient allocation that consumers would unanimously prefer to the inefficient one
 - * At an inefficient allocation of goods and inputs, the economy is not getting all that it can get from its resources
- Competitive equilibrium needs to satisfy three conditions if it is to be efficient
 - (1) Exchange Efficiency
 - A characteristic of resource allocation in which a fixed stock of consumption goods **cannot be reallocated among consumers in an economy without making at least some consumers worse off**
 - (2) Input Efficiency
 - A characteristic of resource allocation in which a fixed stock of inputs **cannot be reallocated among firms in an economy without reducing the output of at least one of the goods that is produced in the economy.**
 - (3) Substitution Efficiency
 - A characteristic of resource allocation in which, given to total amounts of capital and labor that are available in the economy, **there is no way to make all consumers better off by producing more of one product and less of another**

Exchange Efficiency

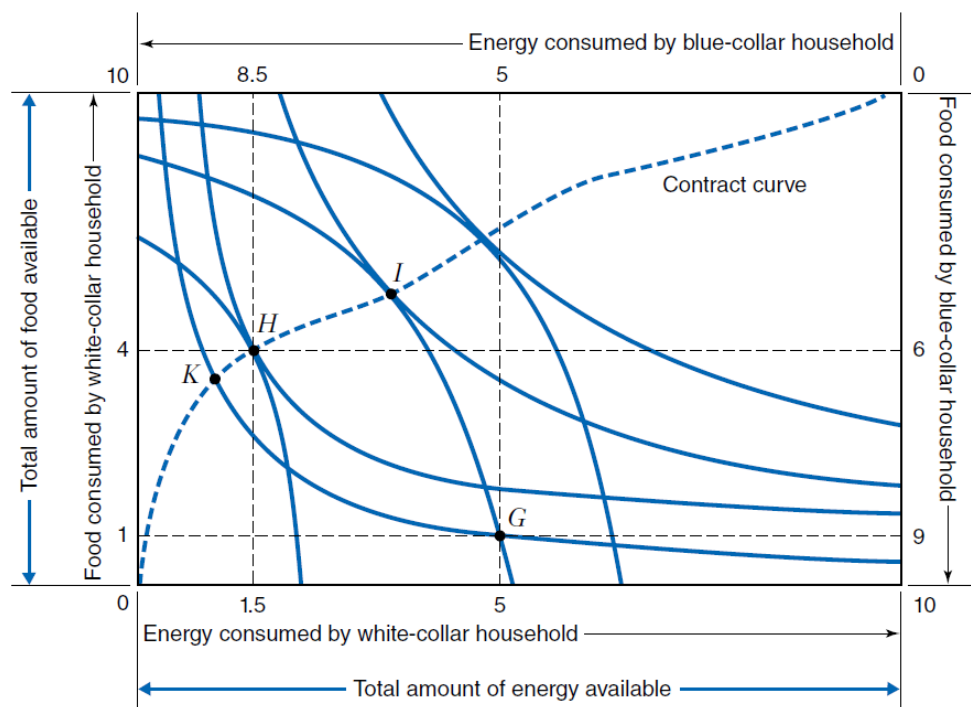
- Edgeworth box
 - A graph showing all the possible allocations of goods in a two-good economy, given the total available supply of each good
 - e.g. two household(A and B), two goods(Energy and Food) Economy
 - * Point G: A consumes 5 units of energy and 1 unit of food, while B consumes 5 units of energy and 9 units of food.



- Describing Exchange Efficiency using the Edgeworth Box
 - Does the point G satisfy the condition of exchange efficiency?
 - * It depends on the preferences of the households
 - * In other words, it depends on their utility functions
 - Point G cannot represent an exchange efficient allocation of the two goods
 - * There are points, such as point H, where both households would be on higher indifference curves
 - * They could gain by exchanging(trading)
 - e.g. Consumer A could give the B 3.5 units of energy in exchange for 3 units of food ($G \rightarrow H$)
 - In this case, both households would be better off
 - At an allocation that is economically inefficient in exchange, there are potential exchanges(trades) among consumers that would benefit all consumers
 - Exchange efficient allocations are represented by points where indifference curves are tangent
 - * Because moving from such a point would make at least one household worse off
 - At an allocation that is economically efficient in exchange, there are no potential trades among consumers that would benefit all consumers.



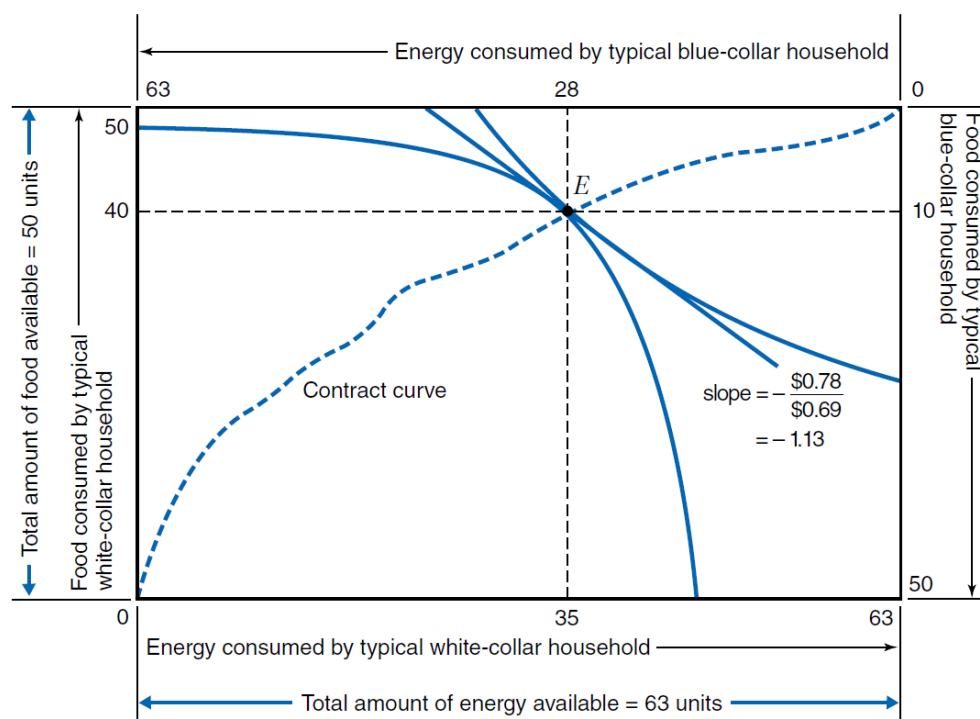
- The contract curve
 - Contract curve
 - * A curve that shows all the allocations of goods in an Edgeworth box that are economically efficient
 - * If the two households were free to bargain and make trades of the two goods and if all their trades were mutually beneficial, they would bargain their way to an allocation that was economically efficient in exchange
 - The exact point they would reach would depend on the *endowment* (initial allocation of goods)
 - * e.g Endowments at Point G
 - They would reach a point on the contract curve between points I and K
 - Between points I and K, both households are at least as well off as they are at point G
 - But below point K or above point I, one household should be worse off



- $MRS_{x,y}^A = MRS_{x,y}^B = \frac{P_x}{P_y}$

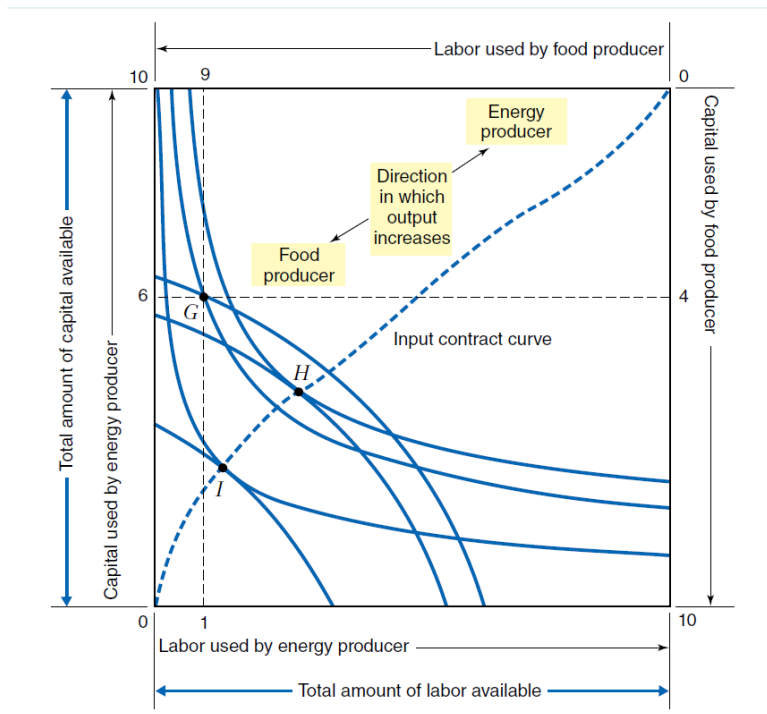
- General Competitive Equilibrium satisfies exchange efficiency

- * At general equilibrium, the $MRS_{x,y}$ equals the slope of the household's indifference curve,
- * And the indifference curves of the two types of households are tangent to each other
- * And tangent to a line whose slope equals the ratio of the equilibrium prices of energy and food
- * Finally, since the indifference curves are tangent, the allocation of energy and food at the equilibrium must be on the contract curve and must satisfy exchange efficiency



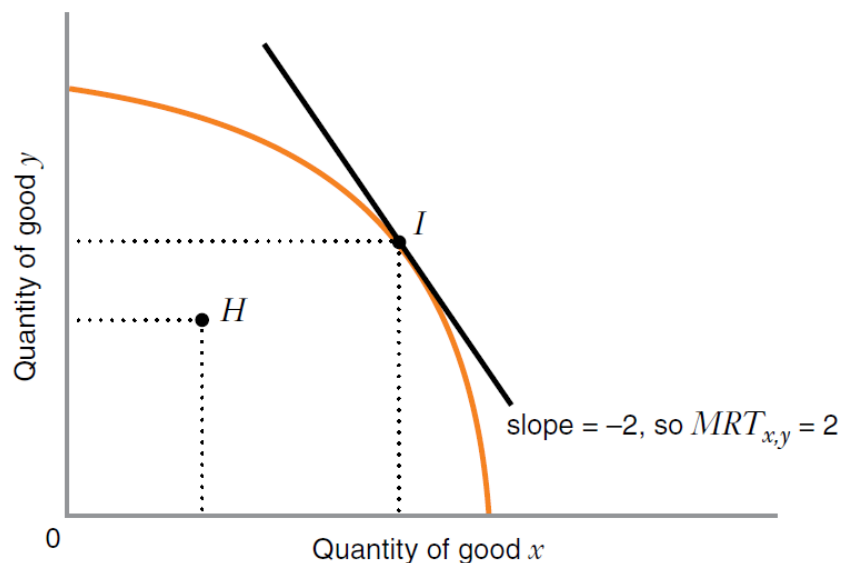
Input Efficiency

- Describing input efficiency using the Edgeworth box
 - Every point in the box for inputs is on two isoquants.
 - * At some points(G), the two isoquants cross
 - It represents economically inefficient allocations of inputs because at such points it is possible to reallocate inputs so as to increase outputs in both industries simultaneously
 - * While at other points(H), the two isoquants are tangent to each other.
 - It represents economically efficient allocations of inputs because no such reallocations are possible
 - Input contract curve
 - * A curve that shows all the input allocations in an Edgeworth box for inputs that are input efficient
- $MRTS_{L,K}^x = MRTS_{L,K}^y = \frac{w}{r}$
 - General Competitive Equilibrium satisfies input efficiency
 - * $MRTS$ are the absolute values of the slopes of the isoquants
 - * And since these slopes are equal(tangent) at a competitive equilibrium, it follows that a general competitive equilibrium satisfies input efficiency
 - * There is no reallocation of inputs across industries that would allow one industry to increase its output without reducing output in the other industry



Substitution Efficiency

- Production Possibilities Frontier(PPF)
 - Definition: A curve that shows all possible combinations of consumption goods that can be produced in an economy given the economy's available supply of inputs
 - Downward slopping
 - * When the allocation of inputs across industries satisfies the condition of input efficiency, if more of good x is produced, less of good y is produced.
- Marginal Rate of Transformation(MRT)
 - Definition: The absolute value of the slope of the production possibilities frontier
 - MRT shows the amount of good y that the economy must give up in order to gain one additional unit of good x
 - e.g. At point I , $MRT_{x,y} = 2$.
 - * It means the slope is -2 so the economy can get one additional unit of good x only by sacrificing 2 units of y
 - $MRT_{x,y} = \frac{MC_x}{MC_y}$
 - * $MRT_{x,y}$ tells us the marginal opportunity cost of good x in terms of forgone units of good y



- $MRT_{x,y} = MRS_{x,y}^A = MRS_{x,y}^B$
 - General competitive equilibrium satisfies substitution efficiency
 - * Household utility maximization: $MRS_{x,y}^A = MRS_{x,y}^B = \frac{P_x}{P_y}$
 - * Profit maximization by competitive firms implies that price equals marginal cost in both the goods: $P_x = MC_x$, and $P_y = MC_y \rightarrow \frac{P_x}{P_y} = \frac{MC_x}{MC_y}$
 - * Therefore, we get $MRT_{x,y} = \frac{MC_x}{MC_y}$
 - * That is, substitution efficiency is satisfied at the general competitive equilibrium

Fundamental Theorems of Welfare Economics

• First Fundamental Theorem of Welfare Economics

- Definition: The allocation of goods and inputs that arise in a general competitive equilibrium is economically efficient. That is, given the resources available to the economy, there is no other feasible allocation of goods and inputs that could simultaneously make all consumers better off
 - * Even though households and firms in our economy behave independently and each pursues its own self-interest, the resulting equilibrium is efficient
 - * However, there is no guarantee that all consumers fare equally well under the equilibrium
 - * The well-being of an individual consumer depends on his or her endowment of scarce economic resources
- Utility possibilities frontier
 - * A curve that connects all the possible combinations of utilities that could arise at the various economically efficient allocations of goods and inputs in a two-consumer economy
 - * At point E , person A enjoys greater utility than the B, while at point F , the distribution of utility is more equal

• Second Fundamental Theorem of Welfare Economics

- Definition: Any economically efficient allocation of goods and inputs can be attained as a general competitive equilibrium through a judicious allocation of the economy's scarce supplies of resources.
 - * It reveals the possibility that an economy could simultaneously attain an efficient allocation and one in which the resulting distribution of utility is in some sense equitable

Reference

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2. Varian, Hal R. *Intermediate microeconomics with calculus: a modern approach*. WW norton & company, 2014.